

SECTION 11.0

PLASTIC AND RECONSTRUCTIVE SURGERY

CHAPTER 11.1

Facial Cosmetic Surgery

SURGEONS

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INTRODUCTION TO COSMETIC FACIAL SURGERY

The presenting signs of the aging face are predictable, based on the effects of gravity, soft tissue atrophy, changes in the bony skeleton, facial expression, ultraviolet radiation exposure, and connective tissue changes. Patients present with concerns about appearing tired, angry, or aged. They may also have functional difficulties, such as difficulty breathing or visual field obstruction due to drooping brows or overhanging eyelids. Other common complaints are wrinkles around the eyes and mouth, and a sagging or fatty chin and neck (turkey waddle).

Cosmetic facial surgery aims to rejuvenate and restore the facial form by surgical manipulation of the hard and soft tissues. The techniques used involve any or all of the following: soft tissue release, resection, plication, augmentation, and resuspension.

Facial aging takes place simultaneously in all areas of the face and neck; combined procedures are not uncommon. Generally in combined procedures, a brow lift would precede a neck lift, followed by a facelift. Blepharoplasty procedures can be performed at different times in the surgical scheme because of their effect on eyelid tissue and brow posture. A rhinoplasty ideally is reserved for last, as it can cause bleeding and swelling that can obscure other facial surgical fields.

FACELIFT AND NECK LIFT

SURGICAL CONSIDERATIONS

FACELIFT (OR MELOPLASTY OR RHYTIDECTOMY)

Description

Facelifts and **midface lifts** are procedures to rejuvenate the face by surgical manipulation of the soft tissues between the inferior orbital rim and the inferior border of the mandible. The lips and nose are generally unaffected. Many types of “facelift” procedures have been developed to address the diverse challenges of facial aging and rejuvenation. Traditional facelift procedures took place in the subcutaneous plane, with some skin resection. Today, three planes of dissection are used ([Fig. 11.1-1](#)). **Subcutaneous dissection** continues to be popular, traversing the adipose tissue below the skin and many of the vessels supplying the skin. The **sub-SMAS** technique develops the plane between the superficial musculoaponeurotic system of the face (SMAS) and the parotid gland.

Finally, **subperiosteal dissection** (midface lift) elevates the periosteum off the overlying bones thereby keeping the skin and underlying soft tissues as one unit. Combinations of these dissection planes have also been described. More recently, greater attention has been paid to minimal access techniques. Surgeries such as the minimal access cranial suspension lift, as well as adjunctive procedures such as the barbed suture lift, are being performed more frequently. See [Figure 11.1-2](#).

Local anesthetics with epinephrine are injected prior to making the incision for the various procedures. A number of subcutaneous infiltration mixtures may be used, including one described by **Klein** that consists of NS 1000 mL with 1.0 mL epinephrine (1:1000) and 50 mL of 1% lidocaine $\pm$ 12.5 mL of 8.5% sodium bicarbonate solution. It has been shown that with the use of this mixture, 5–7 times the traditionally accepted maximum dose of lidocaine with epinephrine can be injected safely into the subcutaneous space. Not only does this solution provide hemostasis and hydrodissection, but decreased operative time and excellent perioperative analgesia also have been attributed to its use.

The traditional incision used begins either in the hair or at the hairline in the temporal region. It then extends inferiorly in front of the ear. If needed, the incision will curve around the earlobe and go superiorly behind the ear and then turn posteriorly into the hair or follow the hairline toward the occiput. The approaches for the subcutaneous and

SMAS techniques resemble those used for bilateral facial palsy and parotid gland operations. Techniques to address the midface may be carried out through intraoral, temporal, and/or lower-lid incisions. These techniques may be performed separately or may be combined with other facelift procedures.

A typical facelift begins with subcutaneous dissection of the facial skin flap ([Fig. 11.1-3](#)) with meticulous hemostasis accomplished with bipolar electrocautery. Then either a plication of the SMAS layer, excision of a strip of SMAS and closure, or a sub-SMA dissection with elevation and resuspension of the SMAS can be accomplished. Some surgeons continue on the same side with skin resection and closure before beginning on the other side, whereas others temporarily pack the first side and perform an identical procedure on the opposite side. In the latter case, a second look for bleeding is made on each side after a waiting period.

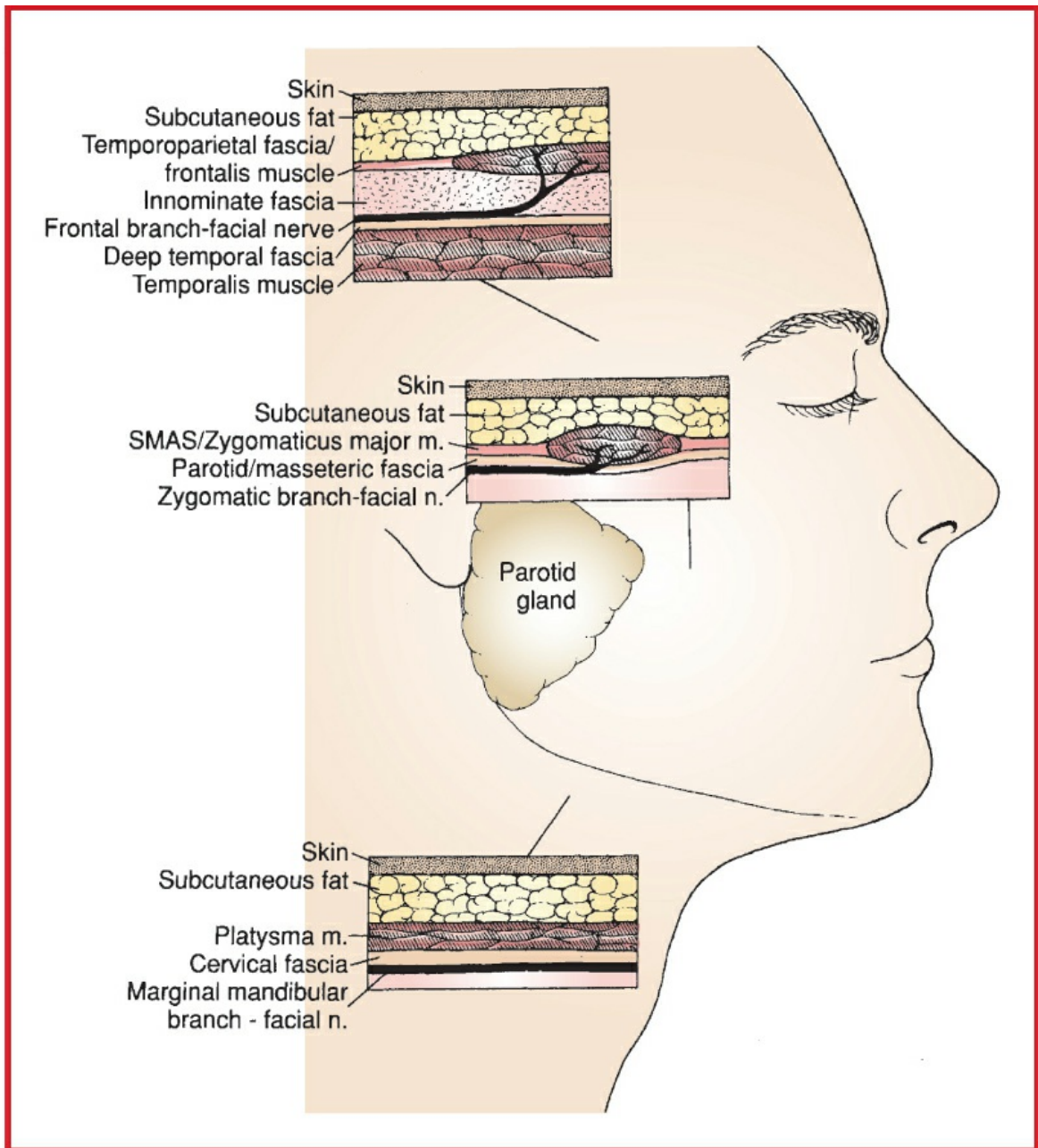


Figure 11.1-1 Anatomic layers of the face. Although the names vary, the arrangement persists, regardless of the area of the face. The facial nerve (CN VII) branches innervate their respective muscles of the SMAS layer via the deep surfaces. (Reproduced with permission from Thorne CHM, Aston SJ: Aesthetic surgery of the aging face. In: Aston SJ, Beasley RW, Thorne CHM, eds Grabb & Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

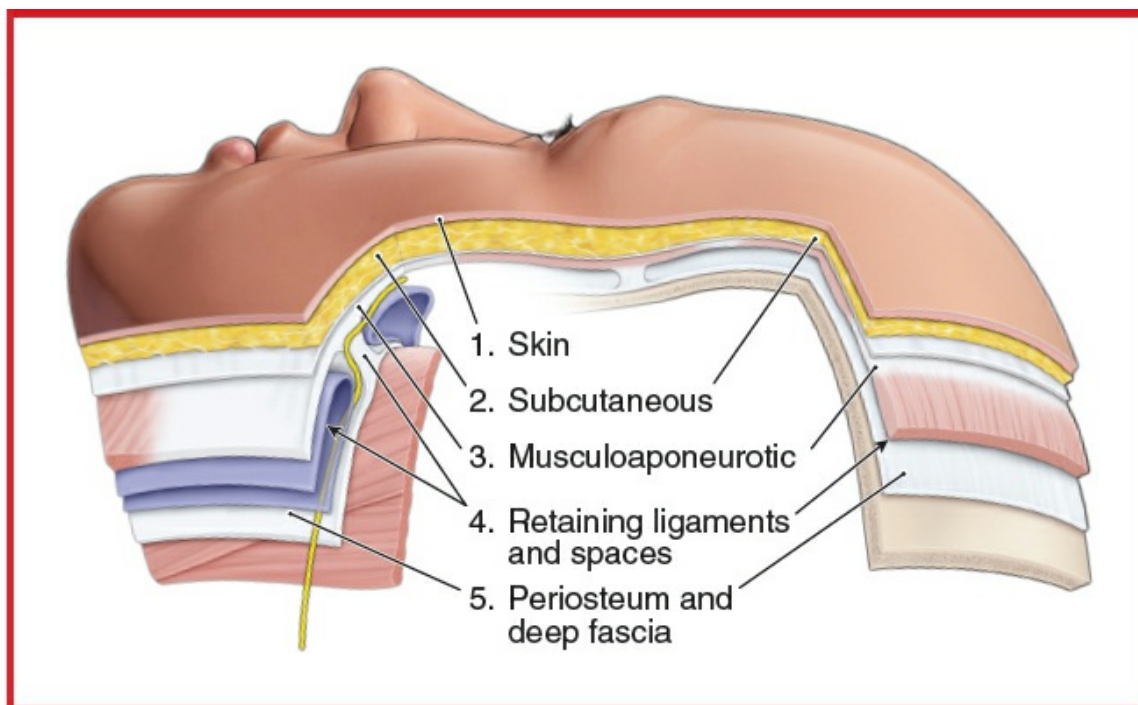


Figure 11.1-2 Tissue layers of the face and neck.

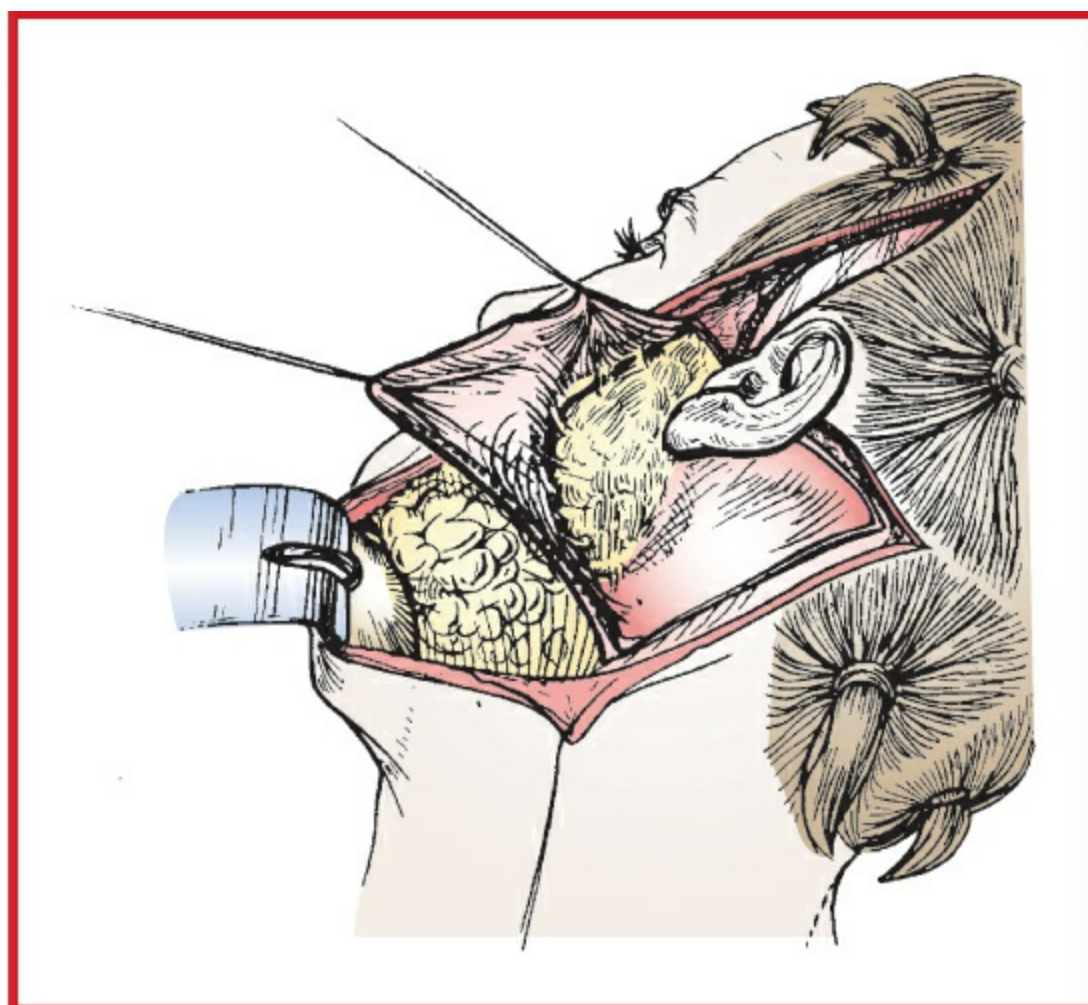


Figure 11.1-3 Dissection of SMAS/platysma flap. If a composite rhytidectomy is performed, the same

plane of dissection is used. If SMAS dissection extends to the zygomaticus major muscle, it is termed “extended.” (Reproduced with permission from Thorne CHM, Aston SJ: Aesthetic surgery of the aging face. In: Aston SJ, Beasley RW, Thorne CHM, eds. *Grabb & Smith’s Plastic Surgery*, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

Hematoma is the most common complication of facelift surgery. Because hypertension is a frequently encountered medical condition in the age group that typically presents for facelift, perioperative hypertension must be anticipated and treated preemptively to avoid bleeding risk and the subsequent development of hematoma. The risk is higher in male patients, perhaps due to increased perfusion of the bearded region, hormonal gender differences, or increased sebaceous gland density. Medications that can increase the risk of bleeding are typically discontinued around the time of surgery (i.e., aspirin, NSAIDs, fish oil, etc.).

Smoking also has been shown to be detrimental to facelift results, especially with regard to skin flap survival. As this is elective surgery, most physicians will have patients stop smoking 2–4 wk prior to and after surgery.

One of the least desirable complications is injury to the facial nerve, which can produce a disastrous result following an elective cosmetic surgery. Many surgeons prefer that no paralytics be used during the procedure to allow for careful monitoring of facial nerve function.

NECK LIFT

Description

Neck lift is the rejuvenation of the area from the inferior border of the mandible to the clavicles. The goal of the procedure is to sharpen the chin and smooth the anterior neck (i.e., improve the cervicomental angle). This procedure often is combined with a facelift procedure where the neck skin is elevated and redraped on the face where it can be addressed by facelift. Patients undergoing isolated neck lifts without a facelift typically have smaller amounts of neck skin laxity/redundancy.

A neck lift is usually achieved by extending the facelift dissection inferiorly through the preauricular incision. A small submental incision may also be used to allow for **submental liposuction, direct fat excision, or platysma muscle modifications** (plication, suspension, resection, or transection techniques). Some platysmal suspension techniques require the facelift incisions to remain open with continuity in the subcutaneous plane laterally. See [Figure 11.1-4](#).

Variant procedures or approaches

Laser resurfacing, especially in the perioral and periorbital regions; **blepharoplasty** and **brow lift** are common adjunct procedures.

Usual preop diagnosis

Facelift: facial rhytids (wrinkles/creases); solar or senile elastosis; jowling; deep nasolabial folds; tear troughs; nasojugal folds; malar bags. Neck lift: “turkey gobbler” neck; platysmal bands; cervical laxity; cervical rhytids

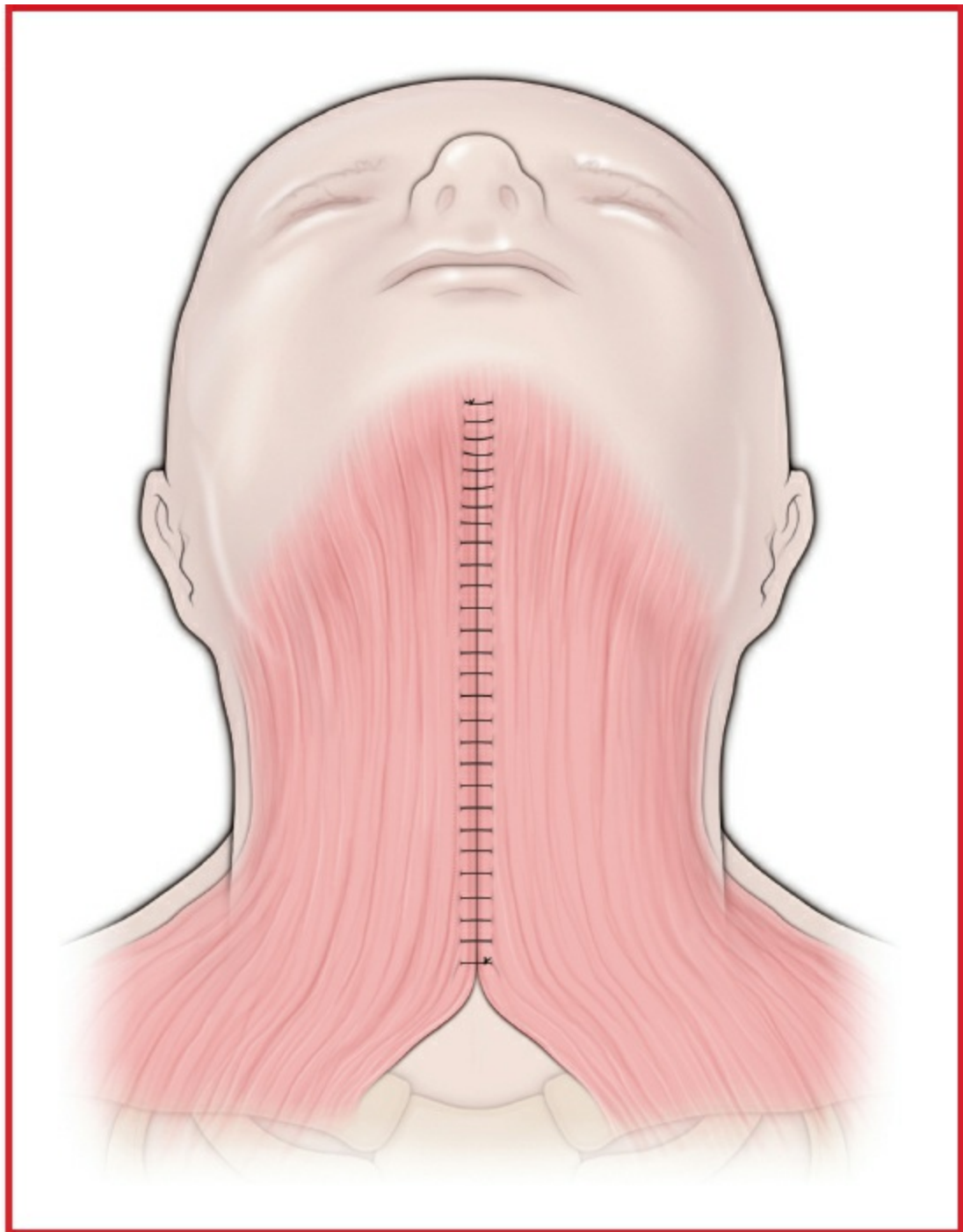


Figure 11.1-4 Platysmaplasty.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, preop carb loading, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Facelift	Neck Lift	Midface Lift
Position	Supine, reverse Trendelenburg	⇐	⇐
Incision	Preauricular, scalp	Postauricular extension of facelift incision + submental incision	Intraoral ± subciliary or inferior lid
Special instrumentation	Infiltration equipment for tumescent techniques; endoscopic equipment (more frequent with brow lifts)	⇐ ± liposuction instrumentation	—
Unique considerations	Oral intubation; ability to move ETT side to side. Consider securing ETT with floss or suture around teeth. Watch for oculocardiac reflex (OCR), retrobulbar hematoma with periorbital approaches. If laser is used: • Special fire-retardant ETT and	⇐	⇐

- drapes
 - Laser eye protection for all in room
 - Cannula-administered O₂ should be far away from laser (fire safety).
 - Smoke evacuation system (See Facial Laser Resurfacing).
- Infiltration of large volumes of local anesthetic with epinephrine in facelift and liposuction procedures

Antibiotics + other meds

Cefazolin 2 g iv (± methylprednisolone 125 mg iv).
Antivirals periop if laser used (e.g., acyclovir 2400 mg × 2 d preop and 14 d postop)

⇐

⇐

Surgical time

4–6 h

1–2 h

2–4 h

Closing considerations

Trendelenburg for final hemostasis ± drains

—

—

Some surgeons prefer gentle ↑ in BP during hemostasis.

⇐

⇐

Tissue thrombin

⇐

—

	agents may be used between the elevated flaps.		
	Sensory nerve blocks by surgeon	⇐	⇐
	± Full head/face wrap before patient awakens	—	—
	Gentle, nonagitated awakening to prevent sudden ↑ BP		
EBL	<50 mL	⇐	⇐
Postop care	Monitor for hematoma: most common complaint is pain; therefore, r/o hematoma before increased analgesia or sedation.	⇐	⇐
	Lightly compressive dressing, ± drains: both removed at 24 h	⇐	⇐
	2–4 wk no aspirin, moderate activity	⇐	⇐
Mortality	Rare	⇐	⇐
Morbidity	Early hematoma:	⇐	Very rare
	• Large expanding: 1–15% (return to OR)		
	• Small (<30 mL): 10–15% (± aspiration in office)	⇐	
	Late hematoma	⇐	—

(average = 9 d
postop; 2° to
exertion or aspirin
use; from
superficial temporal
vessels)

Infection: 0–0.33% ⇐

⇐

Nerve injury:

- Motor
(temporal/marginal
mandibular
branches)

Very rare

- Temporary: 0.1–
2.6% ⇐

- Permanent: 0–
0.66% ⇐

- Motor (spinal
accessory nerve):
Rare ⇐

—

Sensory injury:

—

- Great auricular
nerve → ↓
sensation in lower
½ of ear ± painful
neuromas or
paresthesias

- Lesser occipital
nerve painful →
neuroma ⇐

—

Alopecia: 0.4% ⇐
(most temporary,
along the incision)

—

Skin slough: 14%
(especially in
retroauricular area)
(12.5 × greater in
smokers)

—

Dehiscence: 0.1–0.35%	⇐	⇐
Parotid cysts: Rare	⇐	—
Poor cosmetic result:	⇐	⇐
• Hyperpigmentation • Telangiectasia (preexisting lesions may worsen) • Hypertrophic scarring • Keloids: Very rare • “Pixie” (pulled-down earlobe) deformity (technique dependent) • Hairline shifts		
Ectropion (midface, approached via lid incisions only): Up to 3%		

Pain score	3	3	3
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■ PATIENT POPULATION CHARACTERISTICS

Age range	>35 yr
Male:Female	1:9 to 1:5 (increased from 1:17 in the 1970s)
Incidence	131,106 facelift procedures in the United States (2016); 5th most common cosmetic plastic surgery procedure 55,227 neck lift procedures in the United States (2016); 10th most common cosmetic plastic surgery procedure
Etiology	Facial rhytids 2° solar elastosis, senile elastosis, facial expression

Associated conditions

Cancers of the skin (basal cell carcinoma, squamous cell carcinoma and precursors, and melanoma), in solar elastosis cases, especially fair-skinned patients

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Brow Lift and Blepharoplasty.

Suggested Readings

1. American Society of Plastic Surgeons: Website: www.plasticsurgery.org
2. American Society of Plastic Surgeons: Website: www.plasticsurgery.org/news/plastic-surgery-statistics
3. Bloom JD, Immerman SB, Rosenberg DB: Face-lift complications. *Facial Plast Surg* 2012; 28(3):260–72.
4. Chuang J, Barnes C, Wong BJF: Overview of facial plastic surgery and current developments. *Surg J (N Y)* 2016; 2(1):e17–28.
5. Derby BM, Codner MA: Evidence-based medicine: face lift. *Plast Reconstr Surg* 2017; 139(1):151e–67e.
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3. Rasko YM, Beale E, Rohrich RJ: Secondary rhytidectomy: comprehensive review and current concepts. *Plast Reconstr Surg* 2012; 20(4):262–6.
9. Warren RJ, Aston SJ, Mendelson BC: Facelift. *Plastic Reconstr Surg* 2001; 128:747–64.

BROW LIFT AND BLEPHAROPLASTY

SURGICAL CONSIDERATIONS

BROW LIFT (OR FOREHEAD LIFT)

Description

Brow lift is the elevation of the eyebrows that softens forehead rhytids and redrapes the skin that can hood over the lateral orbit to restore the youthful appearance of the upper face. This procedure is often performed in conjunction with an upper lid blepharoplasty. Patients presenting for brow lift usually have specific concerns about lateral brow hooding, forehead wrinkles, and a glabellar creases that give them an angry appearance.

Like facelift procedures, brow lifts have been performed in the subcutaneous plane, but the relatively avascular subgaleal and subperiosteal planes are more commonly used. The subgaleal and subperiosteal approaches have become more popular with the incorporation of **endoscopic techniques**. A coronal incision is used for an open brow lift. A more limited lateral temporal brow lift uses only a portion of this incision for each side. For an endoscopic technique, three to five small, interrupted access incisions are made along the hairline or within the hair-bearing scalp ([Fig. 11.1-5](#)).

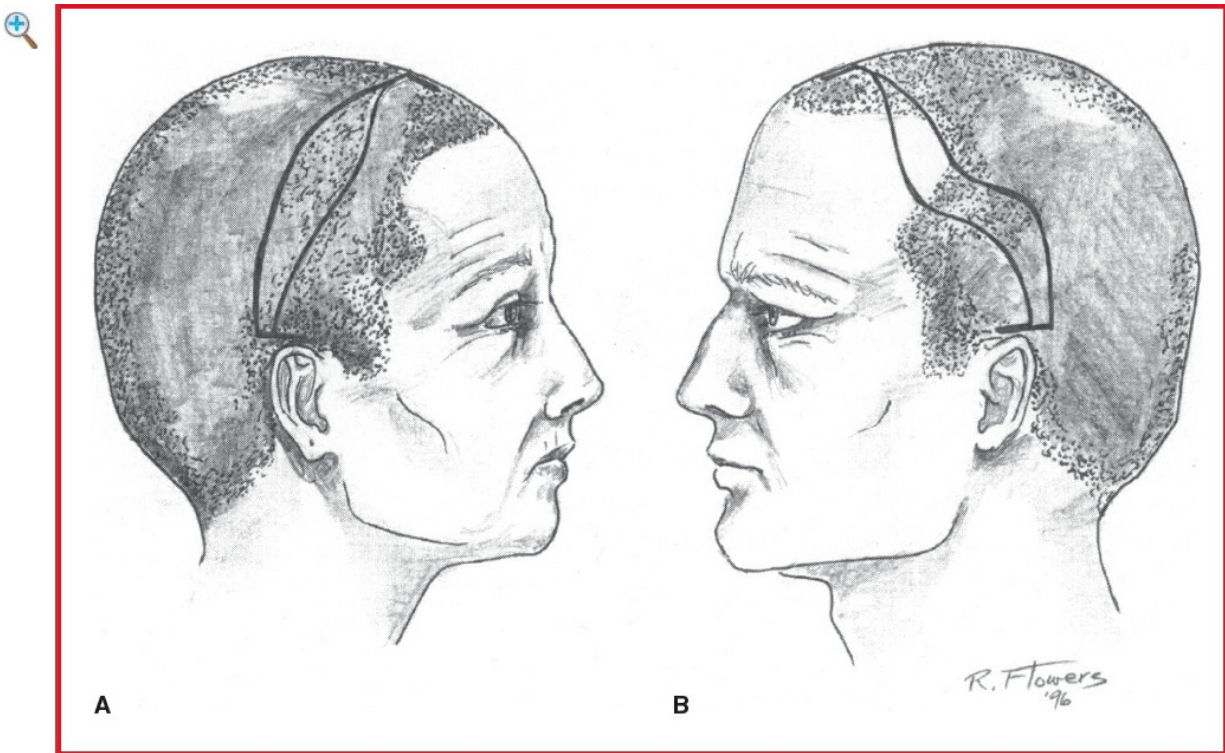


Figure 11.1-5 Incisions for forehead/brow lifting. Consistent blepharoplasty results demand appropriate frontal lifting technique. **A:** Standard coronal incision. **B:** Male and female balding incision. Note the posterior displacement of the ascending incision for maximum camouflage. Hair perimeter incisions are rarely necessary. The central brow corrects nicely from only parietotemporal scalp excisions (after appropriate suprapariosteal release). (Reproduced with permission from Flowers RS, DuVal C: Blepharoplasty and periorbital aesthetic surgery. In: Aston SJ, Beasley RW, Thorne CHM, eds. *Grabb & Smith's Plastic Surgery*, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

In the open technique, the bicoronal flap is dissected off the upper face ([Fig. 11.1-6](#)). The brows are elevated by **scalp resuspension ± resection**. Closure of the scalp anchors this resuspended position. The soft tissues may also be fixated directly to the cranium with screws, resorbable fixation devices, or sutures to the temporal fascia to maintain their new positions. Release of the periosteum along the superior orbital rims is a prerequisite to adequate resuspension when using a subperiosteal approach. Elimination of the upper facial rhytids (i.e., glabellar wrinkles) is achieved by resection of the medial brow musculature (corrugator and procerus) from beneath the elevated

flap. Muscular bleeding is controlled with bipolar electrocautery.

Variant procedures or approaches

The brow lift has become the facial plastic surgery procedure most adaptable to the techniques of **endoscopy**. Multiple smaller (1–1.5 inches) incisions are used within the scalp for access, and small, elliptical excisions also may be used to achieve the desired effect. The muscle resection is accomplished endoscopically with very small biting forceps from beneath the flap. See [Figure 11.1-7](#).

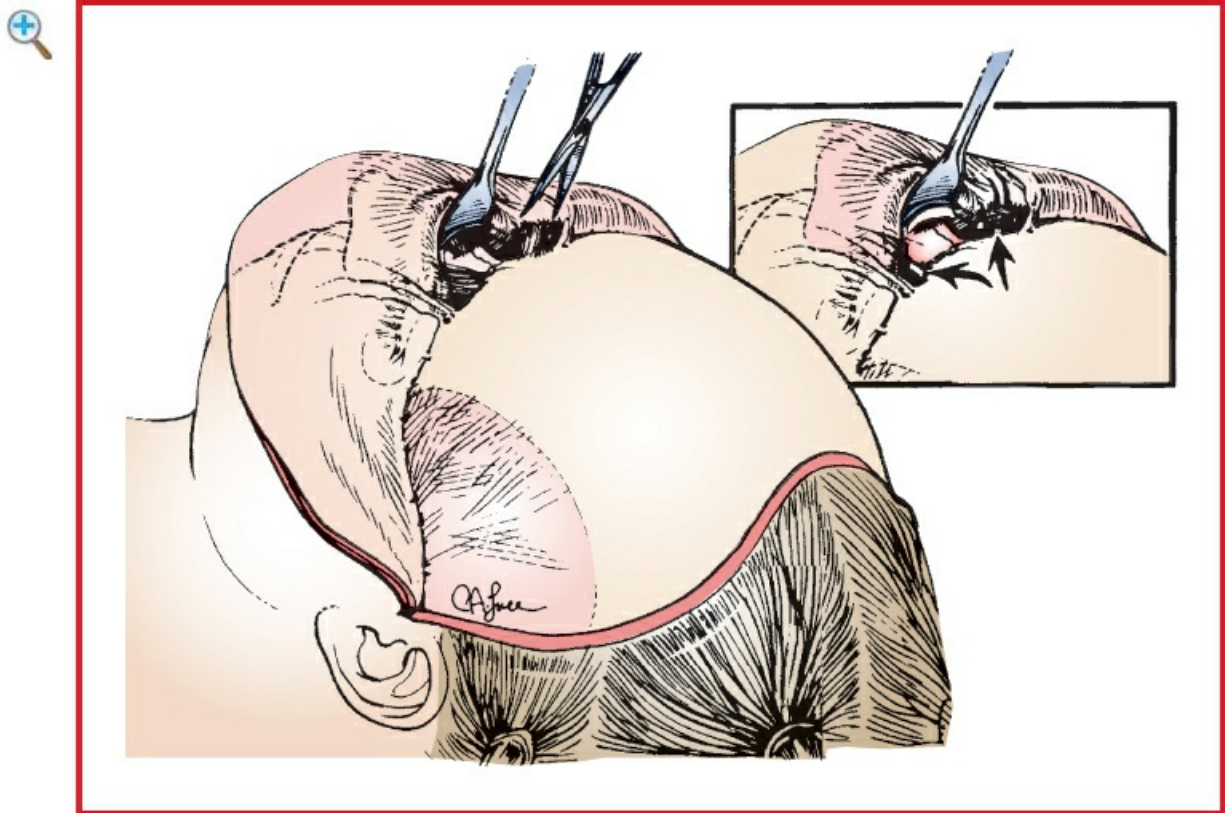


Figure 11.1-6 Exposure after subgaleal, suprapariosteal dissection of the forehead. The supraorbital nerves can be seen easily, but the supratrochlear nerves are more superficial and are hidden by the corrugator muscles. Scissors are used to tease through the corrugator muscles to locate the supratrochlear nerve branches. The muscle is then aggressively resected, preserving the sensory branches. (Reproduced with permission from Thorne CHM, Aston SJ: Aesthetic surgery of the aging face. In: Aston SJ, Beasley RW, Thorne CHM, eds. Grabb & Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

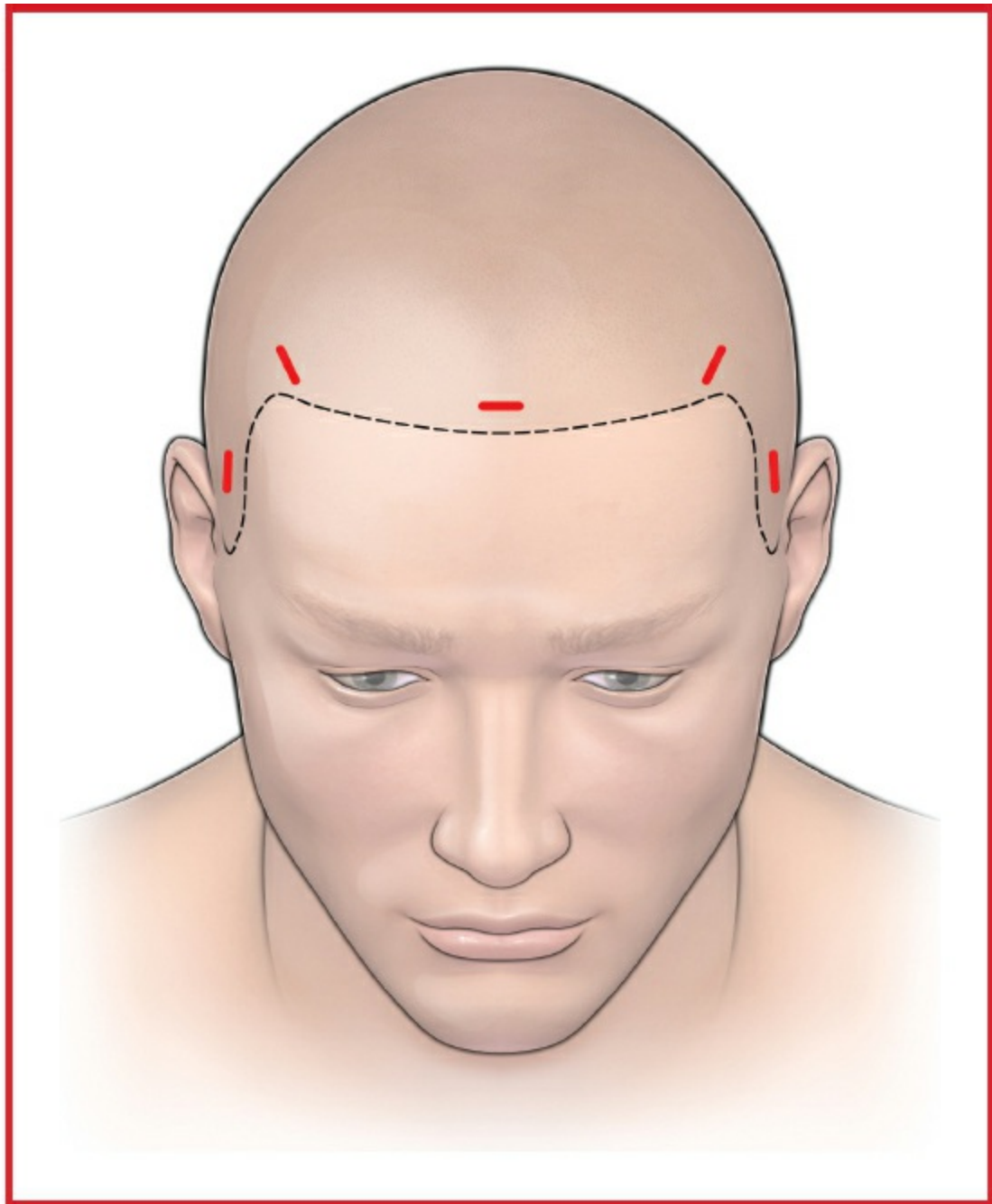


Figure 11.1-7 Typical endoscopic portals for facelift.

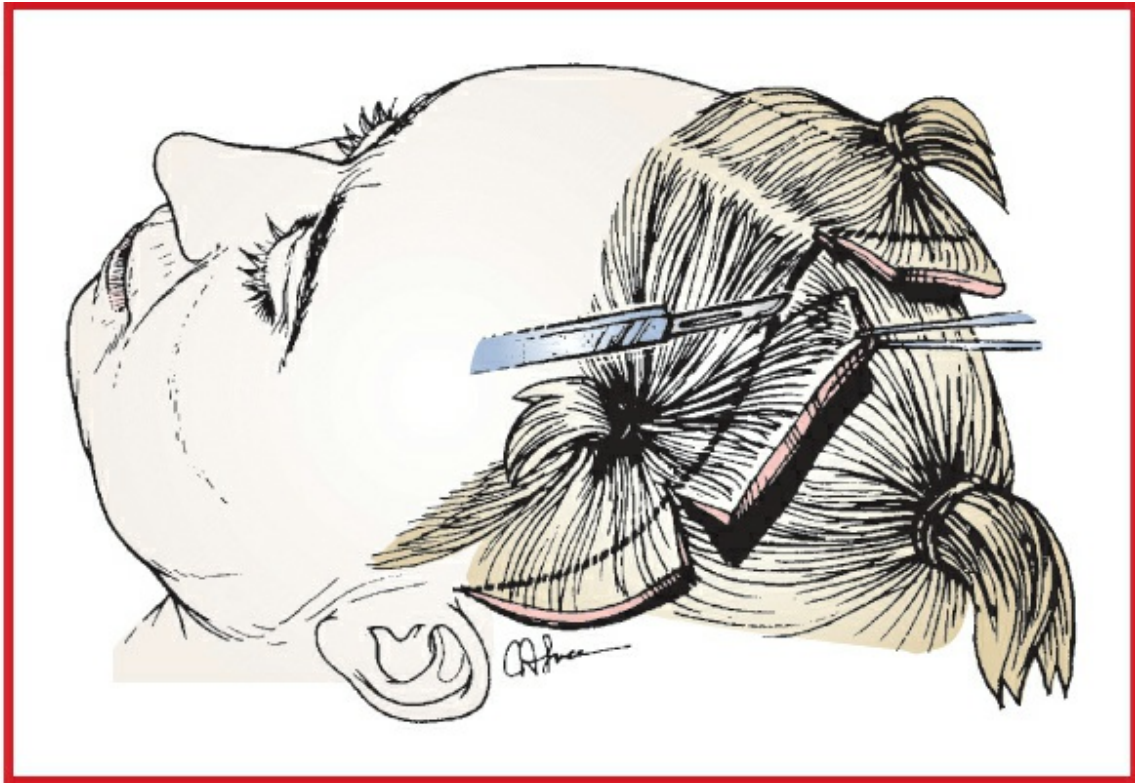


Figure 11.1-8 Redraping the forehead/brow using “key” fixation sutures. Maximal tension is placed laterally to elevate the lateral brow to a greater extent than the medial brow. (Reproduced with permission from Thorne CHM, Aston SJ: Aesthetic surgery of the aging face. In: Aston SJ, Beasley RW, Thorne CHM, eds. Grabb & Smith’s Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

Usual preop diagnosis

Brow ptosis; brow droop; upper facial rhytids (wrinkles or creases)

BLEPHAROPLASTY (OR LID LIFT)

Description

Blepharoplasty ([Fig. 11.1-9](#)) is the surgical rejuvenation of the periorbital region to eliminate the tired and aged appearance of the eyes. Westernizing the Asian eyelid also has become quite commonplace. Presenting complaints include excess lid skin, prominent periorbital fat, and absence of upper lid folds. Blepharoplasty can involve resection of skin, muscle (orbicularis oculi), and fat. Patients presenting for this procedure may also benefit from a simultaneous brow lift to reestablish the baseline position of the brows, revealing the true amount of upper-lid redundancy. Eyelid ptosis repair also can be achieved in the same surgery.

Although a seemingly benign procedure, the manipulation of periorbital fat can have very serious consequences. Retrobulbar hematoma and blindness can occur postop, and

the oculocardiac reflex (OCR) can complicate the intraop course with bradycardia and hypotension. This generally resolves with elimination of the stimulus.

Blepharoplasty, as an isolated procedure, is often performed with local anesthetic and intravenous sedation so that patients can open and close their eyes during the surgery. This helps to tailor the amount of eyelid skin to be removed without tension thus limiting the risk of lagophthalmos (inability to close the eyelids) and scleral show. Having the patient able to respond to commands can be useful if a ptosis repair is also planned.

Variant procedures or approaches

Once more, popular CO₂ **laser blepharoplasty** techniques have proven effective but must be done under the safety parameters of eye protection and fire and burn prevention. With this technique, the fat and skin resections are achieved with a laser, replacing the use of a scalpel. Using the laser to gain some of the skin tightening associated with blepharoplasty has also been described.

Usual preop diagnosis

Blepharochalasis; periorbital fat; blepharoptosis; dermatochalasis; supratarsal fold absence; Asian eyelid

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, preop carb loading, and postop pain management options. See [Appendix K, p. K-1](#).

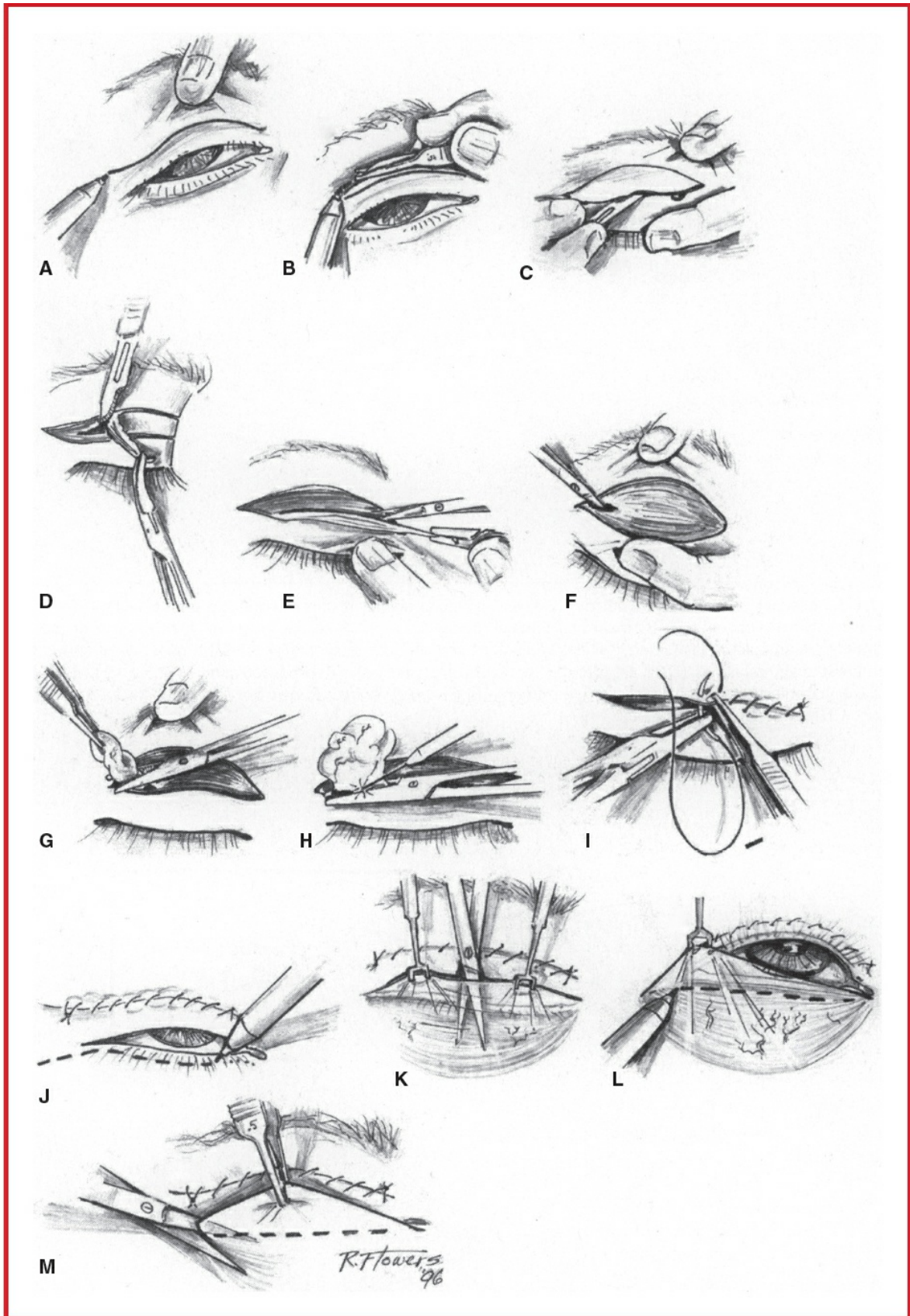


Figure 11.1-9 Traditional blepharoplasty technique. **A:** The caudal margin of the excision is marked and the upper eyelid skin is pinched (**B**). Skin and muscle are excised (**C, D, E**); excess or herniated fat is removed from medial and lateral compartments (**F, G, H**); and the wound is closed (**I**); on the lower lid, the traditional approach is flap elevation, consisting of skin or skin with attached muscle (**J, K**). The

skin is draped upward and outward so the surgeon can assess and remove excess skin (**L, M**). (Reproduced with permission from Flowers RS, DuVal C: Blepharoplasty and periorbital aesthetic surgery. In: Aston SJ, Beasley RW, Thorne CHM, eds. Grabb & Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

■ SUMMARY OF PROCEDURES

	Brow Lift	Blepharoplasty
Position	Supine; table rotated 90° or 180°	⇐
Incision	Hairline, coronal; multiple scalp for endoscopic procedure	Upper, tarsal fold; lower, subciliary, transconjunctival
Special instrumentation	Fibrin glue, screws, or resorbable fixation devices for suspension; endoscopic equipment	Bipolar electrocautery
Unique considerations	Local anesthetic with epinephrine	Retrobulbar hematoma; OCR; local anesthetic with epinephrine
Antibiotics	Cefazolin 2 g iv ± methylprednisolone 125 mg iv	⇐
Surgical time	1–2 h	⇐
Closing considerations	Place dressing before arousing patient. Gentle arousal from anesthesia (↑ in BP or emesis → risk of hematoma)	No dressing (ointment)
EBL	50 mL	Minimal
Postop care	PACU → room/home Lightly compressive dressing Head of bed elevated	⇐ Cool packs ⇐ Ophthalmic lubricant

Mortality	Rare	Vision checks ⇐
Morbidity	Hematoma: Rare	⇐
	Alopecia: Rare (more common with bicoronal approach)	—
	Infection: <1%	⇐
	Frontalis paralysis: Rare (usually transient)	—
	Poor cosmetic result	⇐
	Sensory nerve dysfunction	⇐
	Lagophthalmos	⇐
		Blindness: Extremely rare Ectropion Entropion
Pain score	3	2–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	Most ≥35 yr
Male:Female	Blepharoplasties: 1:4 Brow lifts: 1:7.33
Incidence	Blepharoplasties: 209,020 (4th most common cosmetic procedure in the United States, 2016) Brow lifts: 43,038 (11th most common cosmetic procedure in the United States, 2016)
Etiology	Facial rhytids secondary to solar elastosis, senile elastosis, facial expression, and increased muscle resting tone; Asian eyelids
Associated conditions	Cancers of the skin (basal cell carcinoma, squamous cell carcinoma and precursors, and melanoma) in solar elastosis cases, especially fair-skinned patients

Suggested Readings

1. American Society of Plastic Surgeons: Website: www.plasticsurgery.org
2. Angelos PC, Stallworth CL, Wang TD: Forehead lifting: state of the artFacial Plast

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3. Mack WP: Blepharoplasty complications. Facial Plast Surg 2012; 28(3):273–87.
4. Matarasso A: The oculocardiac reflex in blepharoplasty surgery. Plast Reconstr Surg 1989; 83(2):243–50.
5. Ridgway JM, Larrabee WF: Anatomy for blepharoplasty and brow-lift. Facial Plast Surg 2010; 26(3):177–85.
5. Yang P, Ko AC, Kikkawa DO, et al: Upper eyelid blepharoplasty: evaluation treatment, and complication minimization. Semin Plast Surg 2017; 31(1):51–7.

ANESTHETIC CONSIDERATIONS

(Procedures covered: facelift, neck lift; brow lift, blepharoplasty)

PREOPERATIVE

Cosmetic facial surgery is elective and should be performed preferably on ASA I or II patients. Often, several cosmetic procedures (including facial laser resurfacing) are performed during the same surgical session. A preop discussion with the surgical team is important to help define the anesthetic plan. The above procedures are usually done under GA in the hospital but can also be done under MAC with local anesthesia.

Airway	A careful inspection of the airway should be performed. Surgeon may request intraop manipulation of the oral ETT from side to side or may prefer a nasal ETT.
Cardiovascular	A thorough cardiovascular evaluation should be performed. HTN is the most common medical condition in this patient population. Many procedures involve the use of significant amounts of local anesthetic with epinephrine, placing the patient at higher risk for iatrogenic HTN, dysrhythmias, and coronary artery spasm. In addition, consider patient suitability for the use of mild controlled ↓ BP (particularly the facelift patient). Tests: ECG, if indicated from H&P
Hematologic	✓ for recent aspirin/NSAID use and h/o easy bleeding, bruising, frequent epistaxis, etc. If possible, stop aspirin 1 wk before surgery. Tests: CBC, if indicated from H&P

Laboratory	Other tests as indicated from H&P
Premedication	Preop sedation with midazolam usually is appropriate. Preop steroids (dexamethasone 4–8 mg) also may be used to reduce PONV and swelling.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss possibility of nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po for patients with h/o PONV.

INTRAOPERATIVE

Anesthetic technique

Cases are usually done under GA, using an ETT or LMA, as appropriate. Several authors describe the use of a propofol/ketamine MAC or “dissociative anesthetic” in the office-based setting. There are varying descriptions of this technique, generally involving a propofol infusion with incremental ketamine boluses or infusion, resulting in elimination or significant reduction in the administration of iv opiates.

MAC with local anesthetic also is an option and may be advantageous for certain patients (e.g., with Hx of PONV) or for cases that benefit from a patient’s intraop ability to follow commands (e.g., ptosis repair).

Induction	For those procedures done under GETA, a standard induction is appropriate. An oral RAE ETT may be used to minimize intrusion into the surgical field. For cases involving a laser, a shielded ETT manufactured for laser surgery should be used. (Note: No shielded ETT is 100% laser proof; always use standard precautions.)
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia. Consider preincision placement of nerve blocks (see Appendix K , p. K-6).

Maintenance	Standard maintenance with volatile anesthetic ± propofol infusion is appropriate in most cases. Muscle relaxation should be avoided in cases with facial nerve monitoring. Mild, controlled ↓ BP may be requested and used to facilitate hemostasis. HTN should be avoided and treated immediately if it occurs. Maintain anesthesia during application of head/face wrap.	
Emergence	Antiemetic prophylaxis (e.g., ondansetron 4 mg iv + dexamethasone 4 mg iv if nondiabetic ± preop aprepitant 40 mg po) is recommended, as postop emesis greatly increases the likelihood of hematoma formation. Perform thorough oropharyngeal suctioning. A smooth emergence with no notable increase in BP is the goal. Consider nerve block placement.	
Blood and fluid requirements	Blood loss generally minimal IV: 18 ga × 1 BSS @ 1–3 mL/kg/h	Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors	
Control of blood loss	Local infiltration with epinephrine Surgical hemostasis Mild degree of ↓ BP	
Positioning	✓ and pad pressure points. Rotate OR table 90–180°.	Scleral shields ± ophthalmic ointment
Complications	OCR Retrobulbar hematoma Local anesthetic toxicity	Remove inciting stimulus. Consider atropine 0.5 mg and deepening anesthetic. Rx: 20% lipid emulsion: 1.5 mL/kg iv bolus (may repeat × 2) + start infusion @ 0.5 mL/kg/min (max dose in 30 min = 12 mL/kg).

POSTOPERATIVE

Complications	PONV	Vigorous treatment of nausea is important.
Multimodal pain management	Local infiltration + iv/po narcotics, if needed Acetaminophen	R/O expanding hematoma as cause of increasing pain. Consider regional or local blocks.
ERAS	PONV protocol	Encourage early enteral nutrition and early mobilization.

Suggested Readings

1. Bogan V: Anesthesia and safety considerations for office-based cosmetic surgery practice. AANA J 2012; 80(4):299–305.
2. Mustoe TA, Buck DW II, Lalonde DH: The safe management of anesthesia, sedation and pain in plastic surgery. Plast Reconstr Surg 2010; 126(4):165e–76e.
3. Nekhendzy V, Ramaiah VK: Prevention of perioperative and anesthesia-related complications in facial cosmetic surgery. Facial Plast Surg Clin North Am 2013; 21(4):559–77.
4. Ramanadham SR, Costa CR, Narasimhan K, et al: Refining the anesthesia management of the face-lift patient: lessons learned from 1089 consecutive face lifts. Plast Reconstr Surg 2015; 135(3):723–30.

RHINOPLASTY

SURGICAL CONSIDERATIONS

Description

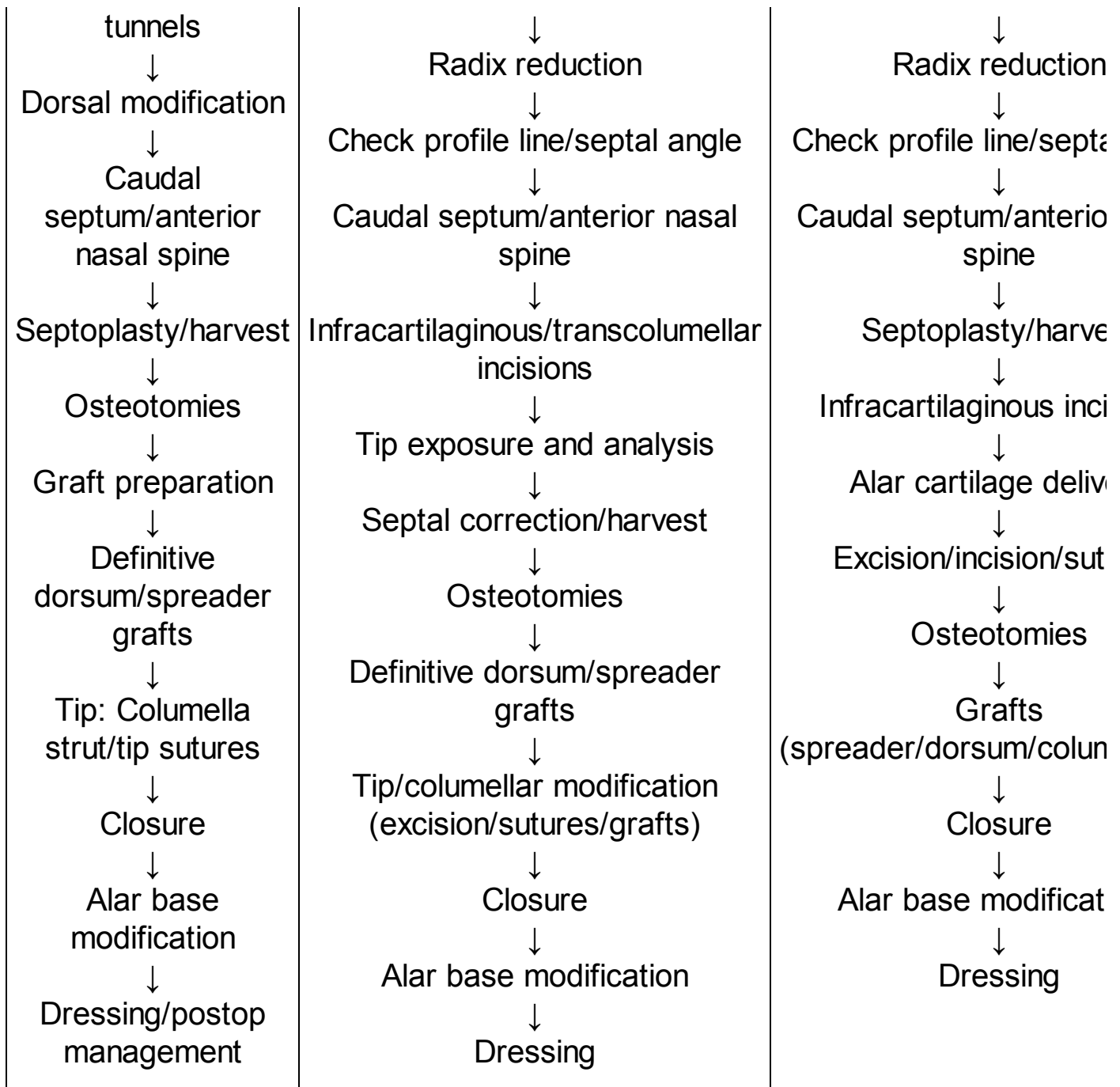
Rhinoplasty is one of the greatest challenges of plastic surgery. It is the surgical manipulation of the nasal form for either aesthetic or functional improvement. In combination with nasal septal surgery, it is called **septorhinoplasty**. Common patient requests are for dorsal hump reduction and improved tip definition. Cosmetic surgery of the nose can address four major components: **the tip, the dorsum, the width (either of bones or alae), and the septum**, in addition to other ancillary procedures to enhance airway function.

Tip and dorsal procedures may be accomplished by either **reduction** or **augmentation**. Augmentation can be performed with either synthetic or biologic grafts. Examples of synthetic grafts include silicone, expanded fibrillated polytetrafluoroethylene polymer (Gore-Tex), porous polyethylene implants (Medpor™), and hydroxyapatite. Examples of biologic grafts include cadaveric or autologous tissue (cartilage, bone, fascia, or dermis). Common donor sites for cartilage are the ear concha (via an anterior or posterior approach), the nasal septum (internal nasal approach), and the ribs. Bone harvest sites may include the outer table of cranium, the iliac crest, and the ribs. Dermal graft is commonly harvested from the groin, and fascial graft harvest is often taken from the temporoparietal region. (Table 11.1-1 shows the range of open and closed rhinoplasty techniques.) Add positive pressure to avoid PTX.

A throat pack is useful to prevent aspiration or ingestion of blood, as significant blood pooling can occur in the naso/oropharynx area, especially with nasal osteotomies used to narrow or straighten the nasal dorsum. Cases where such pooling is expected are safer under GA with a throat pack. Often, rhinoplasties are done with local or regional (nasociliary and infraorbital blocks) anesthesia with sedation. Vasoconstriction may be performed by the administration of oxymetazoline in the preop holding area or in the OR with soaked nasal packs (cocaine vs. epinephrine vs. oxymetazoline) placed before the first incision.

The decision of open versus closed technique is based on patient requirements and surgeon preference. An **open approach** will utilize a transcolumellar incision to allow elevation of a nasal skin flap and degloving of the lower alar cartilages for direct and wide exposure of the nasal framework (see Fig. 11.1-10). **Closed approaches** use intercartilaginous, intracartilaginous, infracartilaginous, rim, hemitransfixion, and transfixion incisions (all hidden within the nose).

Table 11.1-1 Rhinoplasty Techniques		
Open	Closed/Open	Closed
Incisions/skin flap elevation ↓ Tip analysis/cephalic crura excision ↓ Extramucosal	Intercartilaginous/transfixion incisions ↓ Skin elevation/extramucosal tunnels ↓ Rasp bony hump/excise cartilaginous hump	Transcartilaginous/trar incision ↓ Skin elevation/extramucosal tunnels ↓ Rasp bony hump/excise cartilage hump



Note: Only those steps appropriate for the individual case are performed.

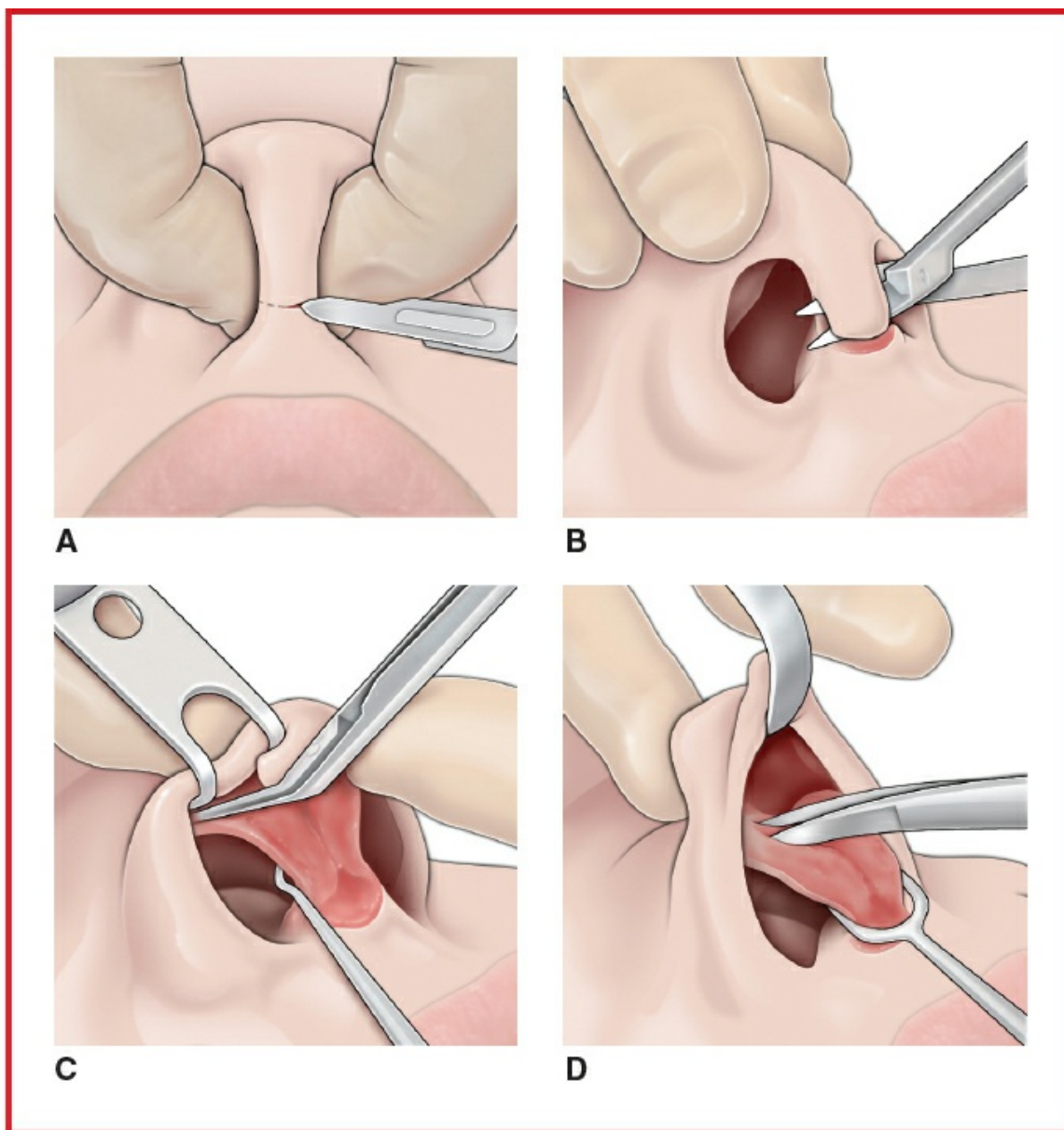


Figure 11.1-10 Initial steps for an open rhinoplasty producing a wide exposure of the nasal framework.

A typical **closed rhinoplasty** (Fig. 11.1-11) begins with dorsal work through intercartilaginous incisions. The dorsum may be reduced using a scalpel and/or rasps beneath the undermined dorsal skin and periosteum. The septum is addressed as necessary through a hemitransfixion incision ($\pm$ cartilage harvest). Tip reduction by scalpel or scissor resection of the lower alar cartilage $\pm$ tip suture is next. Nasal osteotomies with an osteotome and mallet begin at the base of the nasal bones along the piriform aperture. Access points for the osteotomies can be intranasally, externally, or through an intraoral vestibular incision. Digital manipulation completes the fractures, and this is when most of the blood loss occurs. Dorsal and tip grafts are applied as necessary, with alar modifications made last. **Alar reduction** entails wedge resection of the lateral alar base and primary closure.

Depending on the type of rhinoplasty performed, different dressings will be applied at the end of the procedure. When nasal bone osteotomies are used, the patient will require a dorsal nasal splint ± bilateral nasal packing. Nasal packing is generally removed at 24–72 h postoperatively. When septal manipulation is needed, nasal packing or some sort of septal splint may be placed. The packs are generally removed within 3 d, but the splints can be maintained much longer and the nasal airways kept patent with vasoconstrictor nasal sprays.

Variant procedures or approaches

Release of the tip depressor muscle is achieved via intraoral vestibular incisions (behind the upper lip).

Usual preop diagnosis

Posttraumatic nasal deformity (including disordered breathing, “saddle nose,” crooked nose, septal deviation); developmental nasal deformities (bulbous tip, flat tip, drooping tip, broad dorsum, dorsal hump, alar widening, “Pinocchio nose”); congenital nasal malformation (cleft nasal deformities)

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

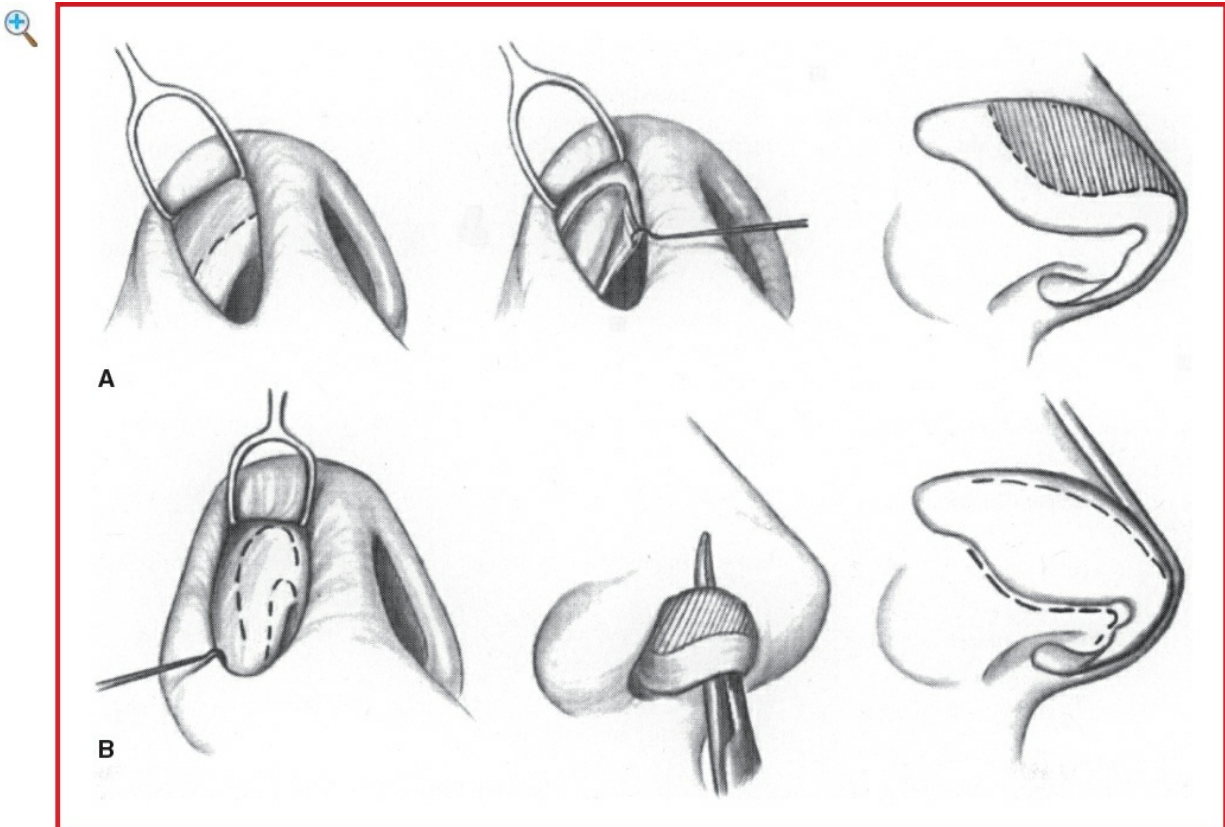


Figure 11.1-11 Closed rhinoplasty. **A:** Transcartilaginous approach using an intracartilaginous incision. **B:** Delivery approach, using a high intercartilaginous incision and a marginal incision to facilitate delivery of the lateral crura. (Reproduced with permission from Daniel RK: Rhinoplasty. In: Aston SJ, Beasley RW, Thorne CHM, eds. Grabb & Smith's Plastic Surgery. 5th edition. Lippincott-Raven, Philadelphia: 1997.)

■ SUMMARY OF PROCEDURE

Position	Supine, table may be rotated 180°. If GA: oral ETT toward foot of bed, shoulder roll, neck extended, scleral lubricant, shields
Incision	External vs. internal nasal incisions
Special instrumentation	Headlight
Unique considerations	Throat pack for expected nasopharyngeal bleeding. Intranasal vasoconstrictors
Antibiotics	Cefazolin 2 g iv
Surgical time	1–2.5 h
Closing considerations	Suction stomach via OG tube at the end of surgery. Remove throat pack prior to extubation. Internal and/or external nasal splints are placed for dressings.
EBL	Tip rhinoplasty: 20 mL Dorsum with osteotomies: 75–150 mL Septoplasty: +50 mL
Postop care	PACU → room (most patients are home the same day). Ensure minimal PONV; elevate head of bed; no pressure on nose (e.g., O ₂ mask).
Mortality	Rare
Morbidity	Infection: >1% Adverse cosmetic result: Alar notching Alar collapse Dorsal irregularity Asymmetry Tip droop

Adverse functional result (i.e., poor airway)
Septal perforation

Pain score 3 (4 with osteotomies)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Most ≥ 15 yr
Male:Female	1:1.7
Incidence	223,018 (third most common cosmetic procedure in the United States, 2016)
Etiology	Developmental; acquired (posttraumatic); congenital (see Secondary Cleft Lip and Nasal Surgery)
Associated conditions	Breathing difficulties; psychosocial issues (body dysmorphic disorder)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Nasal and Sinus Surgery.

Suggested Readings

1. American Society of Plastic Surgeons: Website: www.plasticsurgery.org
2. Angelos PC, Been MJ, Toriumi DM: Contemporary review of rhinoplasty. Arch Facial Plast Surg 2012; 14(4):238–47.
3. Cannon DE, Rhee JS: Evidence-based practice: functional rhinoplasty. Otolaryngol Clin North Am 2012; 45(5):1033–43.
4. Cuzalina A, Qaqish C: Revision rhinoplasty. Oral Maxillofac Surg Clin North Am 2012; 24(1):119–30.
5. Friedman O, Cekic E, Gunel C. Functional rhinoplasty. Facial Plast Surg Clin North Am 2017; 25(2):195–9.
5. Stevens MR, Emam HA: Applied surgical anatomy of the nose. Oral Maxillofac Surg Clin North Am 2012; 24(1):25–38.
7. Tanna N, Nguyen KT, Ghavami A, et al: Evidence-based medicine: current practices in rhinoplasty. Plast Reconstr Surg 2018; 141(1):137e–51e.

OTOPLASTY

See [Chapter 12.8](#), Surgery for Craniofacial Malformations, Otoplasty.

FACIAL LASER RESURFACING

Description

Laser resurfacing is a technique by which a controlled burn is administered to the skin of the face with laser technology, creating a healing process that reduces the signs of aging or acne. **Various wavelengths of lasers are** commonly used with facial cosmetic procedures including CO₂ ($\lambda = 10.6 \mu\text{m}$: mid-infrared), erbium ($\lambda = 2.94 \mu\text{m}$: mid-infrared), and Nd:YAG ($\lambda = 1.064 \mu\text{m}$: near-infrared). Longer wavelength = deeper tissue penetration. These lasers are used widely for the periorbital and perioral creases and wrinkles not addressed by previously described facial cosmetic surgical techniques. Nerve blocks, local anesthesia, intravenous sedation, and GA are all possibilities for laser treatment. The choice of anesthetic depends more on the specific surgical procedures to be performed first, as the laser procedure is usually adjunctive and added at the end. Facial laser resurfacing is done frequently in an office-based setting (see [Chapter 14.0](#), Office-Based Procedures). Because laser resurfacing is usually an adjunct to another facial cosmetic procedure, the following discussion pertains primarily to the unique set of **safety issues** that must be addressed in the OR.

Ocular Hazards

These include direct and reflected injury to the eye. Everyone present, including the patient and all medical personnel, requires laser-specific (i.e., wavelength-specific) safety eyewear. Laser-specific scleral shields must be available for the patient in cases where the patient's eyewear would be in the operative field. Protective eyewear must be undamaged and have

- Permanent labels with wavelength and optical density tolerance
- Side shields
- Damage threshold of >10 sec
- No surface reflection
- Good fit
- Approval from the laser safety officer

Fire and reflectivity hazards

Many items used in the OR (e.g., drapes, sponges, plastic cannulas) are made of materials that can be fire hazards if not kept from interacting with the laser beam. Protection from fire and reflectivity is provided by:

- Having fire-retardant or moist draping
- Having water basin available

- Having a fire extinguisher readily available
- Avoiding all alcohol-containing prep solutions
- Avoiding use of plastic and rubber instruments (may melt or ignite)
- Using special fire-resistant ETTs or wet sponge protection for plastic ETTs to decrease the possibility of tube breach or ignition
- Avoiding open sources of O₂ (nasal cannulas, etc.) or reducing the amount of O₂ delivered temporarily
- Avoiding metal or other reflective materials

Airborne contaminants

The laser destruction of cells releases carbon particles, microbials, DNA, and toxic fumes. Protection for the patient and medical personnel is provided by:

- Utilizing a smoke evacuation system 2 cm from created plume.
- Wearing high-filtration (e.g., N95) masks. Note that these masks become less effective if moistened from perspiration during a long case; if the laser is to be used at the end of a case, changing masks before using the laser may be prudent.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Brow Lift and Blepharoplasty or Office-Based Laser Skin Resurfacing.

Suggested Readings

1. Carniol PJ, Hamilton MM, Carniol ET: Current status of fractional laser resurfacing. *JAMA Facial Plast Surg* 2015; 17(5):360–6.
2. Cillo JE Jr, Finn R: Moderate intravenous sedation for office-based full face laser resurfacing using a continuous infusion propofol pump. *J Oral Maxillofac Surg* 2005; 36(7):903–7.
3. Hamilton M, Campbell A, Holcomb JD: Contemporary laser and light-based rejuvenation techniques. *Facial Plast Surg Clin North Am* 2018; 26(2):113–21.
4. Potter JK, Finn R, Cillo J: Modified tumescent technique for outpatient facial laser resurfacing. *J Oral Maxillofac Surg* 2004; 62(7):829–33.
5. Ramos-Zabala A, Pérez-Mencía MT, Fernández-García R, et al: Anesthesia technique for outpatient facial laser resurfacing. *Lasers Surg Med* 2004; 34(3):269–72.
6. Wald D, Michelow BJ, Guyuron B: Fire hazards and CO₂ laser resurfacing. *Plast Reconstr Surg* 1998; 101(1):185–8.

Suggested Viewing

Links are available online to the following videos:

Live Surgery Split Thickness Skin Graft: <https://www.youtube.com/watch?v=pvbxmm9inoo>

Liposuction Surgery Video: <https://www.youtube.com/watch?v=nfRsBijqS4E>

BSSO & Osseous Genioplasty: <https://www.youtube.com/watch?v=oQI8L5Sxa6w>

LeFort III Osteotomy: https://www.youtube.com/watch?v=Hk_AUbVuDWE

LeFort II Osteotomy: <https://www.youtube.com/watch?v=5YddYjhThgw>

LeFort I Osteotomy: <https://www.youtube.com/watch?v=7z5TNrJcE3U>

TRAM flap narrated: <https://www.youtube.com/watch?v=xcew3ycbG50>

CHAPTER 11.2

Nonfacial Aesthetic Surgery

SURGEON

David M. Kahn, MD

ANESTHESIOLOGISTS

Richard A. Jaffe, MD, PhD

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We gratefully acknowledge the past contributions of surgeon, George W. Commons, MD.

AUGMENTATION MAMMOPLASTY

SURGICAL CONSIDERATIONS

Description

Augmentation mammoplasty is accomplished through the use of saline or silicone gel-filled breast implants and more recently using autologous fat grafting. The surgery may be performed under GA or local anesthesia with sedation. The patient is positioned with the arms either abducted at 90°, with the hands tucked at the sides, or draped across the lower abdomen. Local anesthetic ($\pm$ epinephrine) is infiltrated into the skin at the incision site and under the glandular tissue. Implant insertion can be done through inframammary, periareolar, transaxillary, or transumbilical incisions. The implant is placed in a pocket that is created beneath the mammary gland (subglandular), beneath the pectoralis fascia (subfascial), under the pectoralis muscle (submuscular), or in a dual plane partially subglandular and partially submuscular (dual plane), depending on the surgeon's preference and the thickness of the soft tissue coverage available. An endoscope may be used to assist with dissection of the pocket. When the implant is

placed in the dual plane position, the pectoralis muscle is divided from its insertion along the inframammary fold up to the sternal insertion to allow the muscle to drape over the implant. Regardless of the location of the pocket, the surgical wound is carefully irrigated and inspected for hemostasis. Sizers, either predetermined volumes of silicone gel or adjustable saline- or air-filled temporary implants, may be used to help determine the appropriate final volume and placement. The patient may be placed in the seated position to assess the size, shape, and symmetry of the breasts. The sizers are then replaced with the permanent prostheses. If permanent saline implants are used, they are filled with saline until the desired volume is reached; gel-filled implants do not have alterable volumes. The wounds are closed, and dressings are applied (Fig. 11.2-1).

Augmentation mammoplasty usually is performed as an outpatient procedure, although some patients may want an overnight stay for pain management and antiemetics. PONV is not uncommon, and all efforts should be made to decrease its frequency.

Variant procedure or approaches

The endoscopic transumbilical approach is used much less frequently. Autologous fat transfer is an evolving procedure, with graft survival depending on many factors including fat harvesting, processing, and injection techniques.

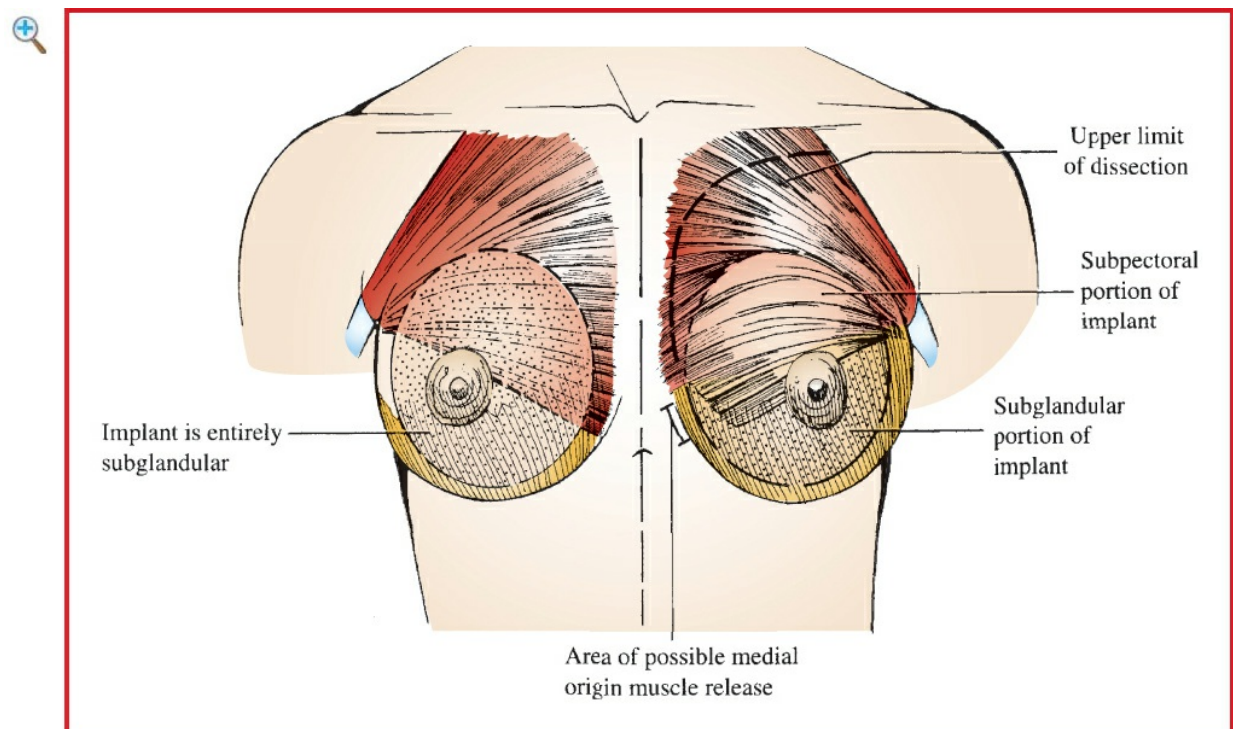


Figure 11.2-1 Breast augmentation. Implants may be placed in a subglandular or subpectoral position. (Reproduced with permission from Spear SL: *Surgery of the Breast: Principles and Art* Lippincott-Raven, Philadelphia: 1998.)

Preop diagnosis

Hypomastia; breast ptosis

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Inframammary, periareolar, transaxillary, or transumbilical
Antibiotics	Cefazolin 2 g iv
Unique considerations	May place patient in sitting position during procedure
Surgical time	1 h
Closing considerations	May need patient in sitting position for application of dressings
EBL	Minimal
Postop care	Outpatient procedure
Mortality	Rare
Morbidity	Prosthesis failure: 5% Capsular contracture: 5% Hematoma: 3% Infection: 2% Wound dehiscence: <1% Anaplastic large cell lymphoma (ALCL): <1% Cosmetic disappointments
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS	
Age range	Typically 17–45 yr but also may be done on the contralateral breast in a patient undergoing breast

	reconstruction
Incidence	290,467 performed in the United States in 2016; the most common cosmetic surgical procedure
Etiology	Developmental; involution after breast-feeding; age-related atrophy or ptosis
Associated conditions	Not common but can be seen with Poland syndrome

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Mastopexy/Breast Lift.

Suggested Readings

1. Al Sufyani MA, Al Hargan AH, Al Shammari NA, et al: Autologous fat transfer for breast augmentation: a review. *Dermatol Surg* 2016; 42(11):1235–42.
2. Coopey SB, Specht MC, Warren L, et al: Use of preoperative paravertebral block decreases length of stay in patients undergoing mastectomy plus immediate reconstruction. *Ann Surg Oncol* 2013; 20(4):1282–6.
3. Ducic I, Zakaria HM, Felder JM III, et al: Nerve injuries in aesthetic breast surgery: systematic review and treatment options. *Aesthet Surg J* 2014; 34(6):841–56.
4. Graf RM, Bernandes A, Rippel R, et al: Subfascial breast implant: a new procedure. *Plast Reconstr Surg* 2003; 111(2):904–8.
5. Nahabedian MY, Patel K: Management of common and uncommon problems after primary breast augmentation. *Clin Plast Surg* 2009; 36:127–38.
5. See Suggested Readings Mastopexy/Breast Lift.
7. Tan P, Martin MS, Shank N, et al: Comparison of 4 analgesic regimens for acute postoperative pain control in breast augmentation patients. *Ann Plast Surg* 2017; 78(6S Suppl 5):S299–304.
8. Tebbetts JB, Thorne CH: An evidence-based approach to augmentation mammoplasty. *Plast Reconstr Surg* 2012; 126:2184.
9. Vyas KS, Rajendran S, Morrison SD, et al: Systematic review of liposomal bupivacaine (Exparel) for postoperative analgesia. *Plast Reconstr Surg* 2016; 138(4):748e–56e.

REDUCTION MAMMOPLASTY

SURGICAL CONSIDERATIONS

Description

Breast reduction surgery can be done as an outpatient procedure or with an overnight stay. One might choose to admit the patient overnight in a hospital setting to monitor for hematoma formation and evidence of decreased blood supply to the nipple-areola complex. For these patients, the pain from this procedure is relatively low; therefore, PONV tends to be the greater issue in the immediate postoperative period.

The traditional type of breast reduction performed in the United States is the **inferior pedicle technique** using a Wise pattern (“anchor-type” scar) for the skin excision (Fig. 11.2-2). Markings are made with the patient upright in the preop holding area. The patient is then brought to the OR for the procedure. The areola is marked circumferentially with an areola sizer and incised. The remaining incision lines are scored with a scalpel. Next, the inferior pedicle (the inferior portion of the breast), which contains the neurovascular supply to the nipple-areola complex, is de-epithelialized. Excess skin and breast tissue are excised, preserving the pedicle of tissue that will compose the breast mound. The resected tissue from each breast, which can range from 200 to 2000 g, is weighed as to an adjunctive method of ensuring equal resections. Temporary skin closure with staples allows the patient to be placed in a sitting position so that the breasts can be evaluated for symmetry. When the surgeon is satisfied with the appearance of the breasts, they are closed with sutures. Drains may be placed, depending on surgeon preference (Fig. 11.2-2). After the skin has been closed, the location of the nipple and areola is marked and excised, and the nipple-areola complex is delivered and sutured into position. Soft, supportive dressings are placed.

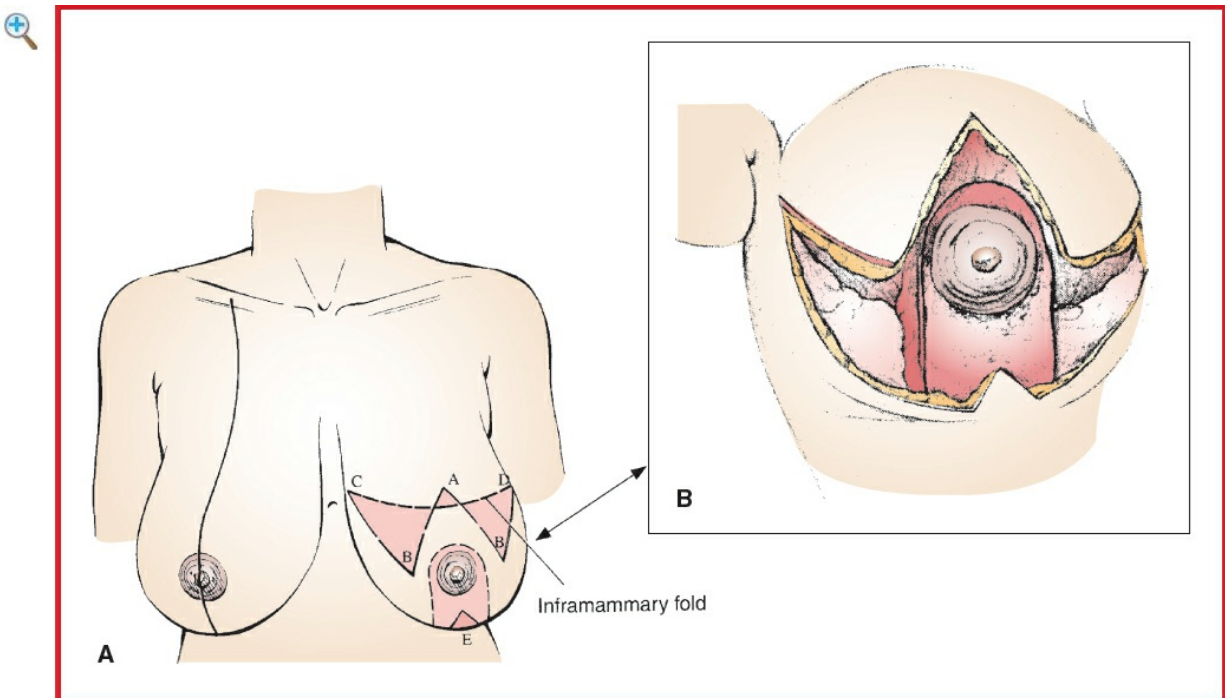


Figure 11.2-2 Reduction mammoplasty using an inferior pedicle technique. **A:** The skin and breast tissue on the medial and lateral sides of the pedicle are resected. **B:** The medial and lateral skin envelopes are sutured at the midline, leaving an inverted T-shaped scar. (Reproduced with permission from Spear SL: *Surgery of the Breast: Principles and Art*. Lippincott-Raven, Philadelphia: 1998.)

A technique that has gained in popularity recently is the **vertical reduction mammoplasty**, which shares the fundamental principles of excision of excess breast tissue and preservation of blood flow to the nipple-areola complex but differs in choice of skin incision and pedicle. Due to the shorter length of the incision, the wound closure time is greatly decreased (resulting in a “lollipop-type” scar) compared with the traditional Wise-pattern anchor-type scar.

Variant procedure or approaches

Liposuction may be used in combination with this procedure. Reduction mammoplasty using liposuction alone has increased in popularity.

Preop diagnosis

Macromastia; gigantomastia; mammary hypertrophy

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine, arm abducted 90°
Incision	Marked preop. Most have circumareolar incision with an inferior anchor-shaped extension (Fig. 11.2-2) or a short vertical component (“lollipop” scar).
Antibiotics	Cefazolin 2 g iv
Unique considerations	May place patient in sitting position during procedure to assess symmetry; Foley catheter; SCDs
Surgical time	2–5 h, depending on volume of reduction and technique
Closing considerations	May place patient in sitting position for application of dressings
EBL	100–200 mL

Postop care	Outpatient or 24-h stay for pain/nausea management. Avoid the use of Toradol in the first 24 h because of the large, raw surface created between the skin flaps and breast tissue.
Mortality	Minimal
Morbidity	Dehiscence: 5% Infection: 1% Seroma/hematoma: <1% (may require reexploration) Skin flap necrosis Loss of nipple-areola complex (may require urgent reexploration) Loss of nipple-areola sensation Hypertrophic scarring Cosmetic disappointments
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	25–65 yr
Incidence	39,148 performed in the United States in 2016 (asthetic patients only)
Etiology	Developmental; childbearing; obesity
Associated conditions	None common

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Mastopexy/Breast Lift.

Suggested Readings

1. Greco R, Noone B: Evidence-based medicine: reduction mammoplasty. *Plast Reconstr Surg* 2017; 139(1):230e–9e.
2. Palmieri B, Benuzzi G, Costa A, et al: Breast reduction and subsequent cancer: a prophylactic perspective. *Breast* 2006; 15(4):476–81.
3. See Suggested Readings Mastopexy/Breast Lift.
4. Shestak KC, Davidson EH: Assessing risk and avoiding complications in breast reduction. *Clin Plast Surg* 2016; 43(2):323–31.

MASTOPEXY/BREAST LIFT

SURGICAL CONSIDERATIONS

Description

Mastopexy procedures reduce the volume of the skin envelope to match the volume of the breast gland. Depending on the degree of ptosis (“droopy breasts”) and the wishes of the patient, the ptosis may be treated by augmentation alone to increase the volume of the breast, by skin excision alone to reduce the skin envelope appropriately, or by a combination of a mastopexy and an augmentation.

The operation itself resembles a reduction mammoplasty, except that breast tissue is generally excised minimally or not at all, and an implant may be added (**augmentation/mastopexy**). The patient is marked before surgery in the upright position. After the induction of anesthesia, the arms are positioned either tucked at the patient’s sides or abducted 90°. The procedure begins with the areola being marked circumferentially with an areola sizer and then incised. Next, the skin flaps are elevated. The breast tissue is moved to a higher position on the chest wall, and the skin is redraped and tailor-tacked closed. The patient is placed in a sitting position to assess for symmetry and nipple location. The nipple-areola complex is then brought out into its new position, and dressings are applied.

Preop diagnosis

Breast ptosis

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURE

Position	Supine
Incision	Circumareolar ± inferior vertical extension or complete anchor
Antibiotics	Cefazolin 2 g iv
Unique considerations	May place patient in sitting position during procedure

Surgical time	2–4 h
Closing considerations	May place patient in sitting position for application of dressings
EBL	<100 mL
Postop care	Outpatient or 24-h stay
Mortality	Minimal
Morbidity	Wound healing: 5% Infection: 1% Seroma/hematoma: <1% Hypertrophic scarring Cosmetic disappointments
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	35–65 yr
Incidence	101,264 performed in the United States in 2016
Etiology	Age-related ptosis; involution after breast-feeding; weight loss
Associated conditions	None common

■ ANESTHETIC CONSIDERATIONS FOR MAMMOPLASTY/MASTOPEXY

▲ PREOPERATIVE

Typically, three patient populations present for mammoplasty: (a) healthy individuals, for breast reduction, augmentation, breast lift, or removal of an implant; (b) morbidly obese for breast reduction; and (c) breast cancer patients, for reconstruction after mastectomy or a symmetry procedure on the noncancer breast (breast reduction, or augmentation). (For preop considerations in the morbidly obese patient, see Anesthetic Considerations for Abdominoplasty.) Breast cancer patients undergoing mastectomy with immediate reconstruction typically will not have had either chemotherapy or radiation. The following considerations are for breast cancer patients undergoing

delayed reconstruction post-chemotherapy.

Respiratory	Pulmonary fibrosis may complicate chemotherapy. Alkylating agents (e.g., cyclophosphamide and melphalan), used to treat breast cancer, have some pulmonary toxicity. Consider pulmonary fibrosis in a patient reporting dyspnea, nonproductive cough, and fever. Tests: Consider CXR; ABG, PFTs as indicated from H&P.
Cardiovascular	Cardiomyopathy and CHF may result from chemotherapy, especially doxorubicin (Adriamycin) >550 mg/m². Tests: Consider ECG; echo, if indicated from H&P.
Neurologic	Note any preexisting injury to long thoracic nerves, as evidenced by winged scapula deformity.
Musculoskeletal	Avoid upper extremity iv and BP cuff on mastectomy side.
Hematologic	Leukopenia, thrombocytopenia, and anemia from chemotherapy may be present. Tests: CBC; Plt count
Renal/Hepatic	Methotrexate can produce some renal and hepatic dysfunction. Tests: Cr; LFTs
Laboratory	Other tests as indicated from H&P, prior chemotherapy
Premedication	Midazolam 1–2 mg iv immediately preop. Surgeon may want to mark the patient's skin preop, with patient standing. Delay premedication until this has been done.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss possibility of nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	Standard induction. ✓ with surgeons regarding use of a nerve stimulator during dissection (and the need to avoid muscle relaxants). Consider intratracheal lidocaine 4% LTA to minimize coughing during position changes.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia. Consider preincision placement of nerve blocks (see Appendix K , p. K-6).	
Maintenance	Standard maintenance. Surgeons may want patient sitting for part of the procedure. Pneumothorax should be considered with any change in lung inflation pressure, O ₂ sat, or BP.	
Emergence	During some of the procedure and for application of dressing, patient may be moved to sitting position, with consequent coughing, bucking, etc. (Rx: deeper anesthesia, e.g., propofol 0.5 mg/kg or lidocaine 1 mg/kg). Watch BP carefully and treat orthostatic hypotension if it occurs, usually with a fluid bolus if the patient does not have h/o CHF or renal failure. Consider nerve block placement or reinforcement if initially placed >4–6 h ago (see p. 1239).	
Blood and fluid requirements	IV: 16–18 ga × l BSS @ 4–8 mL/kg/h	Minimal blood loss for simple reconstruction, augmentation, or reduction; larger blood losses anticipated for combined procedures (e.g., mastectomy with immediate reconstruction)

		or flap reconstruction). Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors	Arterial line in the morbidly obese may be necessary to obtain accurate BP measurements.
Positioning	Patient may need to be sitting for application of dressing.	Avoid HTN, bucking, and straining; these may cause or exacerbate bleeding at reconstruction site. Careful padding of arms to protect them during position change.

▼ POSTOPERATIVE

Complications	Pneumothorax	
Multimodal pain management	PCA Regional nerve blocks Acetaminophen NSAID Gabapentin	Consider multimodal therapy including gabapentin, opiates, and paravertebral/intercostal nerve blocks. Regional or local blocks. Continuous infusion catheters may be beneficial.
ERAS	PONV protocol	Encourage early enteral nutrition, Foley catheter removal, and early mobilization.

Suggested Readings

1. Massenburg BB, Sanati-Mehrizy P, Ingargiola MJ, et al: Flap failure and wound

complications in autologous breast reconstruction: a national perspective. *Aesthetic Plast Surg* 2015; 39(6):902–9.

2. O'Halloran N, Potter S, Kerin M, et al: Recent advances and future directions in postmastectomy breast reconstruction. *Clin Breast Cancer* 2018; 18(4):e571–85.
3. Temple-Oberle C, Shea-Budgell MA, Tan M, et al; ERAS Society: Consensus review of optimal perioperative care in breast reconstruction: enhanced recovery after surgery (ERAS) society recommendations. *Plast Reconstr Surg* 2017; 139(5):1056e–71e.
4. Terkawi AS, Tsang S, Sessler DI, et al: Improving analgesic efficacy and safety of thoracic paravertebral block for breast surgery: a mixed-effects meta-analysis. *Pain Physician* 2015; 18(5):E757–80.

BRACHIOPLASTY

SURGICAL CONSIDERATIONS

Description

Brachioplasty is performed as outpatient surgery for patients who note laxity or a “bat-wing” appearance of their upper arms. Markings are done in the preoperative holding area with the patient upright, arms abducted and flexed. The arms are positioned on arm boards or the hand table. An incision is made starting along the chest wall or in the axilla, extending onto the upper arm, and stopping before the elbow. Excess skin and soft tissue are excised, and the incision is closed, typically over a drain. Dressings are applied, often followed by a compression garment or ACE wraps. See [Figure 11.2-3](#).

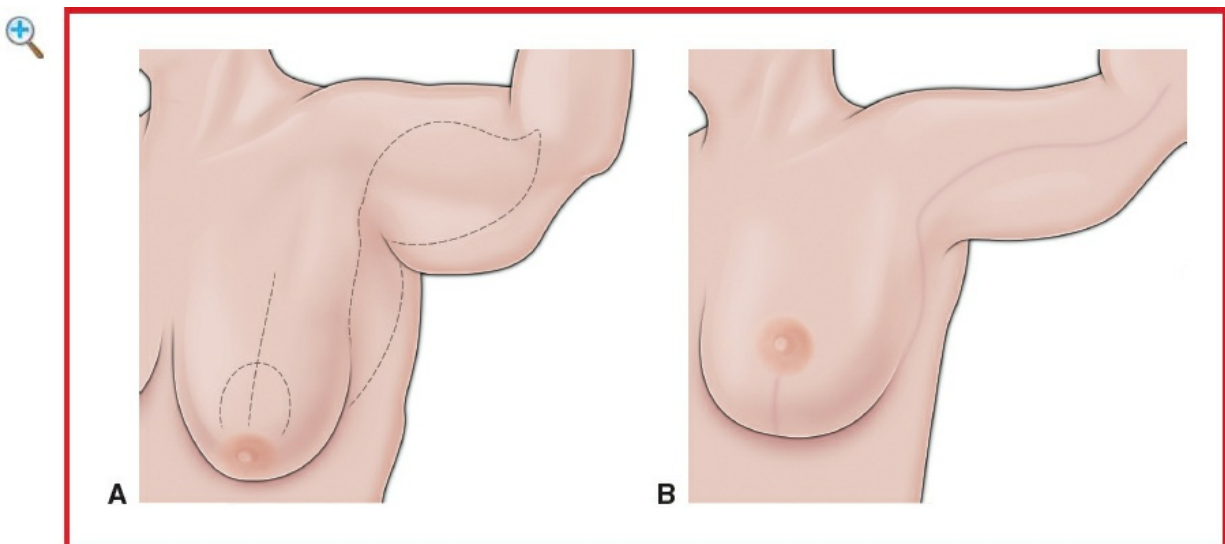


Figure 11.2-3 Preop incision planning (A) and postop results (B).

Usual preop diagnosis

Upper arm laxity or redundancy; lipodystrophy

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine, arms abducted
Incision	From the chest/axilla, extending along the lower inner arm
Unique considerations	Both arms must be mobile and prepped circumferentially; SCDs.
Antibiotics	Cefazolin 2 g iv
Surgical time	1–2 h
EBL	Minimal
Postop care	Avoid ketorolac in the first 24–48 h.
Mortality	Minimal
Morbidity	Seroma: 10% Hypertrophic scarring: 10% Infection: 5–7% Nerve injury: 5%
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS	
Age range	20–65 yr
Male:Female	1:50
Incidence	17,860 performed in the United States in 2016
Etiology	Overweight; aging; massive weight loss
Associated conditions	Obesity

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for brachioplasty have often undergone gastric bypass surgery with massive weight loss. IV access may be difficult, and the arms may be prepped all the way down to and including the hands. Consider the need for iv access in the lower limb or neck due to bilateral arm surgery. (See Anesthetic Considerations in Morbid Obesity.)

INTRAOPERATIVE

Anesthetic technique

GETA, routine monitors

Procedure may be combined with other plastic procedures, necessitating position changes.

Induction	Standard induction for healthy patients. Special considerations for the morbidly obese include prophylaxis for aspiration, followed by rapid-sequence induction in an appropriately positioned patient (see Fig. 7.2-6). If mandibular and cervical mobility are decreased by excessive soft tissue, an awake fiberoptic intubation with the patient sitting may be necessary. Anticipate rapid O ₂ desaturation (2° ↓ FRC) during periods of hypoventilation, even with adequate preoxygenation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia. Consider preincision placement of nerve blocks (see Appendix K, p. K-6).
Maintenance	Standard maintenance. Calculate most drug doses on basis of lean body mass. In the obese, controlled ventilation with larger TVs, PEEP, and high inspired O ₂ concentration is recommended. Positioning may

	be difficult, and care must be taken to give adequate padding and support.	
Emergence	Give antiemetics (e.g., ondansetron 4 mg) 20 min before conclusion of surgery. Consider nerve block placement or reinforcement if initially placed >4–6 h ago (see p. 1239).	
Blood and fluid requirements	Minimal blood loss, 18 ga × 1 BSS @ 4–8 mL/h	Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors	

▼ POSTOPERATIVE

Multimodal pain management	PCA Oral medications for outpatients Acetaminophen	Consider regional or local blocks
ERAS	PONV protocol	Encourage early enteral nutrition and early mobilization.
Tests	As indicated by patient condition	

Suggested Readings

1. Hurwitz D: Brachioplasty. Clin Plast Surg 2014; 41(4):745–51.
2. Knoetgen J III, Moran SL: Long-term outcomes and complications associated with brachioplasty: a retrospective review and cadaveric study. Plast Reconstr Surg 2006; 117(7):2219–23.

ABDOMINOPLASTY

■ SURGICAL CONSIDERATIONS

Description

Patients who present for **abdominoplasty** have laxity in the abdominal wall musculature and excess skin and adipose tissue. This laxity may be associated with rectus muscle diastasis. **Liposuction** is sometimes performed in conjunction with abdominoplasty to

remove additional adipose tissue and improve contour.

Incision lines are marked on the patient preoperatively in the upright position (Fig. 11.2-4). If the procedure will include liposuction of the flanks, the procedure may start in the prone position. Once completed, the patient will then be turned to the supine position. The umbilicus is circumscribed, with care taken to preserve its blood supply. An incision is made above the pubic hairline and extended bilaterally to each anterior-superior iliac spine. Electrocautery is used to raise a flap of skin, subcutaneous tissue, and fat at the level of the abdominal wall fascia. The dissection extends cephalad to the costal margin. The operating table is flexed to place the patient in the semi-Fowler position. The elevated flap is pulled down to overlap the inferior incision, and the redundant soft tissue is excised in a tailor-tack fashion (Fig. 11.2-5). The surgical area is inspected for hemostasis and irrigated. Sutures may be placed to plicate the abdominal wall musculature if there is laxity. Fibrin sealant may be sprayed to aid in hemostasis. The wound is closed typically over drains, and the umbilicus is brought out through a new incision. Dressings, which may include an abdominal binder, are applied. The patient is maintained in the semi-Fowler position during transfer from the operating table. The patient may elect to have the procedure as an outpatient or with an overnight stay in a monitored facility.

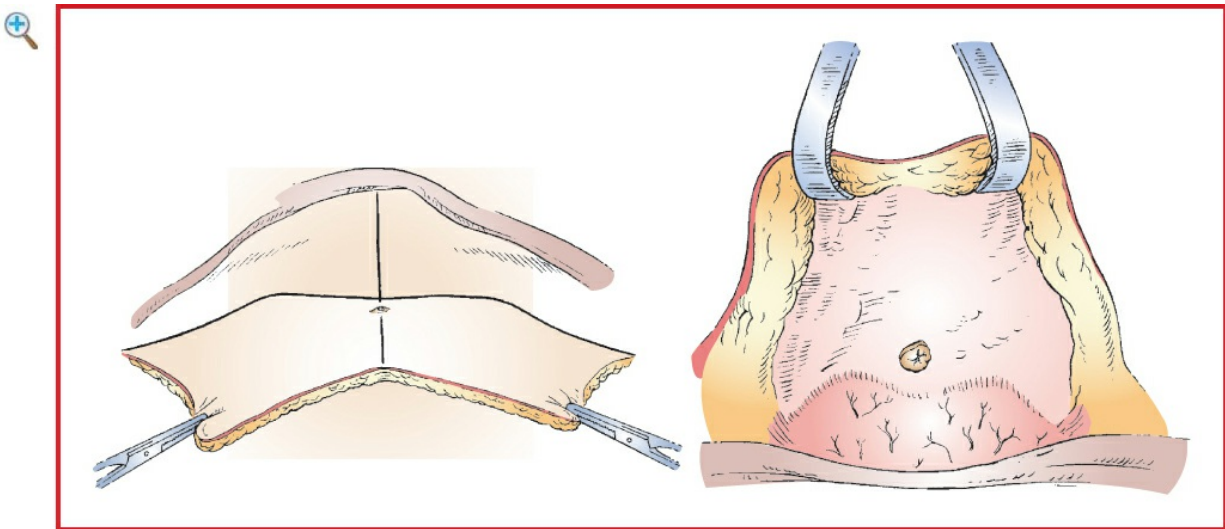


Figure 11.2-5 After the lower incision has been made and the abdominal flap has been elevated, the flap is pulled down to overlap the inferior incision, and the redundant soft tissue is excised in a tailor-tack fashion. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CHM, eds. Grabb & Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

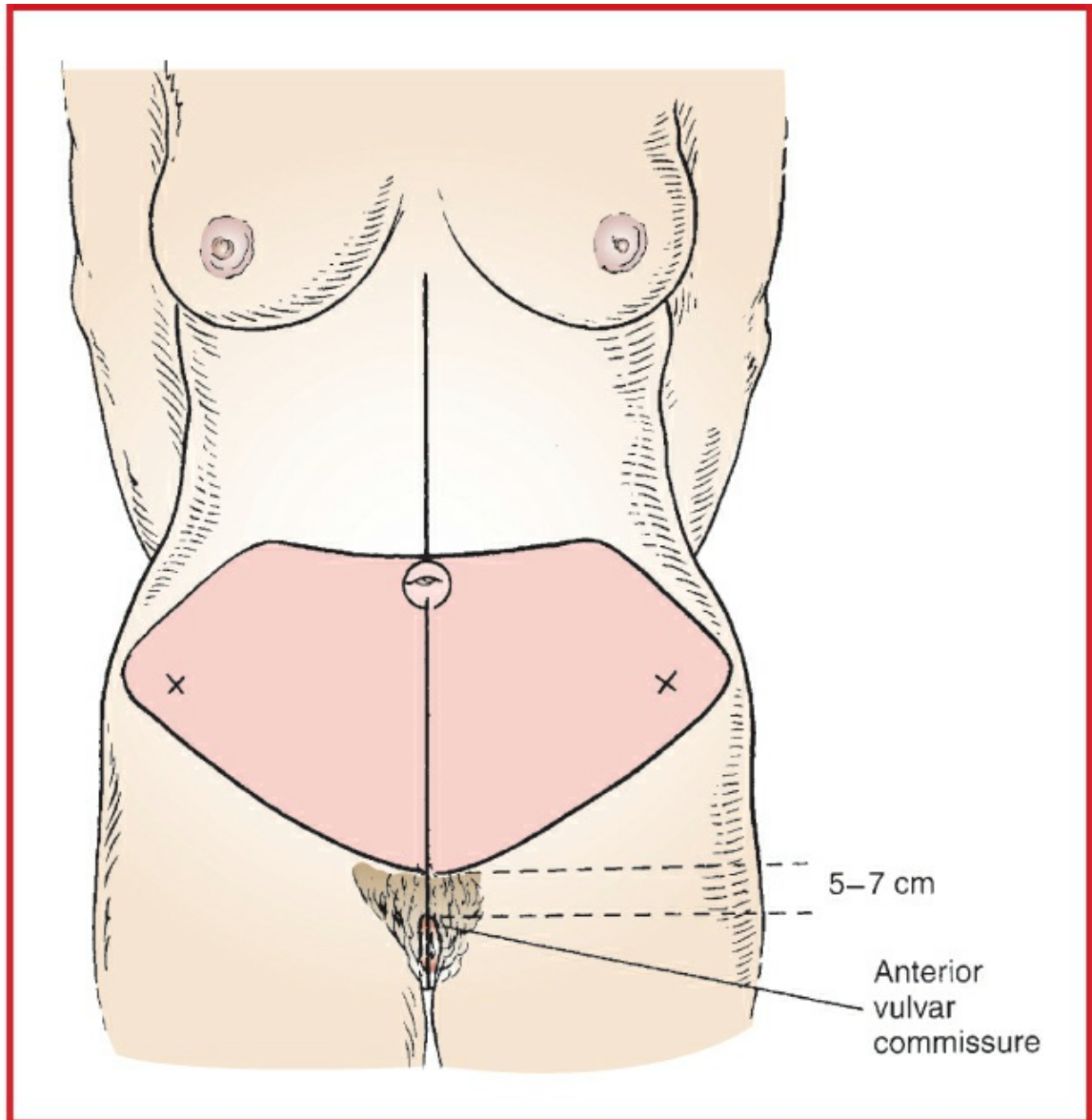


Figure 11.2-4 Abdominoplasty, markings for incisions. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CHM, eds. Grabb & Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

Variant procedure or approaches

Patients who have laxity above the umbilicus may require additional reduction with a fleur-de-lis abdominoplasty (vertical midline extension of incision and scar), or this can be addressed secondarily with a reverse abdominoplasty with the final scar underneath the breasts. Panniculectomy (simple resection of overhanging skin and soft tissue) can be performed alone without the fascial plication, particularly in the case of morbidly obese patients. Miniabdominoplasty may be performed in those patients who require less extensive dissection. Panniculectomy and abdominoplasty can also be combined with hernia repairs and gynecologic procedures.

Usual preop diagnosis

Abdominal wall laxity; rectus diastasis; lipodystrophy; redundant skin and soft tissue

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Extended Pfannenstiel's; periumbilical; vertical midline extension if needed
Unique considerations	Foley catheter; SCDs
Antibiotics	Cefazolin 2 g iv
Surgical time	1.5–4 h
Closing considerations	Flex table to facilitate closure; patient must subsequently remain in semi-Fowler position.
EBL	~100 mL, not including blood contained in specimen.
Postop care	Maintain flexed position; avoid ketorolac in the first 24–48 h.
Mortality	0–1%
Morbidity	Ileus: 10% Infection: 2–3% Dehiscence: 1% Fat embolism: 1% DVT: 1% Hematoma Seroma
Pain score	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–65 yr
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Male:Female	1:25
Incidence	127,633 performed in the United States in 2016
Etiology	Overweight; laxity of skin after pregnancy; massive weight loss
Associated conditions	Obesity; post–gastric bypass

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Typically, there are two patient populations for abdominoplasty: the generally healthy and the morbidly obese. Some patients have Hx of amphetamine, cocaine, or thyroid hormone abuse and ↑ incidence of hiatal hernia. The following considerations focus on the morbidly obese patient (body weight $\geq 2 \times$ ideal weight. Ideal body weight can be estimated by subtracting 100 (male) or 105 (female) from height in cm.

Respiratory	In the morbidly obese patient, findings include ↑ O_2 consumption, ↑ CO_2 production, restrictive lung disease, ↓ FRC, ↓ ERV, ↓ VC, and ↓ PaO_2. These changes are exacerbated by the supine position. Younger patients may show alveolar hyperventilation in response to hypoxemia; older patients may not and may therefore retain CO_2. Patients may have obesity hypoventilation syndrome (pickwickian syndrome) and sleep apnea, with intermittent airway obstruction, hypoxemia, and hypercarbia during sleep, which may → pulmonary HTN. Obese patients are at ↑ risk of pulmonary aspiration due to ↑ incidence of hiatal hernia, GERD, and ↑ gastric volumes (typically >25 mL with pH < 2.5). See Premedication, below, for aspiration prophylaxis. Tests: Consider CXR, room-air ABG, and PFT (helpful, but may not predict postop complications, e.g., atelectasis, pneumonia).
Cardiovascular	↑ CO and ↑ blood volume → LVH. Chronic hypoxia and pulmonary compromise may produce right heart

	failure, ↓↓ exercise tolerance, ↑ risk of CAD, and pulmonary HTN. Patients with LVH may have ↑ dysrhythmias. Some patients may have previously taken fenfluramine alone or in combination with phentermine for weight loss. Those patients should be evaluated for pulmonary HTN (e.g., dyspnea, central cyanosis, right axis deviation, and CXR changes) and valvular heart disease. Tests: ECG (↑ HR, conduction abnormalities, LVH); CXR (cardiomegaly)
Metabolic	Increased incidence of diabetes, hypercholesterolemia, hypertriglyceridemia, liver abnormalities; ↓ plasma folate and B₁₂; ↑ incidence of cholelithiasis, nephrolithiasis. Determine whether electrolyte abnormalities are present in patient s/p ileojejunal bypass. Tests: As indicated from H&P
Hematologic	Polycythemia suggests chronic hypoxemia (see Respiratory, above). Tests: Hb/Hct
Laboratory	Other tests as indicated from H&P
Premedication	Sedative premedication is usually avoided in the morbidly obese due to their pulmonary compromise and increased sensitivity to sedative agents. Aspiration prophylaxis is useful: ranitidine 50 mg iv 30–60 min before surgery, plus nonparticulate antacid (Na citrate 0.3 M, 30 mL po) preinduction. In addition, metoclopramide 10 mg iv may be given, although it has not been shown to be more effective in combination with an H₂-blocker than the H₂-blocker alone. For the healthy outpatient, midazolam 1–2 mg iv immediately before transport to the OR may lessen anxiety.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po for patients with h/o

severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: Morbidly obese patients may not tolerate the supine position for an extended period of time. Consider placement of thoracic epidural for postop pain control.

Induction	Standard induction for healthy patients. Special considerations for the morbidly obese include prophylaxis for aspiration (see above), followed by rapid-sequence induction in an appropriately positioned patient (see Fig. 7.2-6). If mandibular and cervical mobility are decreased by excessive soft tissue, an awake fiberoptic intubation with the patient sitting may be necessary. Anticipate rapid O ₂ desaturation (2°↓FRC) during periods of hypoventilation, even with adequate preoxygenation (e.g., 4 VC breaths of 100% O ₂ in 30 sec).
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia. Consider preincision placement of nerve blocks (see Appendix K , p. K-6).
Maintenance	Standard maintenance. Calculate most drug doses on basis of lean body mass. In the obese, increased plasma fluoride concentrations are found after anesthesia with volatile anesthetics; controlled ventilation with larger TVs, PEEP, and 50% O₂ is recommended. Because epinephrine infiltration generally is used to decrease blood loss, isoflurane is recommended as the least dysrhythmogenic of the inhalation agents. NB: Midazolam has a prolonged half-life in obese patients, but awakening times from inhalational or narcotic-based anesthetics are comparable to those of nonobese patients.

Emergence	Smooth emergence with minimal bucking, coughing, or retching to minimize tension on the suture line; give antiemetics (metoclopramide 10 mg and ondansetron 4 mg) 20 min before conclusion of surgery. Maintenance of flexed position will minimize tension on suture line. Small additional doses of narcotic (e.g., meperidine 10–20 mg) may be titrated to RR if patient is allowed to resume spontaneous respiration before the end of the case. Consider iv acetaminophen if po dose is given >4–6 h ago.	
Blood and fluid requirements	Moderate blood loss IV: 16–18 ga × 1 BSS @ 6–10 mL/kg/h	Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors ± Arterial line ± CVP line	Additional monitoring for the morbidly obese patient may include arterial and CVP lines.
Positioning	Flexed position Pillows under knees ✓ and pad pressure points ✓ eyes	Flexed position minimizes tension on suture line. Morbidly obese may require two OR tables side by side. Supine position may be poorly tolerated; monitor ventilation closely.
Complications	Fat emboli Rhabdomyolysis	More common during liposuction Increased risk with prolonged surgery

POSTOPERATIVE

Complications	Patients may have postop ileus of 1- to 2-d duration.	The morbidly obese should not be outpatients; they have ↑
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		incidence of wound infection, DVT, PE, and postop pulmonary complications. Provide supplemental O ₂ for the first 2 d postop. Keep patient in semisitting or flexed position to avoid undue stress on wound.
Multimodal pain management	Epidural narcotics or PCA may be used. Acetaminophen.	Monitor patient for postop respiratory depression. Consider regional or local blocks.
ERAS	PONV protocol	Encourage early enteral nutrition and early mobilization.
Tests	Pulse oximetry	Maximum reduction of arterial saturation may occur on postop day 2–3.

Suggested Readings

1. Aldenkortt M, Lysakowski C, Elia N, et al: Ventilation strategies in obese patients undergoing surgery: a quantitative systematic review and meta-analysis. *Br J Anaesth* 2012; 109(4):493–502.
2. Joshi GP, Ankichetty SP, Gan TJ, et al: Society for ambulatory anesthesia consensus statement on preoperative selection of adult patients with obstructive sleep apnea scheduled for ambulatory surgery. *Anesth Analg* 2012; 115(5):1060–8.
3. Kirkham L, Thomas M: Anaesthesia in obese patients. *Br J Hosp Med (Lond)* 2011; 72(9):515–20.
4. Peterson PL, et al: The transversus abdominis plane block: a valuable option for postoperative analgesia? A topical review. *Acta Anaesthesiol Scand* 2010; 54(3):529–35.
5. Schumann R: Anaesthesia for bariatric surgery. *Best Pract Res Clin Anaesthesiol* 2011; 25(1):83–93.
6. Staalesen T, Elander A, Strandell A, et al: A systematic review of outcomes of abdominoplasty. *J Plast Surg Hand Surg* 2012; 46(3–4):139–44.
7. Winocour J, Gupta V, Ramirez JR, et al: Abdominoplasty: risk factors, complication rates, and safety of combined procedures. *Plast Reconstr Surg* 2015; 136(5):597e–

BODY LIFTS

SURGICAL CONSIDERATIONS

Description

With heightened awareness of the importance of nutrition and exercise as well as the advent of improved gastric bypass and restriction techniques, more patients with massive weight loss are presenting to plastic surgeons for treatment of the resultant skin excess and laxity. Patients frequently have multiple areas of concern, from the face (see **Facelifts**) to the breasts (see **Mastopexy**) to the abdomen and thighs. Often patients will require circumferential torsoplasty (combining **Abdominoplasty** with excision of the lower back roll) and/or extensive lower body work also lifting the medial and lateral thighs. All of these body lift procedures may be combined with **liposuction** for additional contouring.

The patient is marked in the standing position in the preoperative holding area. Depending on surgeon preference, the initial operative position may begin supine, lateral decubitus, or prone. Incisions are made, and the marked excess skin and soft tissues are elevated and excised. The patient's position is changed as needed to allow for access to all of the surgical areas. Drains are placed. During wound closure, care is taken to close in several layers, beginning with the strength layer of the superficial fascial system. Dressings are applied, and compression garments or ACE wraps may also be used. The patient may elect to have the procedure as an outpatient but frequently chooses to stay overnight in a monitored facility. See [Figure 11.2-6](#).

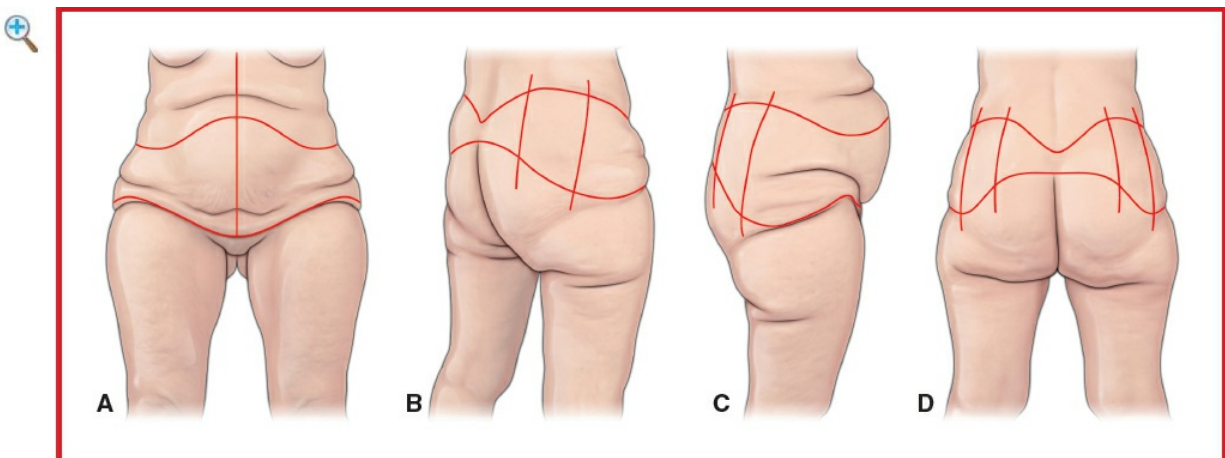


Figure 11.2-6 A–D: Preop incision planning.

Usual preop diagnosis

Thigh and buttock laxity; lipodystrophy; redundant skin and soft tissue

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Multiple: supine, lateral decubitus, prone
Incision	Depending on the combination of procedures; may include circumferential waistline, transverse groin crease, and possible medial thigh vertical extension
Unique considerations	Foley catheter; knee-high SCDs; appropriate pressure point padding, repositioning the patient during the case
Antibiotics	Cefazolin 2 g iv
Surgical time	2–6 h
EBL	50–100 mL, not including blood contained in specimen
Postop care	Avoid ketorolac in the first 24–48 h; maintain semi-Fowler position for any procedures involving abdominoplasty.
Mortality	Minimal
Morbidity	Seroma: 1–16% Dehiscence: 1–32% Skin necrosis: 1–10% Infection: 0.5–3.5% DVT/PE: 2–3%
Pain score	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–65 yr
Male:Female	1:4
Incidence	Thigh lift: 10k performed in the United States in 2016 Lower body lift: >11k performed in the United States in 2016
Etiology	Overweight; massive weight loss
Associated conditions	Obesity; post–gastric bypass

ANESTHETIC CONSIDERATIONS

Patients may have had massive weight loss or may be morbidly obese. (See Anesthetic Considerations for Abdominoplasty). Body lift procedures may be combined with liposuction. (See Anesthetic Considerations for Liposuction.)

Suggested Readings

1. Hamra ST, Small KH: Cosmetic body lift. *Plast Reconstr Surg* 2016; 137(2):453–61.
2. Hurwitz DJ, Ayeni O: Body contouring surgery in the massive weight loss patient. *Surg Clin North Am* 2016; 96(4):875–85.
3. Shermak MA: Body contouring. *Plast Reconstr Surg* 2012; 129(6):963e–78e.
4. Shermak MA: Pearls and perils of caring for the postbariatric body contouring patients. *Plast Reconstr Surg* 2012; 130(4):586e–97e.

LIPOSUCTION

Body contour surgery remains one of the most commonly performed cosmetic surgical procedures in North America. The surgical techniques used to achieve body contour change continue to evolve. The use of epinephrine-containing tumescent solutions injected in the subcutaneous tissue prior to aspiration of fat has dramatically reduced perioperative blood loss and allowed the surgeon to achieve a more dramatic body contour change. Perioperative surgical and anesthetic complications continue to occur during liposuction surgery. Adverse outcome associated with body contour surgery includes pulmonary embolus, fat emboli, fluid overload, and local anesthetic toxicity. Confirmation of the planned surgical procedure, the planned volume of injection of tumescent solution, and the final composition of the tumescent solution must be done prior to the start of surgery. Local anesthetic dosing errors during liposuction have led

to complications and death during liposuction surgery. Adequate time (10 min or longer) after injection of the tumescent solution must be given for epinephrine-induced vasoconstriction to take effect.

External ultrasound/internal ultrasound/laser

The application of energy to disrupt the integrity of fat tissue prior to aspiration may be used to achieve improvements in body contour. Disruption of fat tissue with ultrasound or laser prior to aspiration reduces the physical work of the surgical procedure. Complications reported with the use of ultrasound or laser include seroma formation, increased blood loss, skin loss secondary to burn, and body cavity perforation. Small incisions are made in strategic locations to hide scars. Prefabricated wound protection inserts may be used to protect the skin from cannula irritation. Postop pain is highly variable depending on surgical area and the magnitude of the surgical procedure. Oral analgesics starting with opioids and rapidly tapering to nonnarcotic analgesics over the course of 1–2 wk are usually satisfactory.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	According to body region (repositioning during surgery is common)
Incision	Incisions may be hidden in skin folds and by clothing. Long infusion cannulas and long liposuction probes help minimize the number of incisions necessary.
Special instrumentation	Specialized laser cannulas, ultrasonic probes, and power assisting equipment. Suction machines and canisters used to aspirate and collect fat
Antibiotics	Cefazolin 2–3 g iv
Surgical time	2–9 h depending on the volume of resection and the number of surgical sites
EBL	2–8% of total aspirate volume when using epinephrine-containing tumescent solution before aspiration

Postop care	PACU for small-volume liposuction; hospitalization for postop monitoring for resection volume >5–6 L of fat depending on body size of the patient and the number of regions of the body undergoing surgery
Mortality	Rare
Morbidity	Pulmonary emboli, fat emboli, fluid overload, local anesthetic toxicity, body cavity perforation, skin burn, and tissue loss from ultrasonic energy application or tissue trauma from lipo cannula
Unique considerations	Tumescent solution must be warmed before infiltration to prevent hypothermia; Foley catheter should be used during larger-volume surgeries to monitor fluid status and prevent bladder distension. TEDs, SCDs used on all cases. Postop use of anticoagulant drugs during the 1st wk may lead to excessive bleeding.

■ PATIENT POPULATION CHARACTERISTICS

Age range	Teens to 70 yr
Female:Male	10:1
Etiology	Quest for eternal youth

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients considering liposuction should be ASA category I or II. The ideal candidate for surgery should be physically active and have maintained a stable body weight for 6 mo to 1 yr. Current techniques use the injection of a tumescent solution into the fat prior to aspiration to reduce blood loss and to deliver local anesthetic for analgesia. A typical tumescent solution contains lidocaine 500 mg, epinephrine 1 mg, and bicarb 12 mEq in 1000 mL NS. Typically, 1 mL of tumescent solution will be injected for each 1 mL of anticipated fat resection. Determining the maximum amount of fat that is safe to remove in one surgical setting is complex. Safety is partially determined by the size of the patient and the number or surgical regions planned to have liposuction. Large-volume (>5000 mL) resection may require overnight medical observation and monitoring (fluid shifts, drop in Hct, fluid overload). Large volumes of tumescent solution are often used

in these procedures and require limiting iv fluid administration. In contrast, small-volume liposuction often requires larger volumes of iv fluid administration because of the small volumes of tumescent solution that would be available for postop redistribution from the fat tissue into the central circulation. Performing several small-volume resections may be preferable than a single very large resection given concerns with fluid shifts, blood loss, respiratory compromise, and local anesthetic toxicity.

Respiratory	Postop discomfort following chest, upper back, and upper abdomen liposuction may interfere with respiration. Restrictive compression garments applied to the chest or upper abdomen may also restrict breathing. Patients with respiratory impairment may not be candidates for this procedure. Careful postop monitoring of O ₂ saturation is mandatory.
Cardiovascular	Patients with a history of cardiac dysfunction will not be candidates for liposuction. Fluid management is based on estimation of the patient's intravascular volume status, the volume of tumescent (tumescent) fluid injected during surgery, and the quantity of iv fluid given. Tumescent solution injected into the subcutaneous tissue is absorbed over 48 h. Postop pulmonary edema has been reported secondary to fluid overload. Regional anesthesia may make perioperative fluid management more difficult in large-volume surgeries secondary to peripheral vasodilation. Large doses of epinephrine may lead to increases in blood pressure and cardiac arrhythmias. After tumescent injection, epinephrine levels peak at 5–6 h vs. 10–12 h for local anesthetic levels.
Neurologic	Preop neurologic exam should be normal. Local anesthetic in the tumescent solution may cause numbness in the surgical region for several hours after surgery. Nerve trauma caused by the liposuction instruments has been reported. Careful body positioning during surgery is mandatory as the patient's position on the operating room table is frequently changed during the course of the surgery.

	Repeated checking of the patient's body position on the table is needed because of the vigorous motion of the liposuction procedure.
Hematologic	Vasoconstriction from epinephrine-containing tumescent solutions reduces blood loss to 2–8% of the total aspirate volume. Heavy bleeding during liposuction should cause the procedure to be halted and the cause of bleeding investigated. Failure to add epinephrine to the tumescent solution as a cause of bleeding has been reported. Blood vessel injury, accidental puncture of a body cavity, or unknown coagulopathy may be the cause of increased blood loss. Tests: Hct, PT, PTT, platelet count
Laboratory	No other preop testing is needed in healthy patients.
Premedication	Midazolam 1–2 mg iv or lorazepam 1 mg po 1–2 h preop
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Local anesthesia may be suitable for smaller-volume liposuction. Regional anesthesia (spinal, epidural) may be used when the surgical regions are appropriate for this type of anesthetic. Concerns have been raised because of vasodilation causing an increase in blood loss and an increased risk of fat embolization. Adding local anesthetics to an epidural catheter in combination with local anesthetic in the tumescent solution require careful planning to avoid local anesthetic toxicity. Contrary to standard recommendations, which limit the total lidocaine (with epi) dose to 7 mg/kg, for tumescent anesthesia, a total lidocaine dose of up to 55 mg/kg seems well tolerated.

GA ensures patient comfort and allows liposuction to be performed on all body regions. Control of the airway and control of ventilation provide safety during surgery,

particularly for patients in the prone position. SCDs or foot/ankle compression devices are used for all patients to reduce the risk of PE. Routine use of one or two forced air thermal blankets to maintain normal body temperature is recommended.

Induction	Standard induction. Steroids (dexamethasone 8 mg) may be used to reduce postop swelling, reduce the incidence of postop nausea, and may be of benefit in the event of fat embolism.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia.	
Maintenance	Standard maintenance with volatile anesthetics or propofol infusion. Neuromuscular blockade as appropriate. GA is maintained during application of compression garments at the end of the procedure.	
Emergence	Antiemetic prophylaxis with combination drug therapy (metoclopramide 10–20 mg) and ondansetron (4 mg) is appropriate. Careful monitoring of respiratory function is necessary when surgery has been performed on the chest, back, or upper abdomen because compression garments and postop pain may limit respiration.	
Blood and fluid requirements	IV: 18 or 20 ga ×1 BSS as appropriate.	Fluid administration determined by the needs of the case. Open surgery (abdominoplasty) performed in combination with liposuction will require careful monitoring of intravascular status and monitoring of urine output during surgery. Transfusion is rarely needed; diuretics (furosemide 5–10 mg) may be needed for patients receiving large

volumes of tumescent solution. Larger intravenous fluid volumes are needed during smaller-volume liposuction because less tumescent solution has been used. IV fluids may need to be limited in patients undergoing larger-volume liposuction because larger volumes of tumescent solution will be absorbed into the circulation. Goal-Directed Fluid Management (see [Appendix K, p. K-4](#)).

Monitoring	Standard monitoring Foley catheter	Urine output monitoring mandatory on all large-volume lipo cases Careful temperature monitoring
Positioning	✓ and pad pressure points. ✓ and protect eyes. Repeat positioning. Check frequently.	Frequent intraop position checks are needed as patient position may change during surgery → ↑ risk of peripheral nerve injury. Document eye checks q15min while patients are in the prone position.
Complications	Local anesthetic systemic toxicity (LAST) Excessive blood loss Volume overload Abdominal cavity perforation Peripheral nerve injury	Lidocaine up to 55 mg/kg in the tumescent solution with epinephrine has been shown to produce safe serum levels when used for tumescent injection during prolonged

Hypothermia

Fat embolism

Pulmonary embolism

liposuction. Peak plasma lidocaine levels occur 10–12 h after infusion. Peak epinephrine levels occur 5–6 h after infusion and may lead to increases in heart rate and/or blood pressure. At the first signs of LAST, lipid emulsion (1.5 mL/kg iv) should be used to rapidly reduce blood levels of lidocaine. Postop hypoxia may be caused by multiple factors, including fluid overload and pulmonary edema, pneumothorax, restricted ventilation, pulmonary embolus, and respiratory depression. Unexplained postop hypoxia will require hospitalization. Anticoagulation for PE may lead to severe blood loss because of large tissue areas that have been injured during liposuction. Blood transfusion must be readily available before beginning anticoagulation therapy in a patient who has recently undergone liposuction. IVC filter may be needed for a patient with PE. Hypertension may be

		caused by fluid overload, delayed epinephrine effect, or surgery-induced pain.
Multimodal pain management	PO analgesics IV opiates Acetaminophen	Patients often will be comfortable 2° to residual local anesthesia that may persist for 8–24 h. Oral analgesics are usually satisfactory for postop pain control.
ERAS	PONV protocol.	Encourage early enteral nutrition and early mobilization.
Tests	Rarely needed, Hct	

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CHAPTER 11.3

Craniofacial Surgery

SURGEON

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ANESTHESIOLOGIST

Richard A. Jaffe, MD, PhD

REPAIR OF FACIAL FRACTURES

SURGICAL CONSIDERATIONS

Description

Facial fractures are classified by location and the involved bones.

Upper and midface region

Frontal sinus fractures may involve the anterior wall alone or also may involve the nasofrontal ducts and/or posterior wall. Nasofrontal duct disruption may require obliteration of the duct and sinus, which is done with an electric burr and loupe magnification to remove all mucosa before grafting the area with bone, fat, or pericranium. A posterior wall disruption is a fracture into the anterior cranial fossa that may require CSF leak repair ± cranialization of the sinus (complete removal of the posterior wall of the sinus). Each frontal bone forms a large component of the orbital roof, and as such, ocular injury or periorbital entrapment must be considered.

Fractures of the maxilla are classified as **Le Fort I**, **II**, or **III**, depending on the level of the fracture ([Fig. 11.3-2](#)). **Le Fort I** is a horizontal fracture, separating the teeth and lower maxillary components from the upper facial structures. **Le Fort II** is a triangular fracture with a fracture line across the nose, below the infraorbital rims, and extending through the entire lower maxillary structures. **Le Fort III** is essentially a disassociation

of the cranium and face. In these cases, the maxilla is usually mobile or impacted posteriorly and occasionally closes off the posterior airway. Further mobility of the segments may be present with a sagittal split of the palate. Associated fractures in the maxillary region include fractures of the zygoma, orbital fractures (most commonly orbital floor), isolated nasal fractures, naso-orbital-ethmoid (NOE) fractures (usually with severe comminution of the upper face), and cranial base fractures with the potential for dural tears and CSF rhinorrhea. Added procedures that may be required to complete the repair of these fractures include **local flap closure** of a CSF leak and **primary bone grafting**, usually from cranium or distant sites, such as the ilium, to highly comminuted areas (e.g., NOE, orbital floors).

Lower face

Fractures of the mandible are classified by the type of fracture and location (Fig. 11.3-3), the most common being the subcondylar fracture. Fractures involving the mandibular body, such as a parasymphiseal fracture, may result in unstable mandibular segments. In cases of bilateral mandibular body fractures associated with symphyseal fractures, the mandible can be flail and fall posteriorly in the supine position, allowing the tongue to block off the airway. All of the fractures involving change in occlusion (Le Fort maxillary and all mandibular fractures) require reestablishment of a normal occlusion by the application of arch bars, temporary anchorage devices (TAD: a small, screw-like device placed between the roots of teeth), or other means of providing support for intermaxillary fixation (IMF). This may be combined with rigid fixation, most commonly internal plates. In some cases, rigid fixation will allow removal of the IMF at the end of the case; in others, IMF may be required for postop healing. Removal of the throat pack prior to final IMF is of paramount importance.

Trismus may be associated with any of the above injuries 2° direct injury to the muscles of mastication but is more commonly associated with fractures of these muscular attachments (e.g., mandible, zygoma). Associated **dentoalveolar fractures** of the maxilla or mandible may require preop wiring in the ER. The intent is to hold steady those segments with tenuous stability and blood supply. Intubation techniques should avoid displacing these segments. Fractures not involving change in occlusion (e.g., orbital zygomatic, nasal fracture) can be orally intubated. Most fractures with a change in occlusion should be nasally intubated with RAE or 60° curved connector. Exceptions include edentulous segments allowing tube to pass versus edentulous patient with a splint fabricated for oral intubation. Another preop consideration is the amount of blood loss at the scene or in the ER. Facial and scalp vessels can bleed profusely (hypovolemia), and patients may arrive in the OR with both anterior and posterior nasal packs in place (difficult ventilation).

The **surgical approach** depends on the extent of fractures and associated lacerations. Periorbital incisions can be external, on or below the lower eyelid and over the brow, or internal, along the lower eyelid conjunctiva. Upper facial repair may include a bicoronal approach ([Fig. 11.3-4](#)) designed to peel the face off the upper facial skeleton via an ear-to-ear scalp incision. Rainey clips are used to minimize scalp bleeding. Of note, periorbital dissection to explore and repair NOE or orbital floor fractures involves some retraction on the globe. This may cause ↓ HR and ↓ BP via the oculocardiac reflex. The mandible can be approached through external, preauricular or inferior border, or intraoral incisions. A panfacial fracture treatment protocol is illustrated in [Figure 11.3-1](#).

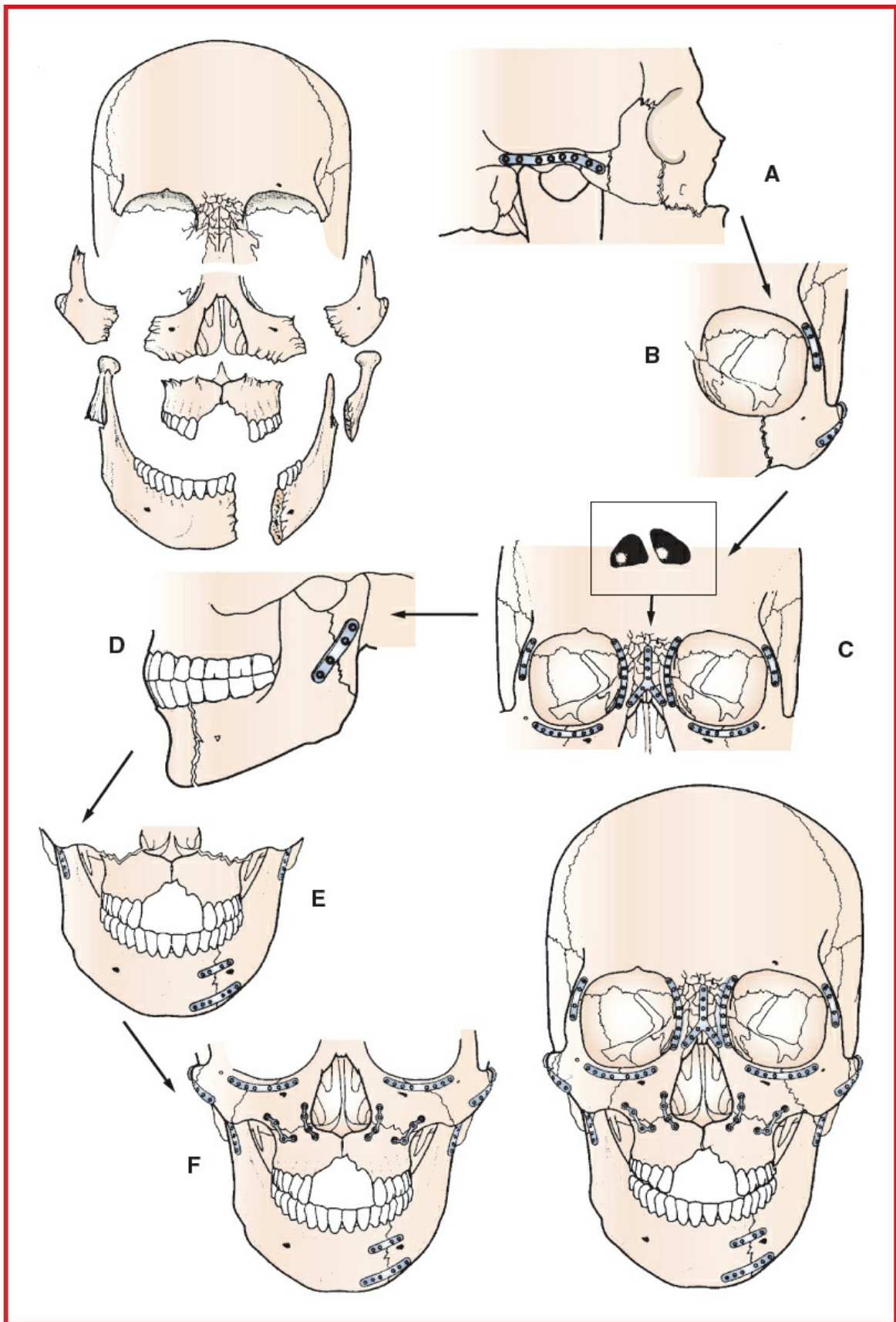
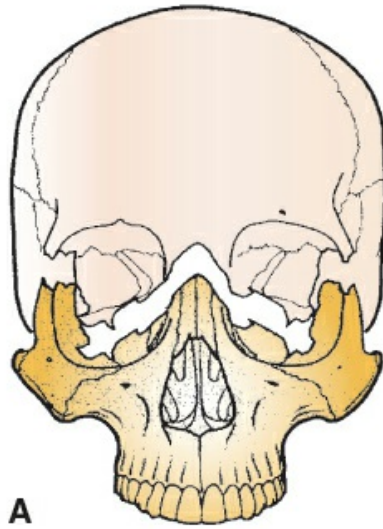


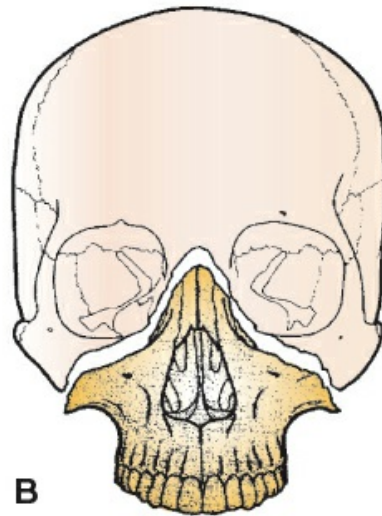
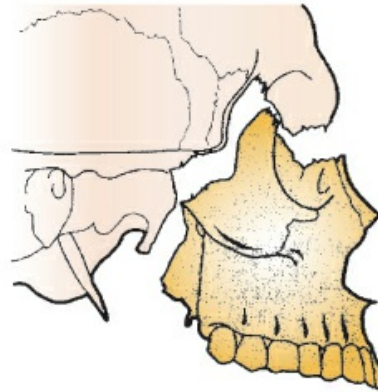
Figure 11.3-1 Panfacial fracture treatment protocol, based on reconstructing load-bearing structures of the facial skeleton. **A:** Projection of the midface is created by reconstructing the zygomatic arches, starting from the stable part of the temporal bone. **B:** The zygomas are fixed to the arches and to the

frontal bone to create the final projection of the midface. **C:** The width of the midface is reconstructed by repositioning the central midface (orbits and nose) to its correct position, in relation to the zygomas and frontal bone. Concomitantly, canthopexy is fixed, and the frontal bone and sinus fractures are treated. (This procedure is independent of the occlusion.) **D:** The posterior vertical height of the face is reconstructed by positioning and fixing the condylar fractures. **E:** Intermaxillary fixation is applied, and the mandible is reconstructed. **F:** Finally, the Le Fort I–level fractures are positioned to natural occlusion. (Reprinted from Booth PW, Schendel SA, Hausamen J-E, edsMaxillofacial Surgery, Vol. 1, 1st edition. Churchill Livingstone, Edinburgh: 1999. Copyright © 1999 Elsevier. With permission.)



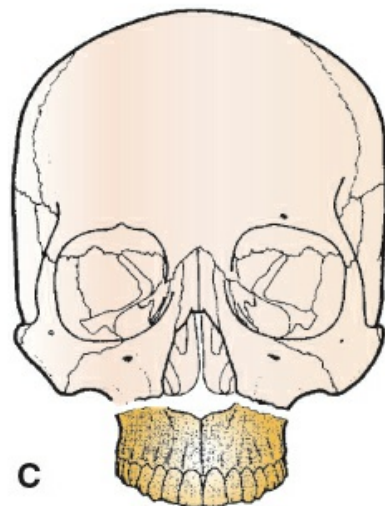
A

Le Fort III fracture



B

Le Fort II fracture



C

Le Fort I fracture



Figure 11.3-2 Schematic of the Le Fort fracture lines. **A:** Le Fort III involves separation of cranium from facial bone structure. **B:** Le Fort II is a pyramid-shaped fracture, including the dentition and nasal structures. **C:** Le Fort I is a horizontal fracture involving mobilization of dentition and maxilla. (Reprinted from Booth PW, Schendel SA, Hausamen J-E, eds *Maxillofacial Surgery*, Vol. 1, 1st edition. Churchill Livingstone, Edinburgh: 1999. Copyright © 1999 Elsevier. With permission.)

Variant procedure or approaches

Endoscopic approaches are being developed for multiple fracture sites. **Resorbable plates and screws**, especially for pediatric cases, can be applied through the same surgical approaches.

Usual preop diagnosis

Facial trauma

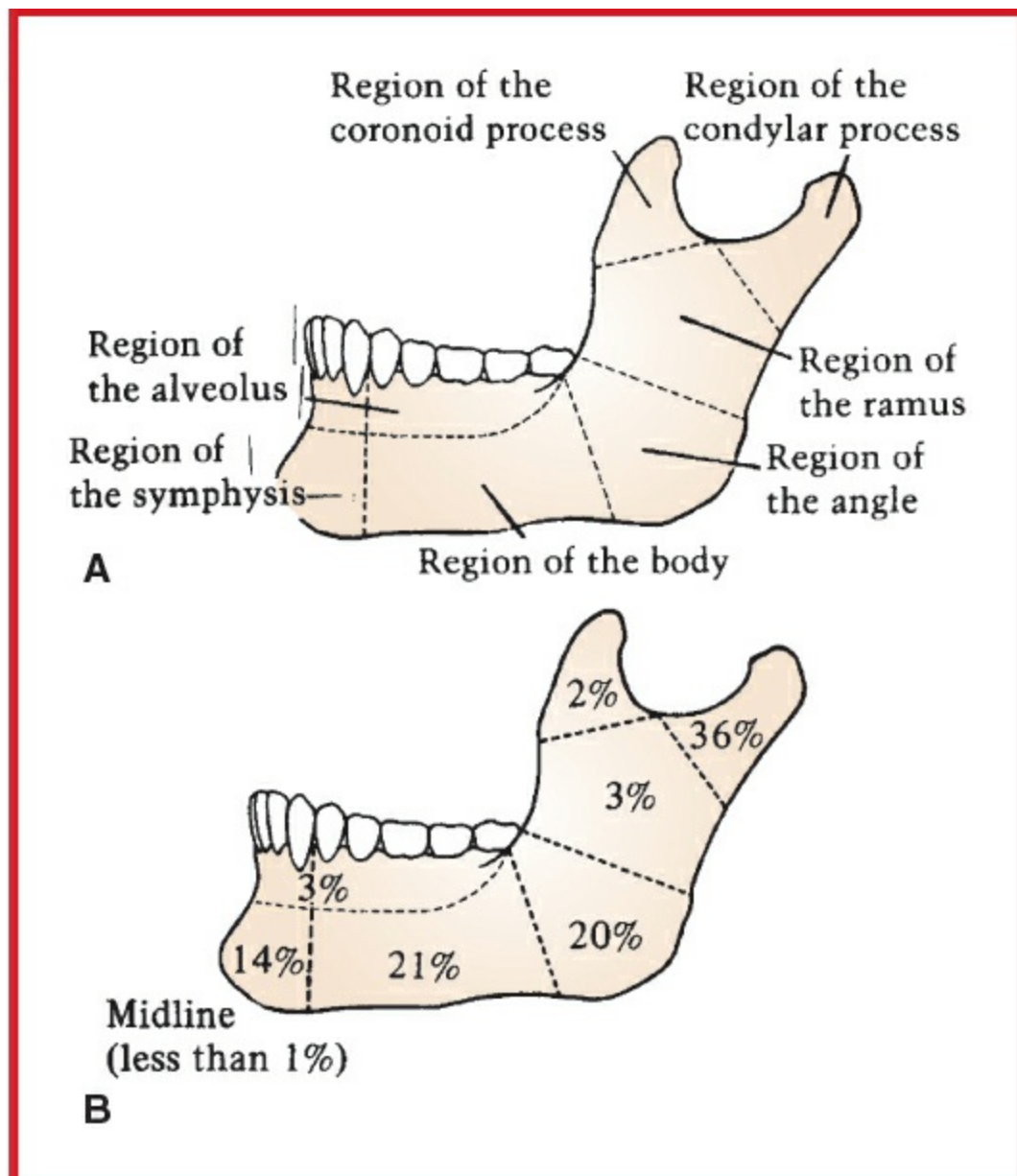


Figure 11.3-3 A: Anatomic regions of the mandible. B: Frequency of fractures in those regions. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds: *Grabb and Smith's Plastic Surgery*, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

ERAS: For elective cases, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status and possible benefits of weight reduction (time permitting), preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

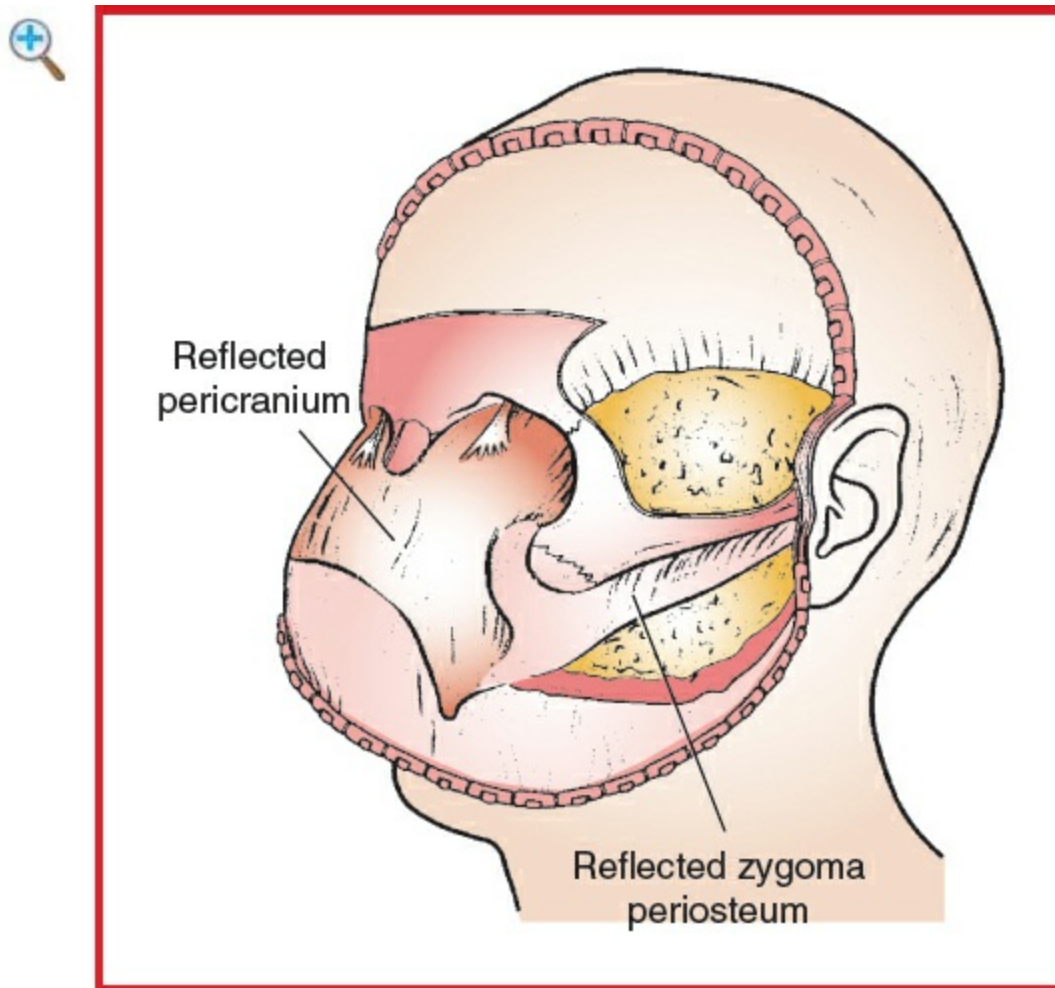


Figure 11.3-4 Bicornal approach. (Reprinted from Booth PW, Schendel SA, Hausamen J-E, eds: *Maxillofacial Surgery*, Vol. 1, 1st edition. Churchill Livingstone, Edinburgh: 1999. Copyright © 1999 Elsevier. With permission.)

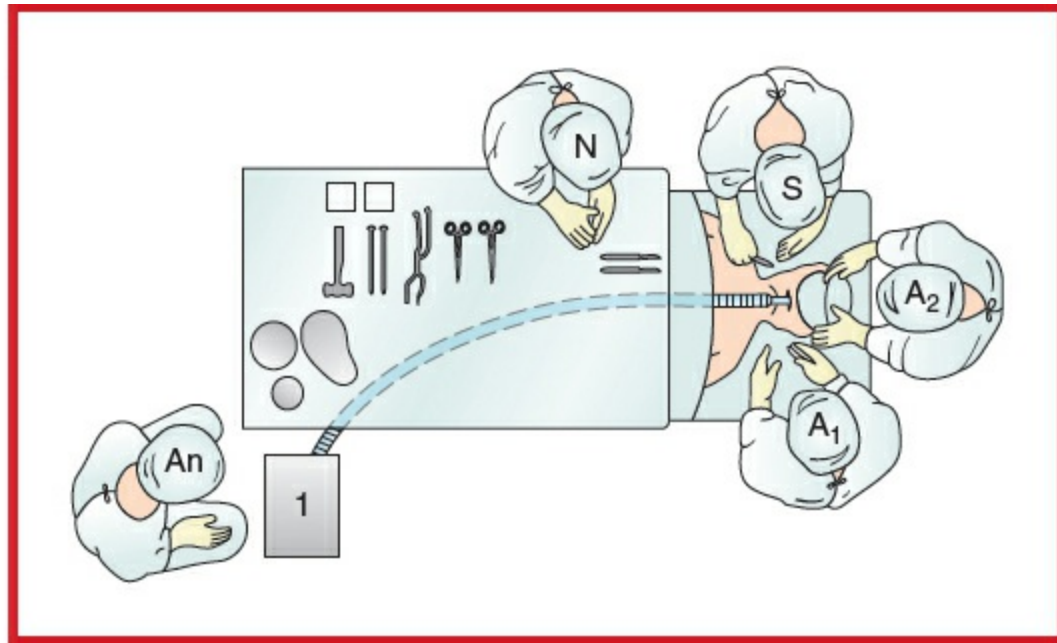


Figure 11.3-5 Standard anesthesia and surgical setup for a maxillofacial surgical procedure and certain craniofacial surgical procedures. The table may be rotated 90°–180° with anesthesia equipment and personnel at the foot or off to one side. (Reprinted from Bell WH, ed:Modern Practice of Orthognathic Surgery, Vol. I. WB Saunders, Philadelphia: 1992. Copyright © 1992 Elsevier. With permission.)

■ SUMMARY OF PROCEDURES

	Mandibular	Maxillary Orbital/Zygomatic	Nasal
Position	Supine	⇐	⇐
Incision	Intraoral; lateral, submandibular	Intraoral, ± subciliary; possibly coronal	Closed reduction; possible intranasal
Special instrumentation	Air power tools; plate fixation; headlight	⇐ + Periorbital malleable retractors; Rowe disimpaction forceps; acrylic dental split	⇐
Unique considerations	Nasal RAE (sutured to dentition or septum); throat pack; corticosteroids periop to ↓ edema	Panfacial smash; airway swelling associated with head injury; burns	—

		or other system involvement. Consider preop tracheostomy. Close monitoring for and removal of stimulus until oculocardiac reflex recovery	
Antibiotics, etc.	Cefazolin 2 g iv	⇐ + Methylprednisolone 125 mg iv (adults)	⇐
Surgical time	1–3 h	1–6 h (bicoronal approach adds 1.5 h)	1 h
Closing considerations	NG at end of procedure, maintained for 24 h; IMF postop; NB: Throat pack removal	⇐	—
EBL	100–800 mL, depending on extent of fractures, need for graft harvest, patient age	⇐	⇐
Postop care	Airway or respiratory status may be compromised (preop and postop swelling/aspiration/associated injuries), requiring maintenance of intubation → ICU.	⇐	⇐
Mortality	Minimal (↑ due to associated injuries and blood loss)	⇐	⇐
Morbidity	Trismus	⇐	—
	Poor occlusion	⇐	—
	Drooling	⇐	—
	Poor cosmetic result	⇐	⇐

	Chronic pain/sensation changes	⇐	⇐
	Loss of taste	Loss of smell	⇐
	Visual problems/epiphora	—	
	Sinus problems	⇐	
	Breathing difficulty	⇐	
Pain score	4–8	4–8	4–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	>4 yr
Male:Female	1:1 (varies by age group)
Incidence	The most common are nasal bones, zygoma and arch, mandible, and orbital floor (on the rise due to airbags in autos).
Etiology	Adults: Motor vehicle accidents (MVAs), mostly young males (43–67%); assaults (13–49%); sports related (4%); falls (4%); gunshot wounds (3%) Children: Uncommon: only ~5% of all facial fractures occur in ages <12 yr; falls (33%); MVA (28%); abuse (3%); dog bites (rare). 69% are periorbital or nasal.
Associated conditions	Airway compromise/closed head trauma (40%); extremity fractures (33%); thoracic injury (29%); open or radiographic brain injury (25%); intra-abdominal (12%); globe injuries (11%); oral trauma (11%); pelvic fractures (10%); C-spine fractures (4–10%); T-L spine injuries (4%); shock; multisystem trauma; burns; massive soft tissue loss. Mandibular fractures during MVAs are associated with a 65% incidence of life-threatening injuries and mortality of up to 8%; up to 1/3 of Le Fort fractures will require intubation, most often due to upper respiratory tract blood and secretions.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The forces required to produce facial fractures are considerable and frequently result in other associated trauma (e.g., closed head trauma; spine injuries; thoracic injury, including pneumothorax and myocardial contusion; intra-abdominal bleeding). Soft tissue injury to the tongue or larynx can make airway management difficult. When in doubt, a tracheostomy under local anesthesia or awake intubation should be considered. In mandible or maxillary fractures, nasal intubation is usually best (in the absence of basilar skull fracture or midface instability) because the patient will be placed in intermaxillary fixation (IMF; teeth brought together via wires or rubber bands) at the conclusion of the procedure. In malar or nasal bone fractures, the fixation may be precarious, making it undesirable to use mask ventilation at the termination of the procedure and necessitating awake extubation. Facial nerve monitoring may be required, contraindicating the use of persistent muscle relaxation.

Frequently, the anesthesiologist's first encounter with these patients is in the ER, where prompt airway management decisions are essential—often before diagnostic imaging studies are complete. These patients should be treated with full-stomach precautions and may have already aspirated. Patients may be unable to open their mouths 2° pain or mechanical factors. The cause of limited mouth opening should be determined before induction of anesthesia. Several options exist. Often, the airway can be managed simply by inserting an oropharyngeal airway; failing this, an emergency intubation or tracheostomy will be necessary. Blind nasal intubation should be avoided in patients with CSF rhinorrhea or other evidence of nasopharyngeal trauma, where the potential for creating false passages and additional trauma is significant. An awake oral intubation with topical anesthesia is often the safest approach. Emergency oral intubation may be complicated by an unstable C-spine and limited jaw opening, together with blood and debris in the oropharynx, making visualization difficult if not impossible. Often, the only recourse is tracheostomy under local anesthesia. As with any trauma victim, attention is first directed toward maintaining the airway and restoration of fluid volume. Repair of the facial fracture may be carried out incidental to the primary trauma surgery or, more often, is deferred until the patient's condition is stabilized. The following preop assessment will focus primarily on the patient coming to the OR for semielective repair of a facial fracture.

Airway

Semielective: Usually, facial swelling and intraoral bleeding will have resolved, although mouth opening may be limited 2° pain or mechanical factors. Airway management requires knowledge of the fracture site(s). Patients with a maxillary fracture may benefit from an oral intubation to allow inspection of the

nasopharynx before nasal intubation and definitive repair. The possibility of an awake fiberoptic intubation should be discussed with the patient. Patients with an isolated orbital, zygomatic, or nasal fracture usually do not present airway management problems. The surgeon should be consulted regarding the preferred intubation route.

Trauma: Airway and nasal obstruction following trauma can be extreme, as a result of soft tissue swelling and accumulated blood and secretions. The extent of facial fractures, particularly in the midface, should be identified as they may preclude nasal intubation. Mandibular fractures may make access to the oropharynx difficult. Unstable dentoalveolar fractures may require preop wiring or temporary screw fixation in the ER. In the case of massive trauma to the face, urgent tracheostomy should be considered.

Tests: As indicated from H&P

Respiratory

Trauma: Evaluate for associated trauma and respiratory insufficiency 2° aspiration. ✓ that chest tubes are functioning properly.

Tests: CXR; others as indicated from H&P

Cardiovascular

Semielective: Typically, several days will have elapsed since the initial trauma, and the patient should be hemodynamically stable.

Trauma: Blunt chest trauma may be associated with myocardial contusion, pericardial effusion/tamponade, and aortic tear/dissection.

Tests: ECG; others as indicated from H&P

Neurologic

Semielective: Document any neurologic deficits and altered mental status. Meningitis may occur in patients with persistent CSF rhinorrhea or pneumocephalus.

Trauma: Intracranial injury may be associated with facial fractures. Patients with head trauma may have ↑ ICP; therefore, appropriate methods (e.g.,

	hyperventilation, smooth induction/intubation) are used to prevent further ↑ ICP. Basilar skull fractures preclude passage of nasotracheal and NG tubes. In the presence of otorrhea or rhinorrhea, positive-pressure mask ventilation is inadvisable due to the potential for causing pneumocephalus.
Musculoskeletal	Tests: Review skull and C-spine x-rays. Semiselective: May be associated with other fractures and soft tissue trauma that may affect patient positioning. Trauma: C-spine injuries are commonly associated with facial injuries. The C-spine should be cleared by clinical and x-ray exam before transport to OR. If C-spine cannot be cleared, intubation should be done with the head in a neutral position (splinted or with axial stabilization), using direct FOL.
Hematologic	Trauma: Maxillary surgery may be associated with major blood loss. For elective cases, autologous donation should be encouraged. Tests: Hct; others as indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	Standard premedication is appropriate for nontrauma, neurologically intact patients with normal airways. Trauma: Trauma patients should be considered to have full stomachs, and sedative premedications are best avoided. Aspiration prophylaxis with 0.3 M Na citrate (30 mL po), ± metoclopramide 10 mg iv ± ranitidine 50 mg iv, should be considered.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po or iv. Consider aprepitant 40 mg po or fosaprepitant 125 mg iv for patients with h/o severe PONV.

Anesthetic technique

The majority of patients presenting for elective procedures are otherwise healthy and have normal airways. In the case of facial trauma, however, intubation of the trachea may be impossible and a tracheostomy under local anesthesia may be lifesaving.

Induction	If there is any doubt regarding the ease of intubation, an awake FOL should be performed. Nasal intubation is preferred for patients with mandibular and maxillary (Le Fort) fractures involving a change in occlusion. Patients with orbital, zygomatic, or nasal fractures usually are intubated orally. In patients with normal airways, a standard induction is appropriate. Nasal or oral ETTs (RAE) or anode ETTs are commonly used to minimize intrusion into the surgical field. A nasal or oral ETT may not be appropriate in some patients (e.g., with midface fractures who also require maxillomandibular fixation). An alternative to tracheostomy in these patients is submental intubation: following oral ETT placement using a wire-reinforced tube, the surgeon will tunnel the covered connector end of the tube (with pilot balloon) through the floor of the mouth to emerge below the submental area of the mandible where it is reconnected to the ventilator.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia.
Maintenance	Standard maintenance; muscle relaxation usually is required except with facial nerve monitoring. Controlled ↓ BP may be appropriate. Administration of an antiemetic (e.g., ondansetron 4 mg iv ± aprepitant 125 mg iv) is beneficial in patients who have their jaws wired or banded together.
Emergence	Patients with difficult airways or with jaws wired together should be extubated when fully awake.

	Extubation over a tube changer may be appropriate. A wire cutter (or scissors for elastic bands) should be at the bedside at all times. Ensure that any throat packing has been removed before extubation. Thorough oropharyngeal suctioning is essential. NG tube may be used postop, so consider placement prior to extubation. Some patients with multiple trauma or extensive soft tissue swelling may require continued postop intubation and mechanical ventilation. Consider iv acetaminophen if po dose given >4–6 h ago.	
Blood and fluid requirements	Moderate—large blood loss IV: 16–18 ga × 1–2 Trauma: BSS @ 6–8 mL/h Other: BSS @ 2–4 mL/h	Blood loss from facial fractures or orthognathic procedures can be extensive. T&C patient so blood is immediately available in OR. Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors ± Arterial line	Invasive monitoring may be required in patients with intracranial or other trauma or for controlled ↓ BP. Muscle relaxants will interfere with facial nerve monitoring.
Control of blood loss	Surgical hemostasis Tranexamic acid Topical vasoconstrictors	Surgical hemostasis should control most bleeding in these procedures. TXA: 1 g iv bolus + 1 g iv infused during surgery will ↓ need for transfusion (NB: 95% renal elimination). Topical vasoconstrictors, such as phenylephrine or

	Posterior oropharyngeal packing Controlled ↓ BP	cocaine, can be applied on the surgical field. Posterior oropharyngeal packing can keep blood from passing undetected into the upper GI tract. Hypotensive anesthesia often can be achieved simply by increasing volatile anesthetic levels. The lowest acceptable MAP depends on preop BP (MAP) and comorbidities. Usual range = 50–65 mm Hg (MAP).
Positioning	✓ and pad pressure points ✓ eyes	Some surgeons prefer that the OR table be rotated 90° or 180°. Be prepared with long hoses and appropriate connectors. Consider protecting eyes with an ophthalmic ointment.
Complications	ETT damage Oculocardiac reflex ↓↓ HR + ↓ BP	Nasal ETT may be wired inadvertently to the maxilla, making extubation difficult. In case of ETT damage, be prepared to reestablish the airway rapidly by reintubation, usually over an intubating stylet. Notify surgeon (stop surgical stimulus). Consider atropine and increasing depth of anesthesia.

POSTOPERATIVE

Complications	Airway obstruction	Wire cutters (or scissors) should be available at bedside to facilitate emergent reintubation or other form of airway management. Consider retained throat pack.
	PONV	Multimodal treatment of PONV is important.
Multimodal pain management	Acetaminophen NSAID Gabapentin Parenteral narcotics or PCA	Local anesthetic commonly infiltrated at end of procedure by surgeon. Continuous infusion catheters may be beneficial.
ERAS	PONV protocol	Encourage early enteral nutrition, Foley catheter removal, and early mobilization.

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LE FORT OSTEOTOMIES

SURGICAL CONSIDERATIONS

Description

Le Fort osteotomies are used to correct maxillary deformities. The most common is the **Le Fort I**, a transverse osteotomy, above the apices of the teeth, used to correct maxillary retrusion or maxillary vertical excess or deficiency. One of the goals in achieving improved function is to improve the occlusion, as classified by Angle ([Fig. 11.3-6](#)). One or both jaws may be moved to achieve normocclusion. An intraoral vestibular incision is used for this approach, followed by an osteotomy of the maxilla, with either a burr or a saw, and completed with osteotomes. The osteotomy extends through the maxillary sinus toward the ETT, within the piriform aperture. There is potential for nasal ETT damage with this maneuver. Of note, BP usually will increase at the start of the osteotomies, increasing bleeding if anesthesia/analgesia is inadequate. The maxilla is subsequently downfractured ([Fig. 11.3-7](#)) and mobilized with Rowe disimpaction forceps. This is when most of the bleeding occurs. Further interdental osteotomies creating multiple maxillary segments may be required. **Le Fort II** or **Le Fort III** maxillary osteotomies may be used to correct severe midfacial retrusion. In these cases, infraorbital incisions are used with either brow or coronal ([Fig. 11.3-4](#)) incisions. With maxillary advancements, iliac or cranial bone grafts are often necessary to close the bony gap created. Most of these patients have orthodontic appliances and will require IMF, with either wire or elastics at the end of the procedure, with or without a prefabricated splint. The maxilla is fixed rigidly in the new position with miniplates.

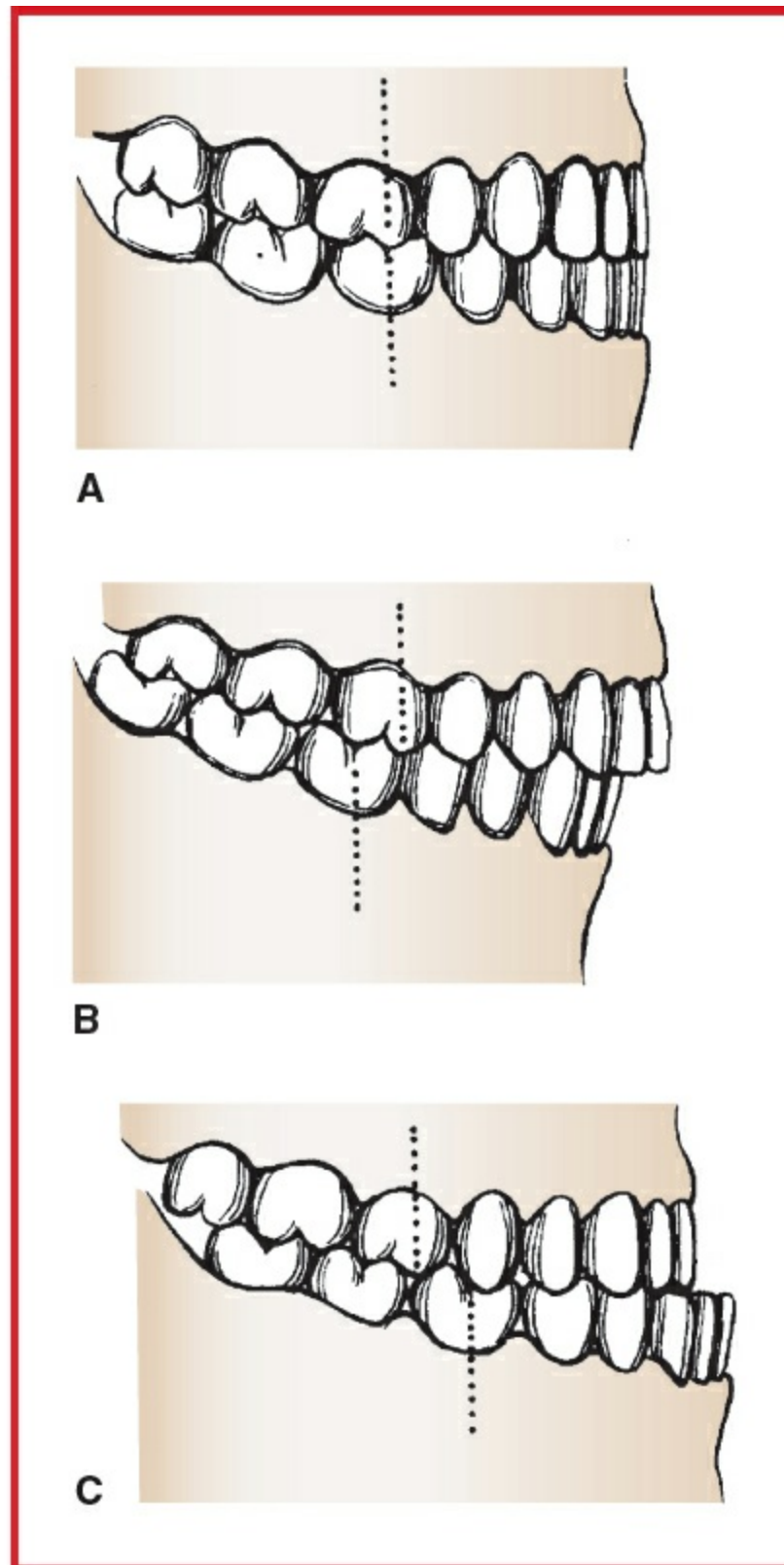


Figure 11.3-6 The Angle classification of occlusion. **A:** Class I, normal occlusion. **B:** Class II, retro-occlusion or mandibular deficiency. **C:** Class III, prognathic occlusion (maxillary deficiency or mandibular excess). The key relationships to be discerned are those of the first molar teeth, cuspids, and incisors. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds Grabb and Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

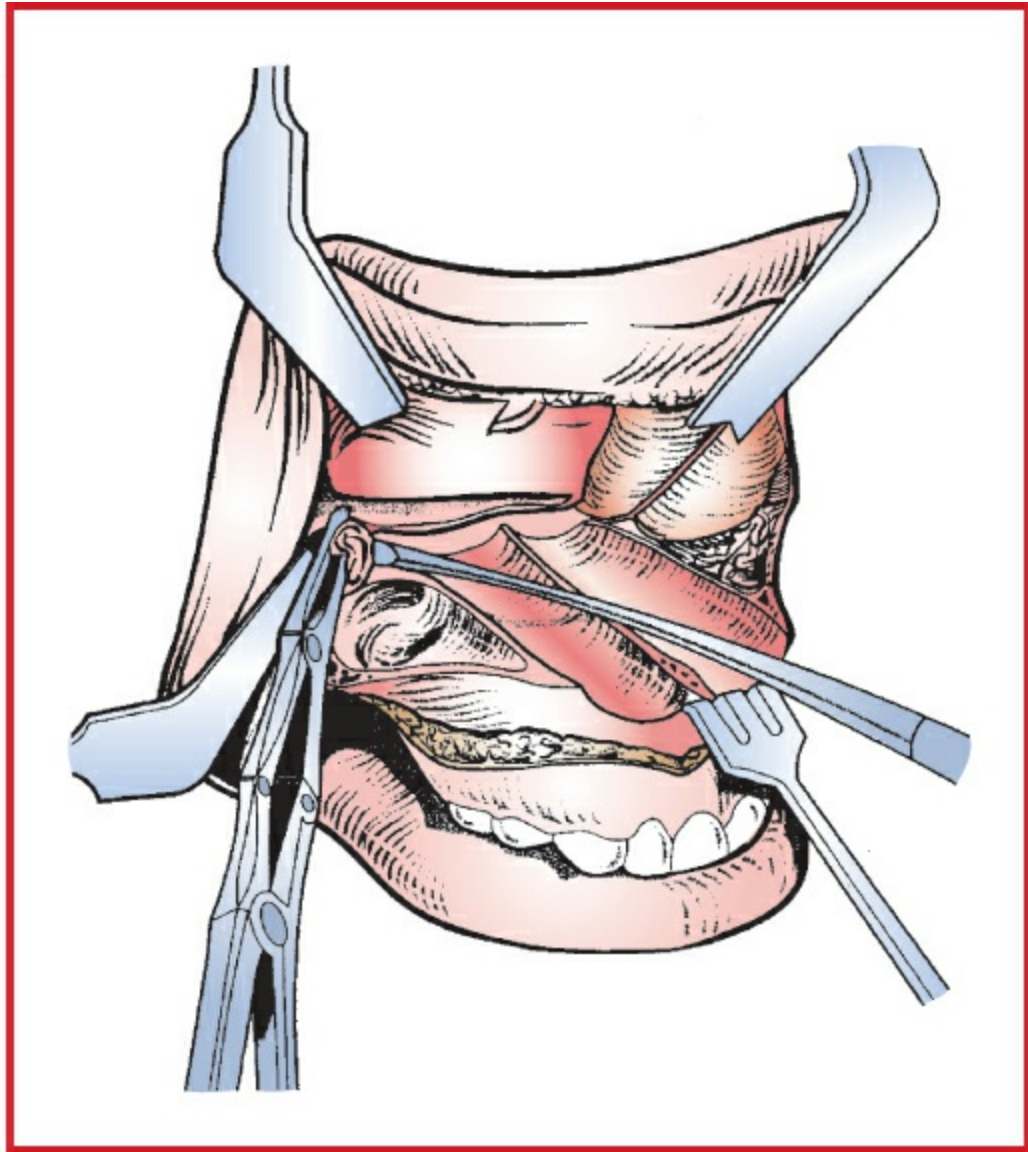


Figure 11.3-7 Removal of bone around the perpendicular plate of the palatine bone. The descending palatine vessels may be ligated. (Reprinted from Booth PW, Schendel SA, Hausamen J-E, eds: Maxillofacial Surgery, Vol. 1, 1st edition. Churchill Livingstone, Edinburgh: 1999. Copyright © 1999 Elsevier. With permission.)

Variant procedure or approaches

Recently, **distraction osteogenesis** after Le Fort osteotomy has been shown to be effective for correction of severe deformity that would be difficult to correct with single movements and internal fixation. This technique creates new bone as the osteotomized bones are slowly separated. No bone grafts are necessary. Distraction hardware is applied either internally or externally. External devices are stabilized via halo and compression bolts into the skull. IMF is not indicated, as the upper jaw must be free to be distracted in relation to the lower jaw. A second operation may be required for removal of distractor hardware. **Absorbable internal devices** also are available.

Usual preop diagnosis

Facial deformities; dentoskeletal dysplasia

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status and possible benefits of weight reduction (time permitting), preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES				
	Le Fort I	Le Fort II	Le Fort III	Distractor
Position	Supine; table may be rotated 90° or 180°	⇐	⇐	⇐
Incision	Intraoral	Intraoral and facial ± coronal	⇐	⇐
Special instrumentation	Miniplates and screws; burr or saws	⇐	⇐	Distractors
Unique considerations	Jaw frequently closed (wires or elastic); nasal RAE or armored tube. NB: Keep SBP < 100 mm Hg during osteotomy closure.	⇐	⇐	No IMF
	With cranial bone graft harvest, the nasal ETT goes opposite to the donor side and is repositioned later.	⇐	⇐	⇐
Antibiotics, etc.	Cefazolin 2 g iv; methylprednisolone 125 mg iv (adults)	⇐	⇐	⇐
Surgical time	3–6 h	⇐	⇐	⇐

Closing considerations	NB: Throat pack removal. NG at end of case × 24 h. NSAIDs contraindicated.	⇐	⇐	⇐
EBL	400–800 mL	⇐	⇐	⇐
Postop care	PACU → room	ICU × 1 d	⇐	⇐
Mortality	Rare	⇐	⇐	⇐
Morbidity	Relapse	⇐	⇐	⇐
	Infection	⇐	⇐	⇐
	Bone/tooth loss: Uncommon	⇐	⇐	⇐
	Severe intraop bleeding: Uncommon	⇐	⇐	⇐
	Blindness: Very rare	⇐	⇐	⇐
	Temporomandibular joint dysfunction	⇐	⇐	⇐
	Nasal obstruction	⇐	⇐	⇐
	Nasolacrimal obstruction	⇐	⇐	⇐
	Oronasal fistulas (multipiece osteotomies, uncommon)	⇐	⇐	⇐
	Poor cosmetic result	⇐	⇐	⇐
	Sensory nerve dysfunction	⇐	⇐	⇐
				Distractor malfunction
Pain score	4	5	5	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	15 yr–adulthood, usually <30 yr
Male:Female	1:1
Incidence	Up to 5% of population
Etiology	Usually developmental in nature; also cleft lip and palate patients Rarely syndromic: Apert (1/100,000); Crouzon (1/25,000)
Associated conditions	Usually none with developmental cases; sleep apnea; or part of a recognized condition associated with congenital anomalies (e.g., cleft lip and palate, proptotic eyes, severe mitten-hand syndactylies)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Mandibular Osteotomies/Genioplasty.

Suggested Readings

1. Adke M, Medonca C: Concealed airway complication during LeFort I osteotomy. *Anesthesia* 2003; 58(3):294–5.
2. Bidoli SJ, Dumont L, Mattys M, et al: A serious anesthetic complication of a Lefort osteotomy. *Eur J Anesthesiol* 1999; 16(3):201–3.
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4. Lo LJ, Hung KF, Chen YR: Blindness as a complication of LeFort I osteotomy for maxillary distraction. *Plast Reconstr Surg* 2002; 109(2):688–700.
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MANDIBULAR OSTEOTOMIES/GENIOPLASTY

SURGICAL CONSIDERATIONS

Description

Mandibular deformities include either a retruded mandible or a prognathic mandible involving a malocclusion (class II or III) and may occur in combination with a small chin (microgenia). Surgical correction of the basic mandibular deformity involves either advancing or retruding the mandible. The most common procedure for this is the **sagittal ramus split osteotomy** (Fig. 11.3-8). An intraoral approach to the mandible is used to sagittally split the mandible (BSSO) with the inferior alveolar nerve solely within the anterior segment, such that the tooth-bearing anterior segment can slide forward or backward to correct the original deformity. A variety of other techniques of mandibular osteotomies exist, often involving the ramus area of the mandible, and are performed via an intraoral approach.

In some very large deformities, an external incision (**Risdon** type) and bone graft placement may be necessary to complete the mandibular ramus reconstruction. Many patients will be treated for dentoskeletal deformities with a combination of Le Fort movement and sagittal split osteotomies of the mandible. When the mandible is set back, mandibular bone is resected and often can be used in combined procedures as bone

graft for the Le Fort osteotomy. This avoids cranial bone harvest and, therefore, avoids repositioning the ETT during the procedure for cranial bone access and avoids donor site morbidity. Occasionally, deformities may be corrected by mandibular body osteotomies.

Rigid fixation of mandibular osteotomies is often accomplished by the use of small miniplates and screws. Fixation (if needed) also can be accomplished with elastic traction (rarely, wire fixation) between mandible and maxilla (IMF), placed either at the time of surgery or several days following, and held in position for 1–2 wk. These procedures may be combined with a **genioplasty** to correct the chin deformity, or genioplasty may be performed as an isolated procedure. The most common type is a horizontal osteotomy of the inferior mandible, with the chin segment repositioned with internal fixation ([Fig. 11.3-9A](#)). Augmentation of the chin also is accomplished with alloplastic onlay materials ([Fig. 11.3-9B](#)) placed via either the oral route or the extraoral route through a small submental incision. As an isolated procedure, genioplasty is performed most frequently with local anesthesia and sedation.

Maxillomandibular advancement, otherwise known as a **bimax procedure**, is frequently used in the correction of obstructive sleep apnea (OSA). The advancement is usually large in the neighborhood of 1 cm and frequently requires bone grafts. These patients commonly have hypertension and other medical conditions associated with OSA and can be difficult to manage both intraop and postop. Most of these patients will be transferred to the ICU for the first postop night.

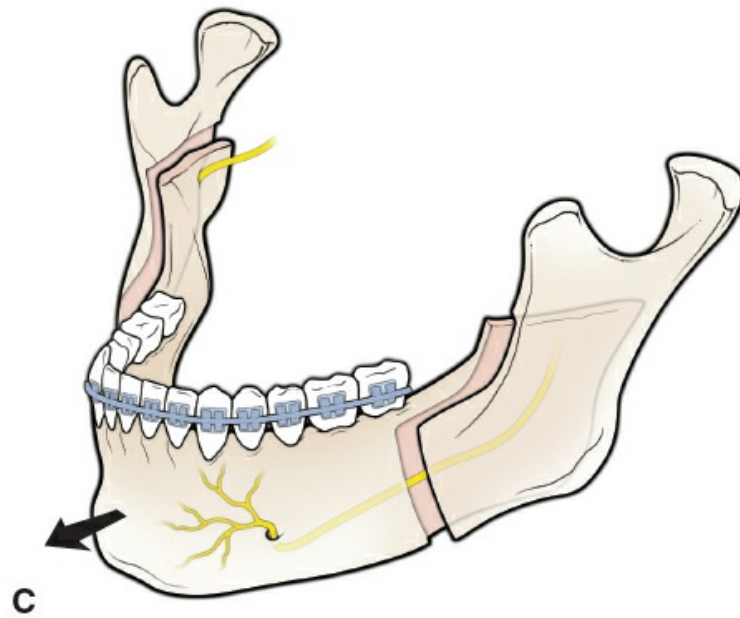
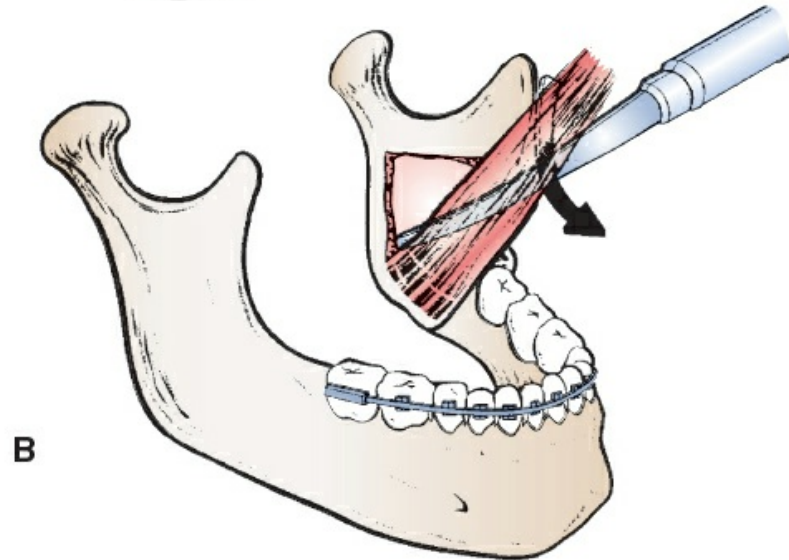
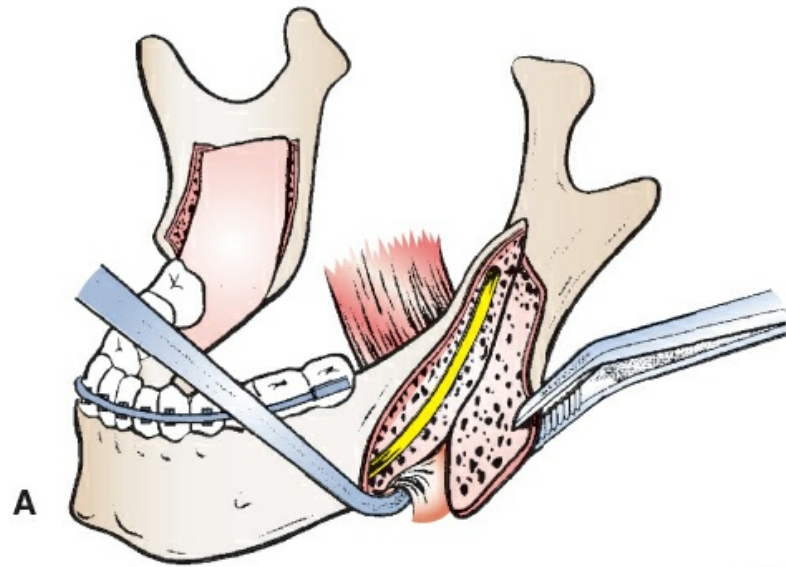


Figure 11.3-8 A, B: “J” stripper used to remove muscle attachments from medial aspect of distal segment. Used through the osteotomy split. (Reprinted from Booth PW, Schendel SA, Hausamen J-E, eds: Maxillofacial Surgery, Vol. 1, 1st edition. Churchill Livingstone, Edinburgh: 1999. Copyright © 1999 Elsevier. With permission.) **C:** Bilateral sagittal split osteotomy (BSSO).

Variant procedure or approaches

Distraction osteogenesis is being used more frequently in treating certain acquired and congenital jaw deformities. This involves a partial mandibular osteotomy and placement of a distraction device (Fig. 11.3-10). Depending on the application, this may be an external or internal device. External devices are visible over the skin with percutaneous pins into the bone. Internal devices are placed beneath the oral soft tissues, with a single adjustment pin exposed.

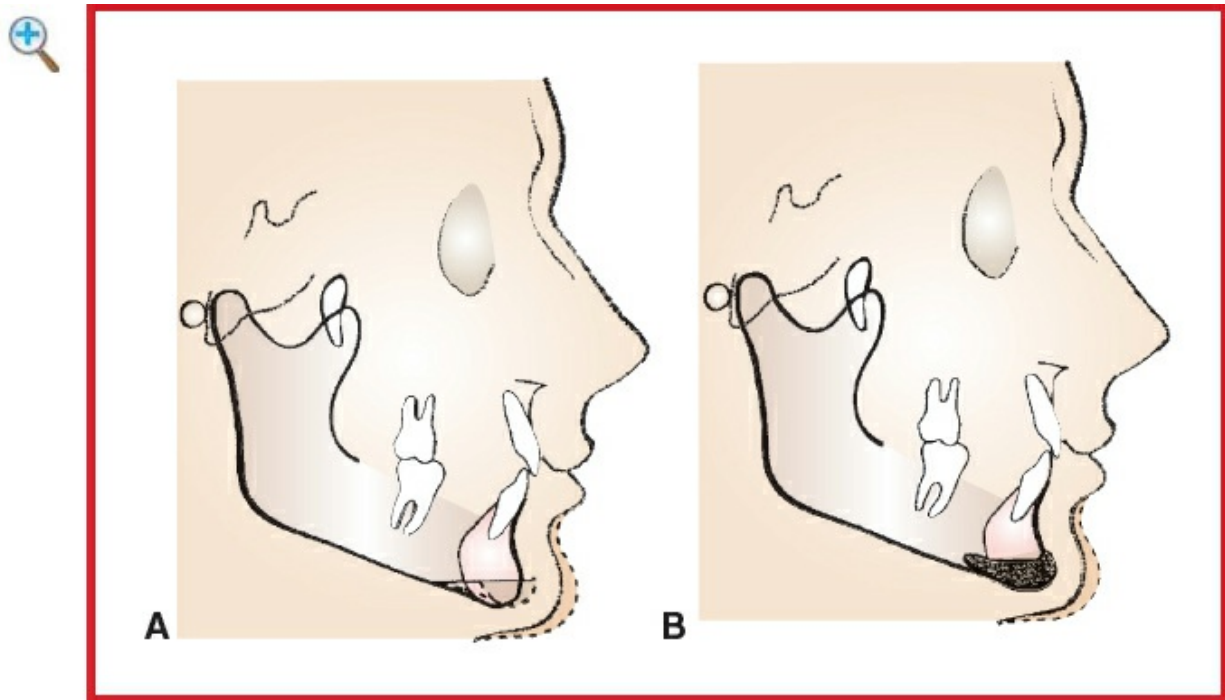


Figure 11.3-9 An osseous genioplasty (A) can be used to augment the chin, move it posteriorly, alter its vertical position, or change the transverse position of the chin. Alloplastic implants (B) can be used to augment the chin anteriorly. These are less effective for vertical augmentation. (Reprinted from Booth PW, Schendel SA, Hausamen J-E, eds: Maxillofacial Surgery, Vol. 1, 1st edition. Churchill Livingstone, Edinburgh: 1999. Copyright © 1999 Elsevier. With permission.)

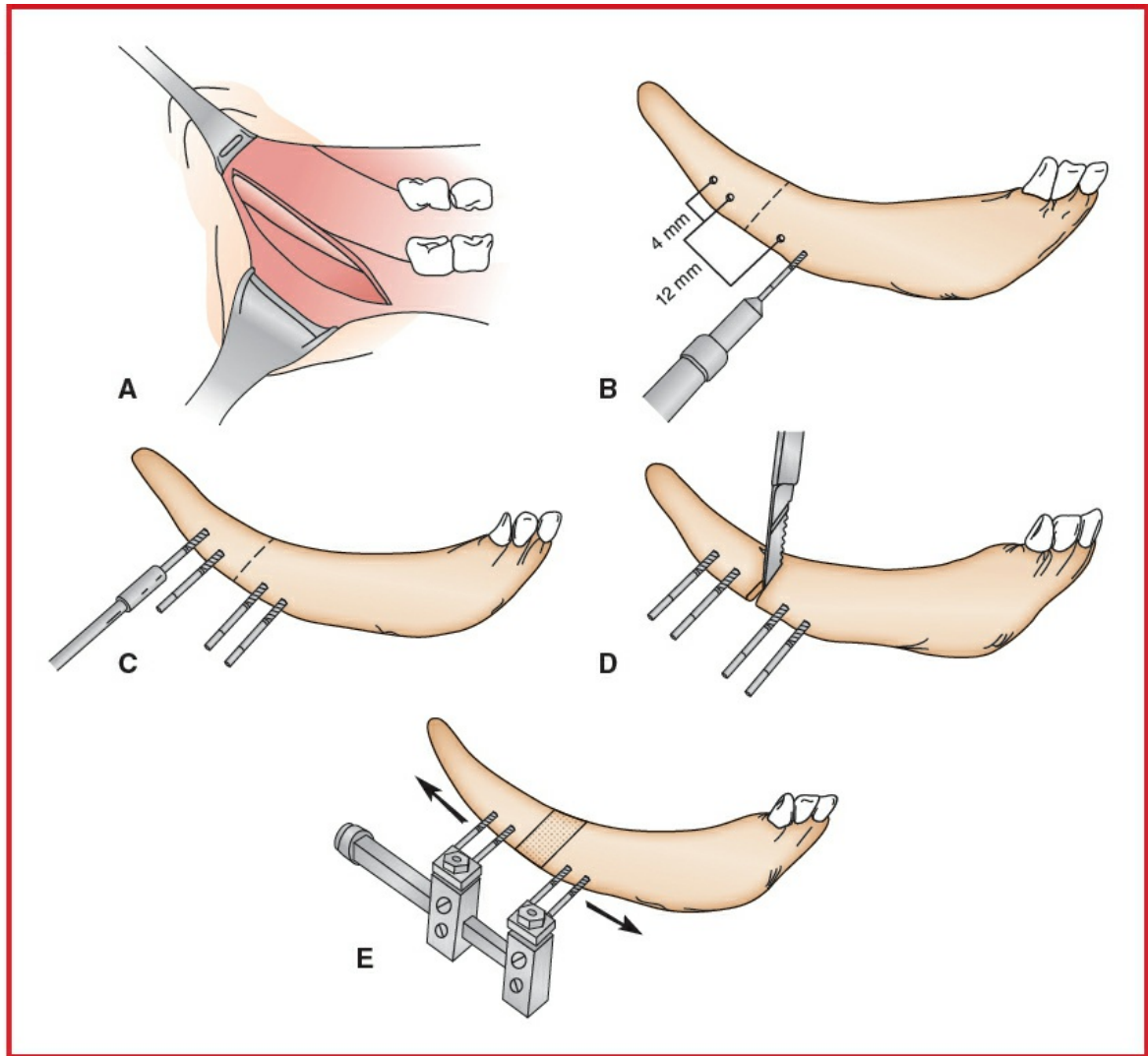


Figure 11.3-10 Mandibular distraction technique. **A:** An intraoral incision is made along the oblique line of the mandibular remnant. **B:** Sites of the pinholes and proposed osteotomy (interrupted line). **C:** Pins have been inserted. **D:** The osteotomy is performed. **E:** Commencement of distraction with the appliance in position. The arrows designate the movement of the mandibular segments with formation of bony regenerate in the resulting gap. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CHM, eds: Grabb and Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

Usual preop diagnosis

Mandibular deformity

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status and possible benefits of weight reduction (time permitting), preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine; table may be rotated 90° or 180°. Nasal RAE tube toward head of bed (sutured through nasal septum), padded, and secured to forehead; shoulder roll; neck extended
Incision	Usually oral but may be external in the submental or posterior mandibular area; preinjection with local anesthetic with epinephrine
Special instrumentation	Miniplates and screws; burrs/saws; osteotomes; distractor for distraction osteogenesis; throat pack
Unique considerations	Scleral lubricant and shields needed. Postop, patient may have IMF with inability to open the mouth (IMF not used with distractor cases). NB: Remove any throat pack before extubation; thorough oro- and nasopharyngeal suction.
Antibiotics, etc.	Cefazolin 2 g iv; methylprednisolone 125 mg iv (adults)
Surgical time	Genioplasty: 0.5–1 h Mandibular osteotomy: 2–4 h
EBL	Genioplasty: 50 mL Mandibular osteotomy: 100–200 mL
Postop care	PACU → room. If sleep apnea present, ICU overnight + extubation
Mortality	Rare
Morbidity	Mandibular relapse: ≤30% Mental nerve paresthesia: 5–20% Lingual nerve injury: 1–16% Infection/nonunion: <1% Facial nerve injury: Rare (most reported cases resolve without treatment) Aseptic necrosis (buccal plate or genioplasty fragment) Worsening TMJ problems Adverse cosmetic result
Pain score	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Infant-adult
Male:Female	Unknown
Incidence	Unknown
Etiology	Developmental (90%); acquired (10%)
Associated conditions	Usually none with developmental cases; sleep apnea; congenitally small mandible may be associated with a large tongue and cleft palate (Pierre Robin syndrome), hemifacial microsomia (which may be associated with heart, vertebral, and other anomalies), Treacher Collins syndrome, or Nager syndrome (many of the congenital cases may be tracheostomy and gastrostomy dependent, with associated TMJ ankylosis).

■ ANESTHETIC CONSIDERATIONS

(Procedures covered: Le Fort osteotomies; mandibular osteotomies; genioplasty)

■ PREOPERATIVE

These surgeries are usually performed on patients with facial disproportion. In general, this patient population is young and healthy; however, many of them will present with challenging airway management problems. In addition, facial disproportion will often accompany congenital anomalies (e.g., Crouzon, Apert, Pierre Robin, or Treacher Collins syndromes). (For discussion of specific syndromes, see Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Limbs.)

Airway	As usual, a careful airway evaluation is essential because many of these patients have abnormal airway anatomy and may be difficult to mask ventilate or intubate. Visual inspection often reveals the reasons for the surgery and allows the anesthesiologist to determine the safest approach to intubation. When a difficult intubation is anticipated, the need for awake fiberoptic intubation should be
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	discussed with the patient.
Respiratory	Consider the anesthetic implications of associated congenital syndromes in this patient population. Obstructive sleep apnea (OSA) also may be associated with patients presenting for mandibular surgery. Tests: As indicated from H&P
Cardiovascular	As above, consider the implications of possible congenital syndromes. Assess whether patient is a suitable candidate for the use of controlled ↓ BP, particularly those undergoing maxillary procedures. Tests: As indicated from H&P
Hematologic	Encourage autologous blood donation for maxillary procedures. Tests: Hct
Laboratory	Other tests as indicated from H&P
Premedication	Standard premedication is usually appropriate.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss possibility of nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	If there is any doubt regarding the ease of intubation or ability to mask ventilate, an awake FOL should be performed. Nasal intubation is preferred for patients undergoing elective mandibular or maxillary osteotomies, as well as genioplasty. In patients with normal airways, a standard induction is appropriate. Nasal or oral ETTs (RAE) or anode ETTs are
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	commonly used to minimize intrusion into the surgical field.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia. Consider preincision placement of nerve blocks (see Appendix K , p. K-6).	
Maintenance	Standard maintenance; muscle relaxation is usually required. Administration of an antiemetic (e.g., ondansetron 4 mg iv ± fosaprepitant 125 mg iv) is essential in patients who have their jaws wired or banded together.	
Emergence	Patients with difficult airways or with their jaws wired (or banded) together should be extubated when fully awake. Perform thorough oropharyngeal suctioning. If not placed previously, an NG tube is often placed before extubation. A wire cutter (or scissors for elastic bands) should be at the bedside at all times. Ensure that all throat packing has been removed before extubation. Consider iv acetaminophen if po dose given >4–6 h ago.	
Blood and fluid requirements	Moderate—large blood loss IV: 16–18 ga × 1–2 BSS @ 5–8 mL/kg/h	Maxillary osteotomies may be associated with major blood loss (e.g., 1500–2000 mL). The majority of blood loss occurs when the maxilla is downfractured. Controlled ↓ BP may be particularly useful during this part of the case (see Control of Blood Loss, p. 1256). Blood should be readily available for this and all maxillary procedures. Goal-Directed Fluid Management (see

[Appendix K, p. K-4\)](#)

Monitoring	Standard monitors ± Arterial line ± CVP line or 2nd iv	Direct arterial pressure measurements are useful for deliberate ↓ BP. Central venous access or a 2nd peripheral iv may be useful for vasodilator infusions.
Positioning	✓ and pad pressure points ✓ eyes	Eyes should be protected with an ophthalmic ointment and eye shields or possible tarsorrhaphy by the surgeons. Some surgeons prefer to have the OR table rotated 90° or 180°. Be prepared with circuit extension tubing.
Complications	ETT damage Hemorrhage	ETT may be cut during maxillary osteotomy, necessitating rapid reintubation.

POSTOPERATIVE

Complications	Airway obstruction PONV	Wire cutters (or scissors) should be available at bedside to facilitate emergent reinduction or other form of airway management. Consider retained throat pack. Multimodal treatment of nausea is important.
Pain management	Parenteral narcotics or PCA with antiemetics Acetaminophen NSAID	Regional or local blocks. Continuous infusion catheters may be beneficial.

	Gabapentin	
ERAS	PONV protocol	Encourage early enteral nutrition, Foley catheter removal, and early mobilization.

Suggested Readings

1. Adke M, Mendonca S: Concealed airway complication during LeFort I osteotomy. *Anesthesia* 2003; 58(3):294–5.
2. Bidgoli SJ, Dumont L, Mattys M, et al: A serious anesthetic complication of a LeFort I osteotomy. *Eur J Anaesthesiol* 1999; 16(3):201–3.
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CHAPTER 11.4

Functional Restoration

SURGEONS

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MICROSURGERY FREE-FLAP RECONSTRUCTION

SURGICAL CONSIDERATIONS

James Chang and Jacinta Leyden

Description

Microsurgical reconstruction involves moving tissue from one site of the body to another (see [Fig. 11.4-1](#), [Table 11.4-1](#)). This may be muscle, skin, bone, or any combination of these tissues. An artery and vein that supply the tissues are connected to an artery and vein at the new recipient site, thereby reestablishing blood flow and ensuring tissue survival. Once transplanted, the tissue is molded and shaped to replace the missing part. Vascular anastomoses are performed on vessels as small as 1.5 mm in diameter. Complex reconstructions may require up to 12 h of GA with significant blood loss and fluid shifts. Patients with comorbid conditions may need additional medical tests before surgery to ensure that they are able to withstand this type of operation.

Peripheral, central, and arterial lines should be placed at the start of the case only after consultation with the reconstructive surgeon. For example, radial forearm flap may be

injured by placement of a peripheral iv in the antecubital fossa or within the body of the flap. The neck is a common site for microvascular anastomoses in head and neck reconstruction. The placement of internal jugular lines should be discussed with the surgeon before surgery. The rectus muscle flap relies on the deep inferior epigastric vessels for its perfusion. Femoral arterial or venous line placement may injure these vessels and is, therefore, contraindicated. In bilateral breast reconstruction with free flaps, the iv lines should be placed in the lower extremities if possible. The reconstructive surgeon will try to operate in conjunction with the extirpative surgeon to minimize operating time. This often necessitates repositioning the patient as the case progresses.

Microsurgical reconstructions are often long operations, and large surface areas of the patient are exposed during the surgery. The patient's core temperature should be monitored closely with the use of a bladder or esophageal temperature probe. Body and fluid warmers should be used routinely to maintain the patient's body temperature. There is no standard practice for anticoagulation in microsurgery. No randomized prospective clinical trial has definitively documented the efficacy of a particular type of anticoagulation in routine reconstructive microsurgery. Aspirin, dextran, and heparin are the most commonly used agents. Close communication between the surgeon and anesthesiologist is critical so that the desired agent is administered at the appropriate time.

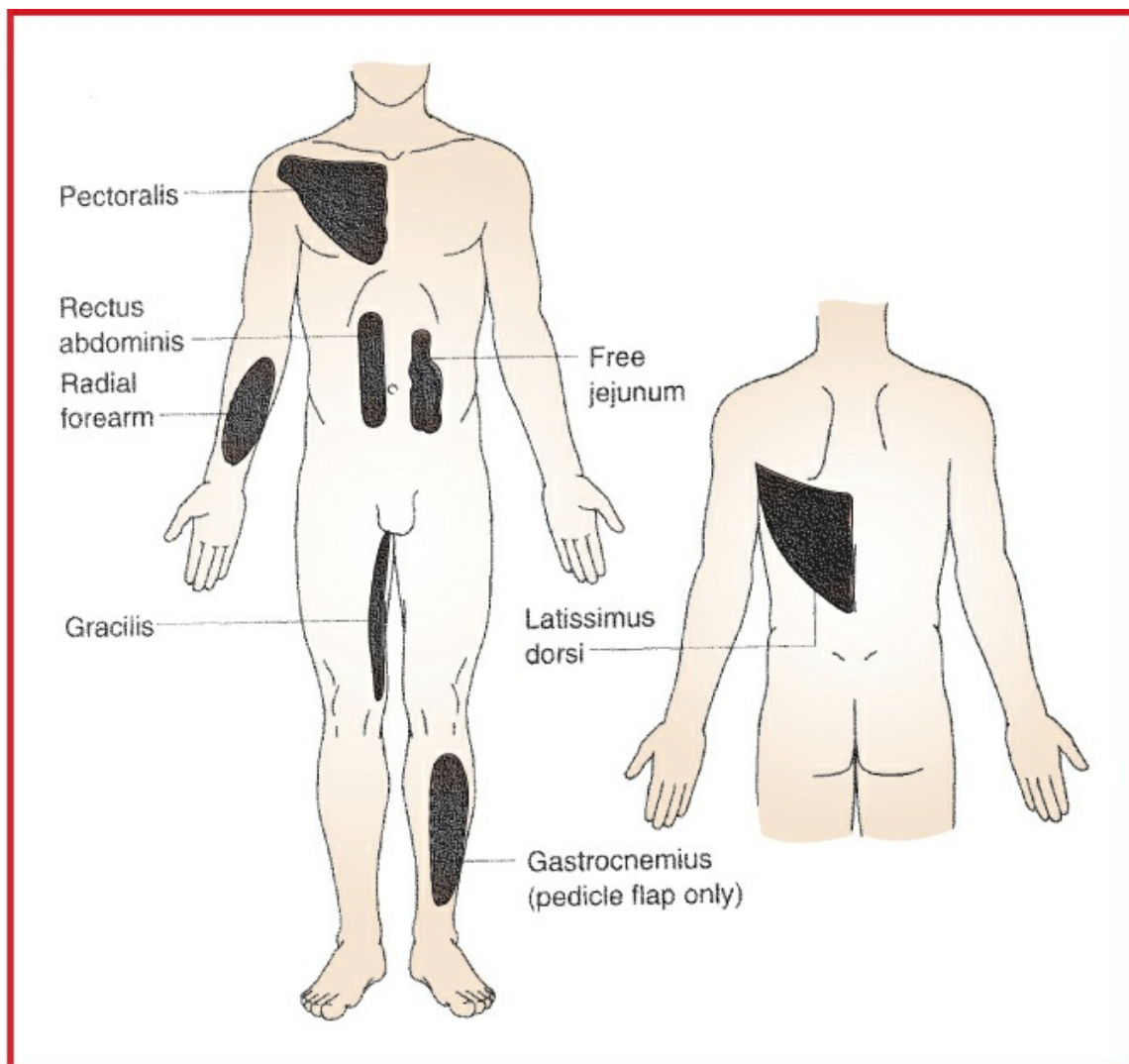


Figure 11.4-1 Locations of commonly used flaps. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Principles and Practice*, 2nd edition. Lippincott-Raven, Philadelphia: 1997.)

Table 11.4-1 Types of Flaps

Reconstruction	Commonly Used Flaps	Location	Positioning
Head and neck	Fibula	Leg	Supine
	Radial forearm	Arm	Supine
	Scapula	Back	Lateral decubitus
	Iliac crest	Hip	Supine
	Anterolateral thigh	Thigh	Supine

Breast	Gracilis TRAM	Thigh Abdomen	Supine Supine
	Gluteal	Buttocks	Lateral decubitus
	Rubens/iliac crest	Hip	Supine
	Tensor fascia lata	Lateral thigh	Supine
	Latissimus dorsi	Back	Lateral decubitus
Lower extremity	Rectus abdominis	Abdomen	Supine
	Latissimus dorsi	Back	Lateral decubitus
	Serratus anterior	Back	Lateral decubitus
	Gracilis	Thigh	Supine

The survival of the flap relies on the patency of the anastomoses. Thrombosis requires reexploration and may lead to complete loss of the tissue. If thrombosis is encountered during reexploration, tissue plasminogen activator (tPA) may sometimes be administered to help reestablish patency. Anastomotic patency is diminished by vasospasm or vascular constriction. The common causes of vasoconstriction include dehydration, hypothermia, pain, and administration of vasoconstrictors; therefore, during microsurgery, vasoconstrictors are contraindicated and should be given only when absolutely necessary, after consultation with the reconstructive surgeon. The patient must be adequately hydrated so that there is good perfusion to the transplanted tissue. Diuretics should be avoided. Postop pain control is important to prevent vasoconstriction on emergence from anesthesia. After surgery, patients are transferred to an area of the hospital where the free flap can be monitored. In many hospitals, this is an ICU.

Usual preop diagnosis

Trauma, cancer, chronic wounds, congenital anomalies, and burns are some of the common diagnoses that result in the need for microsurgical reconstruction.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status and possible benefits of weight reduction (time permitting), preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Requirements of the recipient site take precedence; optimal position for the donor site is then considered.
Incision	Each site will have specific incision.
Special instrumentation	Microscope; tourniquet for extremities
Unique considerations	Multiple surgical teams; avoid hypothermia; $\pm$ anticoagulation; if intraop nerve stimulation is planned, muscle relaxants should be avoided.
Antibiotics	Cefazolin (2 g iv) in uncomplicated cases; broader coverage in complicated circumstances
Surgical time	Simple flap: 4–6 h Complex cases: 8–12 h
Closing considerations	Splints and dressings should be secured before emergence.
EBL	Skin flap: 200 mL Muscle flap: 200–500 mL Bone flap: 500–1000 mL
Postop care	ICU for flap perfusion monitoring
Mortality	<2% (usually associated with coexisting disease)
Morbidity	Soft tissue complications: 20–30% Vascular complications requiring reexploration: 5–10% Flap failure: 5%
Pain score	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	2–90 yr
Male:Female	1:1
Incidence	50–200/yr/major center
Etiology	Malignancy; trauma; chronic infection Complications of underlying disease (e.g., anemia)

Associated conditions

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

These surgeries are carried out on patients who have sustained major soft tissue losses and require flap procedures to cover the defects. There are four typical patient populations presenting for this surgery: (a) those presenting for reconstruction following cancer surgery (e.g., radical neck dissection and mastectomy); (b) patients following trauma, usually with upper or lower limb defects; (c) patients with congenital defects; and (d) previous burn victims. In general, these patients should present few problems for the anesthesiologist. In a patient with a congenital lesion, however, it is prudent to look for evidence of CHD, musculoskeletal deformities, and airway problems.

Respiratory

Exposure of the thorax to radiation may produce pathologic changes in the lungs and related structures, including pneumonitis ($\rightarrow$ dyspnea, hypoxemia) that may progress to fibrosis ($\rightarrow$ $\downarrow$ pulmonary compliance). Tracheal or bronchial fibrosis may $\rightarrow$ partial airway obstruction. Many burn patients have pulmonary pathology 2° smoke inhalation injury. The pathology usually includes edema, inflammation, and loss of ciliary activity but can vary depending on the length and amount of smoke exposure, as well as the composition of the material that burned. Patients with breast cancer may have been treated with chemotherapeutic agents (e.g., methotrexate, cyclophosphamide, bleomycin) that may cause pulmonary fibrosis, interstitial infiltrates, and pleural effusions.

Tests: Consider CXR, PFT, and pulmonary consult.

Cardiovascular

Exposure of the heart to radiation may produce pathologic changes in the heart and related structures, including accelerated atherosclerotic changes, myocardial fibrosis, pericarditis, and valvular dysfunction. Patients with breast cancer may have been treated with doxorubicin (Adriamycin™),

	which may produce cardiomyopathy (usually seen at total doses >550 mg/m ²) → CHF. XRT increases the incidence of clinically significant cardiomyopathy.
Hematologic	Myelosuppression may be present in patients with Hx of chemotherapy. Tests: CBC, with differential and Plt count
Laboratory	Other tests as indicated from H&P
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss possibility of nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40–80 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: Coordinate iv and/or invasive monitoring sites with the surgical team. Adjunctive regional anesthesia is an option for the appropriate patient who will not be anticoagulated. Some authors propose that the resultant vasodilation causes ↑ blood flow through the flap, but this has not been definitely shown in clinical trials.

Induction	Standard induction and intubation. Avoid succinylcholine in burn patients.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia. Consider preincision placement of nerve blocks (see Appendix K , p. K-6).
Maintenance	Standard maintenance with sevoflurane. Muscle relaxation usually is required. The patient must be kept warm and well hydrated to minimize peripheral vasoconstriction, which might impair graft perfusion. Surgeon may request intraop anticoagulation, usually in the form of aspirin, dextran, or heparin.

Emergence	Smooth emergence to avoid disrupting the surgical repair. Patients usually are transported to the ICU for continuous monitoring of flap perfusion. Consider nerve block placement or reinforcement if initially placed >4–6 h ago (see Appendix K, p. K-5). Consider iv acetaminophen if po dose given >4–6 h ago.	
Blood and fluid requirements	Moderate blood loss IV: 16 ga × 1 BSS @ 2–4 mL/kg/h Warm fluids. Humidify gases.	Keep patient warm, and maintain a goal-directed fluid balance. An Hct of 30–35% will provide adequate O₂ transport while minimizing viscosity effects on perfusion. Dextran 40 may be used to further ↓ viscosity, thereby ↑ flap blood flow. Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors UO ± Arterial line	An arterial line may be useful for prolonged procedures where regular ABGs and blood chemistries will be needed.
Positioning	✓ and pad pressure points. ✓ eyes.	These can be very lengthy surgeries, and careful monitoring of pressure points is essential.
Complications	Hypothermia	Maintain normal body temperature with warming blankets and

	Decubitus ulcer	fluid and airway warmers. Pressure necrosis can occur in as little as 2 h. Carefully pad and repeatedly ✓ pressure points.
	Dextran reaction	If the patient develops any immediate cardiopulmonary changes, dextran infusion should be stopped immediately.

POSTOPERATIVE

Complications	Arterial thrombosis	May require reexploration.
	Hematoma	May require reexploration.
Multimodal pain management	Acetaminophen NSAID Gabapentin	Consider a multimodal approach using a combination of gabapentin, iv acetaminophen, opiates, and regional nerve block. Continuous infusion catheters may be beneficial.
	Parenteral opiates PCA	Pain should be treated promptly to minimize reflex peripheral vasoconstriction and impaired graft perfusion.
ERAS	PONV protocol	Encourage early enteral nutrition, Foley catheter removal, and early mobilization.

Suggested Readings

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MICROSURGERY REPLANTATION

SURGICAL CONSIDERATIONS

Description

Patients who require replantation surgery are trauma victims and must be evaluated carefully preop, by both surgeons and anesthesiologists, to ensure that replantation is appropriate and that other injuries are not overlooked. In these cases, time is critical, as the amputated tissue is ischemic and may require immediate revascularization if it is to be salvaged. Coordination between the microsurgeon, anesthesiologist, and trauma team is important to minimize the time between injury and replantation.

As in any microsurgical procedure, it is critical to prevent vasoconstriction in these

procedures. These patients may have experienced significant blood loss at the time of the trauma and require iv hydration and/or blood transfusion. The need for hemodynamic support often indicates another injury that may preclude transplantation. The patient may be hypothermic and require active rewarming. Vasoconstrictors and diuretics should be avoided unless absolutely necessary. During the replantation procedure, other tissues (e.g., skin, vein, bone) are required to aid in the reconstruction. These are routinely harvested from the leg, groin, or foot, which will be prepped into the surgical field.

The amputated stump is initially examined using loupe magnification. The arteries, veins, and nerves are dissected and tagged. The surgeon determines if the part is replantable. The amputated stump is prepared by dissecting the recipient arteries, veins, and nerves. The need for a vein graft may be determined at this time. The sequence of replantation varies; however, a general algorithm is:

Bone fixation → Extensor tendon repair → Flexor tendon repair → Nerve repair → Venous and arterial anastomoses → Skin closure

The replanted tissue must be monitored on an hourly basis to ensure continued viability. In many hospitals, this is performed in a microsurgical unit, whereas in others, the ICU is used. Patients should be kept adequately hydrated, warm, and pain-free to prevent vasoconstriction and subsequent thrombosis. Vascular thrombosis requires immediate exploration and revision of the vascular anastomosis.

Specific variations of replantation procedures have unique features, as follows.

Replantation of fingers and hands

Generally, two surgeons work simultaneously. One surgeon at the back table explores the amputated parts, tagging significant nerves, vessels, and tendons. A second surgeon debrides the amputation sites and identifies the stumps of reparable structures. The surgeons then proceed with replantation. Generally, bone fixation and tendon repairs are performed first. Vessel and nerve repairs are performed next, using a microscope. The need for vein grafting and anticoagulation is determined intraop. Blood transfusions are rarely needed except when using anticoagulants.

Replantation of extremities

Replantation of arms or legs must be handled very efficiently because irreversible muscle damage occurs within 4 h of ischemia. Generally, the sequence of surgery is similar to finger replantation, with the exception being that a temporary arterial circulation (using a dialysis shunt) is established as soon as possible to minimize

ischemia time in an amputated part. Ongoing venous blood loss occurs while skeletal repairs are done, and transfusion is frequently required. Definitive vessel repairs (often requiring vein grafts) and nerve repairs are done under the microscope.

Scalp replantation

Scalp avulsions are caused by entanglement of hair in machinery. These amputations are possibly replantable, sparing the patient a significant aesthetic and functional problem. Initial evaluation should include careful assessment of the C-spine because the patient transiently hangs by the neck until the scalp separates. Initial blood loss can be significant and should be replaced preop. Replantation proceeds by identifying matching vessels at the margin of the defect and the avulsed scalp. The superficial temporal vessels are most commonly repaired, and use of vein grafts should be anticipated. Following the first artery repair, brisk bleeding generally occurs at the scalp margin until vein repairs are completed. This blood loss should be anticipated.

Usual preop diagnosis

Trauma

ERAS: Because of the urgent nature of these procedures preop counseling by the surgeon/anesthesiologist is rarely possible.

■ SUMMARY OF PROCEDURES			
	Fingers/Hands	Extremities	Scalp
Position	Supine, injured arm extended	Supine	Supine or side (depending on vessel position)
Incision	Conventional hand exposure	Extension of injury; fasciotomies may be done.	Preauricular
Special instrumentation	Microscope; hand table; tourniquet	Microscope; tourniquet	Microscope; neurosurgical headrest
Unique considerations	Anticoagulation	⇐	RAE or anode tube; table, turned 180°

Antibiotics	Cefazolin 2 g iv	⇐	⇐
Surgical time	1st finger: 3–4 h; 2 h/subsequent finger Hand: 4 h	4–8 h	4 h
Closing considerations	Splint applied before emergence.	Cast or splint applied before emergence.	Elevate head as much as possible.
EBL	100–200 mL	2–6 U	2–8 U
Postop care	ICU for monitoring	⇐	⇐
Mortality	None	Rare	None
Morbidity	Replant failure: 5– 15%	Failure: 10–20%	Vascular occlusion → reexploration
Pain score	5–6	5–6	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Childhood–old age	⇐	Young adult
Male:Female	>10:1	⇐	1:2
Incidence	250/yr/major center	Rare	⇐
Etiology	Trauma	⇐	⇐
Associated conditions	Other injuries	Other injuries, blood loss	C-spine injuries, blood loss

ANESTHETIC CONSIDERATIONS FOR REPLANTATION

PREOPERATIVE

In general, there are two patient populations for replantation procedures: (a) isolated limb and scalp injury patients (common) and (b) multiple trauma victims (rare). Most patients are otherwise healthy, and the preop workup is routine.

Gastrointestinal

All of these patients should be considered to have full stomachs and, therefore, are at increased risk for

	aspiration pneumonitis. In general, they should receive preop medication to reduce stomach volume and acidity (e.g., metoclopramide 10 mg iv and ranitidine 50 mg iv) 30–60 min before induction, time permitting.
Metabolic	~50% of trauma victims are intoxicated. Anesthesia-related implications of acute ethanol intoxication include ↓ anesthetic requirements, diuresis, vasodilation, and hypothermia.
Neurologic	Assess for possible head or C-spine injury, particularly in the trauma patient presenting with facial or scalp injuries.
Laboratory	As suggested by coexisting disease
Premedication	Standard premedication. Full-stomach precautions: Na citrate 0.3 M 30 mL po immediately before induction of anesthesia

INTRAOPERATIVE

Anesthetic technique

GETA, after rapid-sequence induction. These procedures are often lengthy, and regional anesthesia is usually not appropriate as the primary technique but may be considered as an adjunct.

Induction	Rapid-sequence induction is mandatory in emergency cases, unless awake intubation is performed. C-spine fracture patients or those with facial injuries may require awake fiberoptic intubation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia. Consider preincision placement of nerve blocks (see Appendix K , p. K-6).
Maintenance	Standard maintenance for stable patients
Emergence	Difficult airway or full-stomach cases require awake extubation. Consider nerve block placement or

	reinforcement if initially placed >4–6 h ago (see Appendix K, p. K-5). Consider iv acetaminophen 1 g as part of MMPM.	
Blood and fluid requirements	Significant blood loss possible IV: 16 ga × 1–2 (extremity/scalp) IV: 18 ga × 1 (digit) BSS @ 1.5–3 mL/kg/h + 3 × blood loss Fluid/blood warmers, heating blanket, warmed circuit humidifier	A 16-ga iv catheter in a nonoperated upper extremity should be adequate in hemodynamically stable patients. Keep patient warm and hydrated to maximize perfusion to the replanted site. Vasopressors are not contraindicated and may be preferable to volume overload. HCT > 33% may ↑ viscosity + ↓ perfusion Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors. ± Arterial line, CVP	Invasive hemodynamic monitoring should be considered in cases where large blood loss is anticipated.
Positioning	✓ and pad pressure points ✓ eyes	
Control of blood loss	Tourniquet may be used.	Inflation pressure is typically 100 mm Hg greater than systolic pressure. Maximum “safe” tourniquet time is 1.5–2 h, followed by a 5 to 15-min reperfusion interval if further tourniquet time is necessary.

Special considerations	Tourniquet deflation and limb reperfusion	Mild ↓ BP is common. In patients with moderate-to-severe lung disease, continue controlled ventilation until after the lactic acid that has accumulated in the ischemic limb is metabolized (3–5 min) because these patients may be unable to increase ventilation adequately to buffer this acid load.
Complications	Hemodynamic instability	Previously unrecognized injuries (e.g., pneumothorax, cardiac tamponade, intracranial bleeding) should be considered as a cause of unexplained intraop hemodynamic instability in all acute trauma victims.

POSTOPERATIVE

Complications	Reperfusion failure	May require immediate reexploration.
Multimodal pain management	Acetaminophen NSAIDs Gabapentin PCA Regional nerve block	Regional or local blocks. Continuous infusion catheters may be beneficial.
ERAS	PONV protocol	Encourage early enteral nutrition, Foley catheter

		removal, and early mobilization.
Tests	None routinely indicated	Reimplant perfusion must be monitored.

Suggested Readings

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BREAST SURGERY—INTRODUCTION

David M. Kahn

Patients presenting for plastic surgery of the breast can be grouped into four basic categories along a continuum ranging from amastia/hypomastia to hypertrophy. Plastic surgery procedures are designed to create or make adjustments in the amounts of skin and glandular tissue or to make adjustments in their relationship to each other to create an aesthetic breast. The first type of patient is one who has acquired amastia after undergoing a mastectomy; it is this patient who is featured in this section on functional restoration. In this patient, the goal is to replace the missing tissue, both skin and glandular, with like tissue or an implant. The second type is a person presenting for augmentation mammoplasty. In this situation, the breast is deficient of skin and glandular tissue. The third type is the patient who presents for a reduction mammoplasty. In this situation, there is an excess of glandular tissue and skin. The fourth type is the patient who presents for a mastopexy or breast lift. In this patient, there exists a discrepancy between the amount of glandular tissue present and the volume of the skin envelope, resulting in ptosis.

For all breast procedures, the patient's breasts are marked preop with the patient in either sitting or standing position. This is a necessity, and its importance cannot be overstated. The appearance of the breasts in the supine position versus the upright position is significantly different due to the effects of gravity.

BREAST RECONSTRUCTION

SURGICAL CONSIDERATIONS

Description

The goal of breast reconstruction is to create an aesthetic breast that is symmetrical with the contralateral breast. Typically, this can be accomplished in three ways: expander/implant reconstruction; latissimus flap reconstruction, $\pm$ an implant; and transverse rectus abdominis muscle (TRAM) flap reconstruction, performed using either a pedicled technique or a free tissue transfer. If abdominal tissue is not available, autologous breast reconstruction can also be performed using free flaps from other areas such as the gluteal region (superior gluteal artery perforation flap) or medial thigh (transverse upper gracilis flap). Each type of reconstruction follows the principle of replacing glandular and cutaneous breast tissue. In patients undergoing mastectomy, reconstruction may be performed immediately after the mastectomy or it may be delayed and performed at a later date.

In **expander/implant reconstruction** (Fig. 11.4-2), a tissue expander is placed underneath the pectoralis major muscle and a portion of the serratus anterior muscle, followed by skin closure. Thus, two layers exist above the implant—skin and muscle. The patient returns to the office for expansions, with saline being injected into the implant port. This is continued until the desired size is reached. Then, the patient returns to the OR for the exchange of the expander for a permanent implant that contains either saline or silicone gel as filler material. Except for the psychosocial aspects of patient management, much of the technique and perioperative concerns are similar to those for breast augmentation.

Autologous breast reconstructions

Two types of flaps are used for these procedures—latissimus myocutaneous flap and TRAM flap. The **latissimus dorsi myocutaneous flap** consists of the muscle with overlying skin that is rotated from the back to the anterior chest for the creation of a breast. The flap does not supply sufficient bulk to be used in breast reconstruction unless the contralateral breast is very small. Usually a breast implant is placed between

the latissimus and pectoralis muscles, thus increasing the volume of the reconstruction. The patient is placed in the lateral decubitus position for the latissimus flap harvest. The incision is designed to surround the skin paddle. The ellipse of skin is incised, and the dissection then proceeds along the superficial surface of the latissimus muscle toward its lateral, superior, and inferior borders. Dissection is performed underneath the latissimus muscle to separate it from the deep tissues of the back. The muscle is released from its insertions on the posterior superior iliac crest, medial fascial attachments, and surrounding muscle attachments (i.e., serratus anterior, teres major). The flap is tunneled through the axilla, and the back wound is closed (Fig. 11.4-3). The patient is then returned to the supine position. The muscle is disinserted from the humerus, if necessary, and brought out onto the anterior chest wall. At this point, the muscle is inset into the mastectomy defect. An implant or tissue expander may be placed under the muscle if necessary for size and symmetry. Often, the patient is placed in the seated or semi-Fowler position for closure. The incisions are closed over a drain, and dressings are applied. The skin flap is monitored postop for signs of flap ischemia and congestion, which may necessitate a return to the OR.

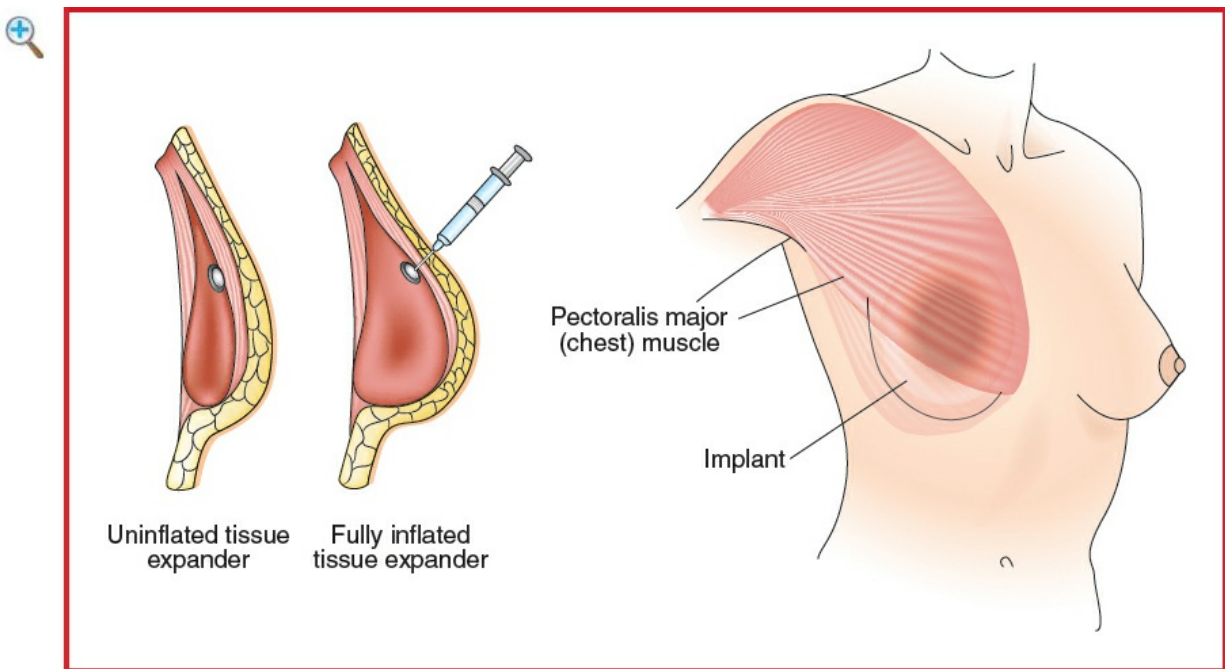


Figure 11.4-2 Expander-implant reconstruction. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practices*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

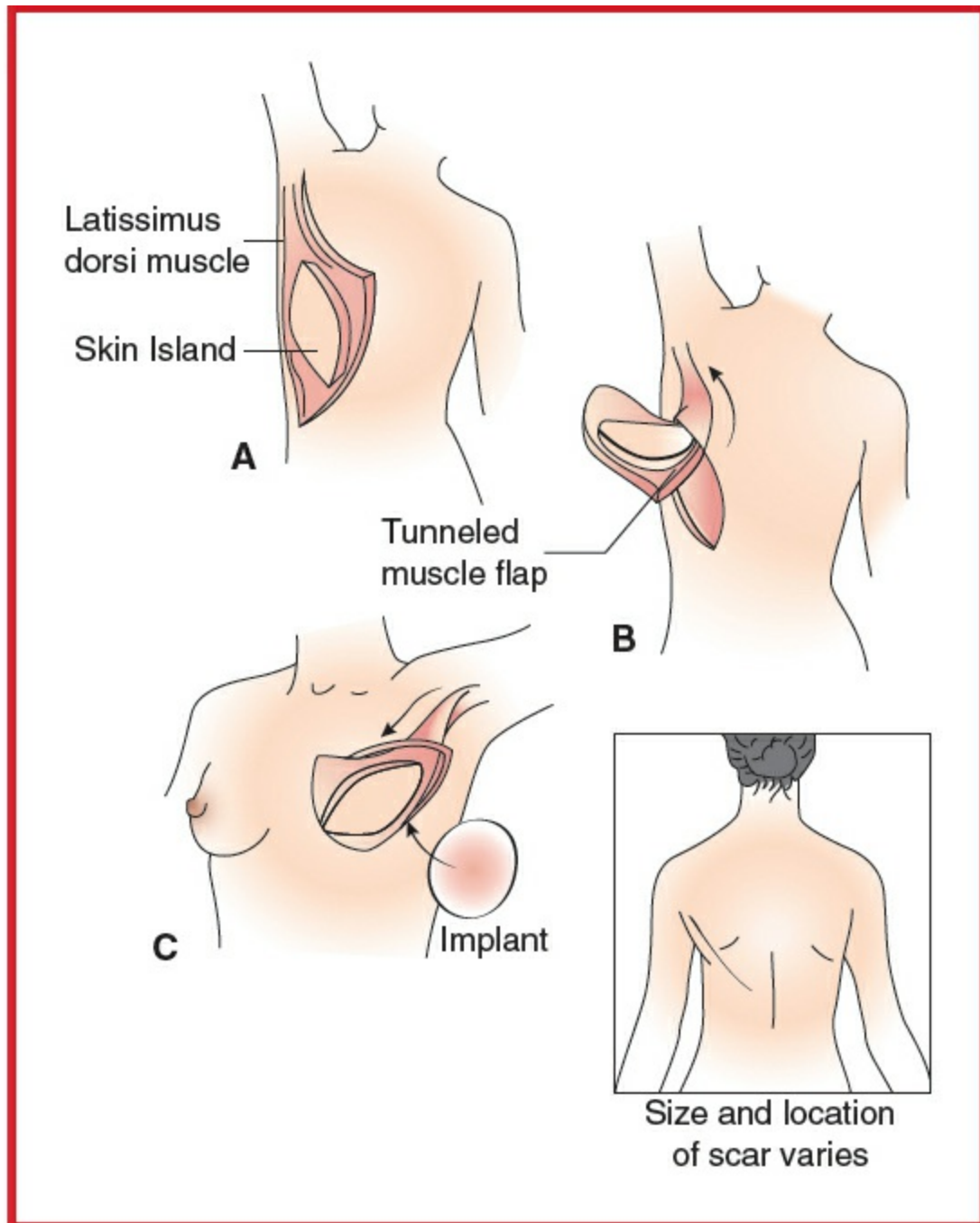


Figure 11.4-3 A–C: Latissimus dorsi flap reconstruction. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practices* 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

The **TRAM flap** procedure replaces the breast with an ellipse of abdominal skin and subcutaneous tissue based on the rectus abdominis muscle (Fig. 11.4-4). There are several variations in the type of abdominal tissue that is transferred. Traditionally, the TRAM flap consists of the entire rectus muscle and overlying subcutaneous tissue and skin. Modified techniques minimize the amount of muscle removed to decrease the occurrence of postoperative hernias. These include muscle-sparing TRAM (MS

TRAM) flaps that take out only a small portion of rectus. Deep inferior epigastric artery perforator (DIEP) flaps involve only the skin and subcutaneous tissue with the perforators dissected from the rectus muscle. See [Figure 11.4-5](#).

DIEP flaps require more tedious dissection. Superficial inferior epigastric artery (SIEA) flaps are based off the SIEA and do not involve any dissection within the rectus muscle. In selecting patients for a TRAM flap reconstruction, the patient must have adequate lower abdominal tissue to make a breast; however, obese patients, smokers, diabetics, and those with a Hx of prior abdominal surgery may have a higher incidence of complications and flap loss. The benefit of this procedure is that it creates a natural appearing breast from the patient's own tissue without an implant. As a bonus, the abdominal donor site is closed as though the patient had undergone abdominoplasty ("tummy tuck"). The myocutaneous perforators that arise from the superior epigastric and inferior epigastric arteries provide blood supply to the flap. This flap can be harvested in either a pedicled fashion, based on the superior epigastric artery, or as a free flap, based on the inferior epigastric artery. The inset of the flap and closure of the donor site are the same in both cases. The basic difference between the two methods of flap transfer is that, in the free TRAM procedure, the flap is removed from the abdomen, brought into the mastectomy wound, and the operating microscope is used to suture the vascular pedicle of the flap to the recipient vessels, either the thoracodorsal or internal mammary arteries. In the case of the pedicled TRAM flap, the flap is passed through a tunnel, created under the skin, maintaining the blood supply via the superior epigastric artery. The flap is brought out into the mastectomy wound where it is sutured into position.

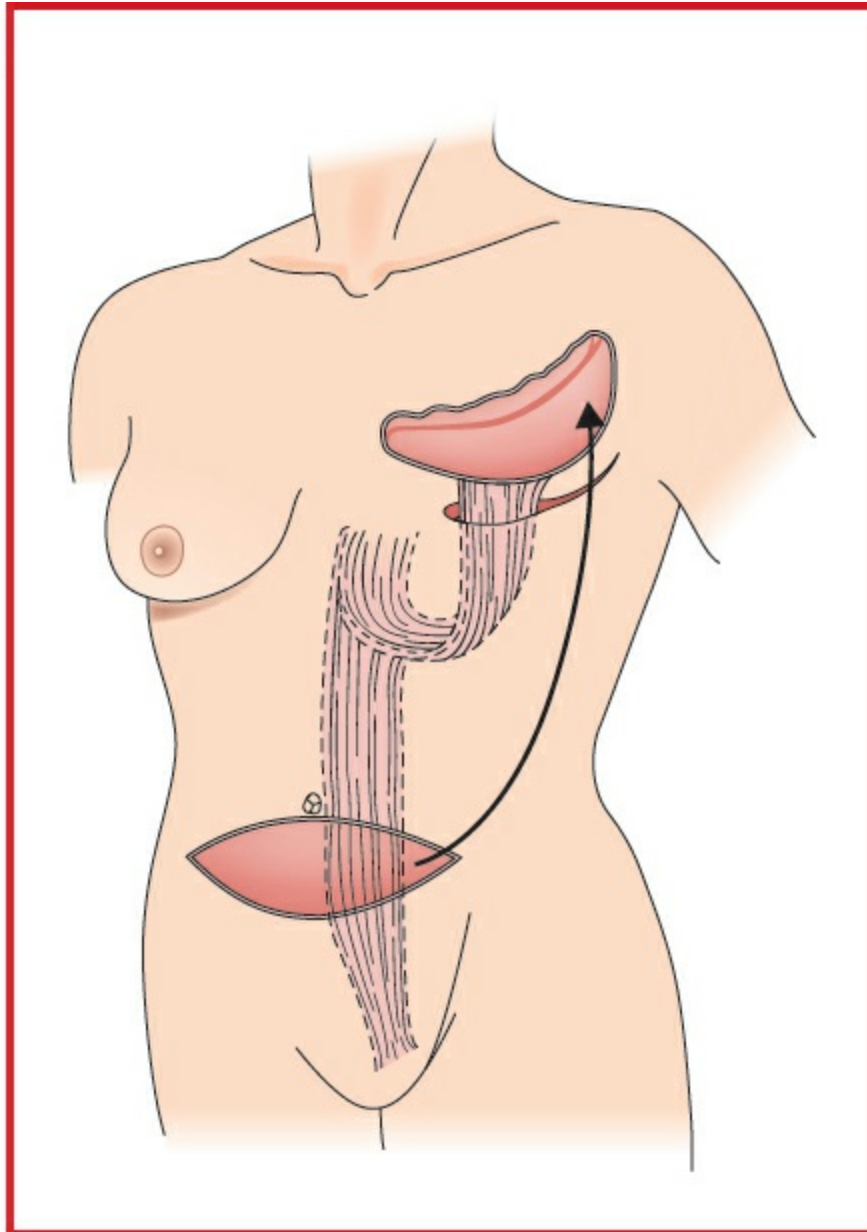


Figure 11.4-4 Breast reconstruction using pedicled TRAM flap. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds *Surgery: Principles and Practice*, 2nd edition. Lippincott-Raven: 1997.)

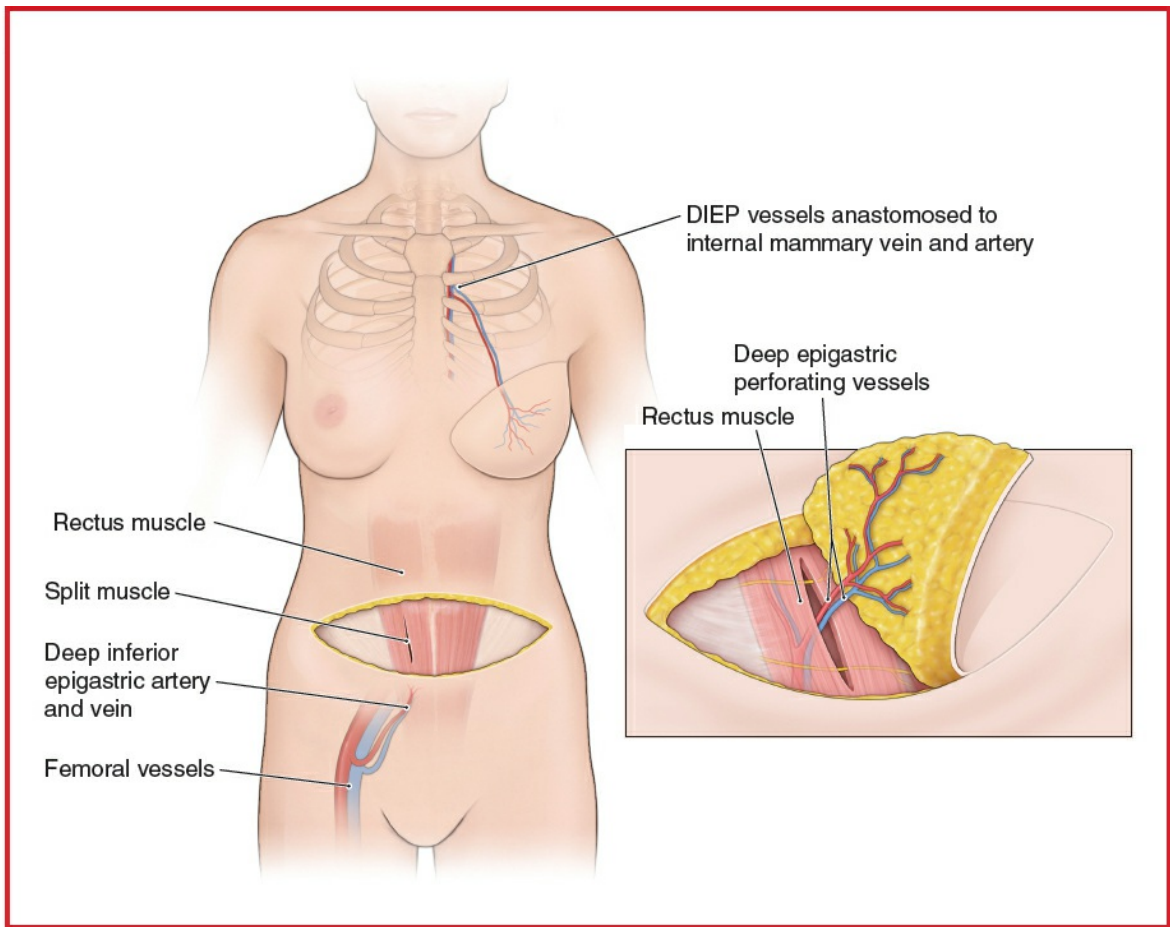


Figure 11.4-5 Unlike a TRAM flap, the DIEP (and SIEA) flap reconstruction spares the rectus muscle.

Pediced TRAM flap

The patient is marked preop in the upright position. The midline and inframammary folds are marked, as is the abdominal ellipse. The arms are placed at 90° abduction. The breasts and abdomen are prepped and draped. A general surgeon performs the mastectomy and, if possible, the TRAM flap harvest is started at the same time. Incising the skin along the superior marking of the abdominal ellipse begins the harvest of the flap. The upper abdominal skin and subcutaneous fat are elevated off the abdominal wall fascia up to the level of the costochondral cartilage, as in an abdominoplasty. The table is then flexed, and the upper skin and subcutaneous tissue are brought to overlap the TRAM flap to ensure that the location of the marked incision at the lower border of the flap will allow for abdominal closure. The patient is returned to the supine position. The skin and subcutaneous tissue of the flap are raised from a lateral to medial direction off the abdominal wall fascia until the lateral border of the rectus muscle is identified. Care is taken to preserve the myocutaneous perforators supplying the flap. The anterior rectus sheath is incised, and the rectus muscle is elevated away from the posterior rectus sheath. The inferior epigastric vascular pedicle is identified and divided,

preserving as much length as possible. The portion of the rectus muscle below the flap is transected so that the muscle, along with the overlying ellipse of skin and subcutaneous tissue, can be rotated into the mastectomy site. A tunnel is created under the skin to connect the abdominal wound and mastectomy site. The flap is passed through this tunnel and rotated into position on the chest wall (Fig. 11.4-6).

In flap reconstructions, the harvested tissue receives its blood supply through a single artery and vein. It is important in these cases to maintain a stable BP that will allow for continued perfusion of the flap tissue. Vasopressors are to be avoided, as they will constrict the artery and thus restrict the inflow into the flap. Paralytics may be requested in the case of DIEP flaps to minimize muscular twitching while the surgeon dissects through the rectus to follow the vascular pedicle. It is preferred, if possible, to maintain a stable BP with volume replacement. Once the flap is in place at the mastectomy site, the table is again flexed as much as 45°–60° for closure. Drains are placed within both the chest and abdominal wound beds. The flap is trimmed and sutured into position to create symmetry with the contralateral breast. The abdomen is closed in a fashion similar to an abdominoplasty. Many surgeons prefer that N₂O (which can distend the abdomen) be avoided during the abdominal closure. The surgeon will evaluate the flap to monitor for signs of ischemia and congestion. This is done both by clinical evaluation (color, temperature, and turgor) and by Doppler. If inflow or outflow is inadequate for flap survival, blood flow may be supplemented by performing a microvascular anastomosis between the inferior epigastric pedicle and the thoracodorsal vessels. In some cases, the surgeon may choose to convert the pedicled flap to a free flap.

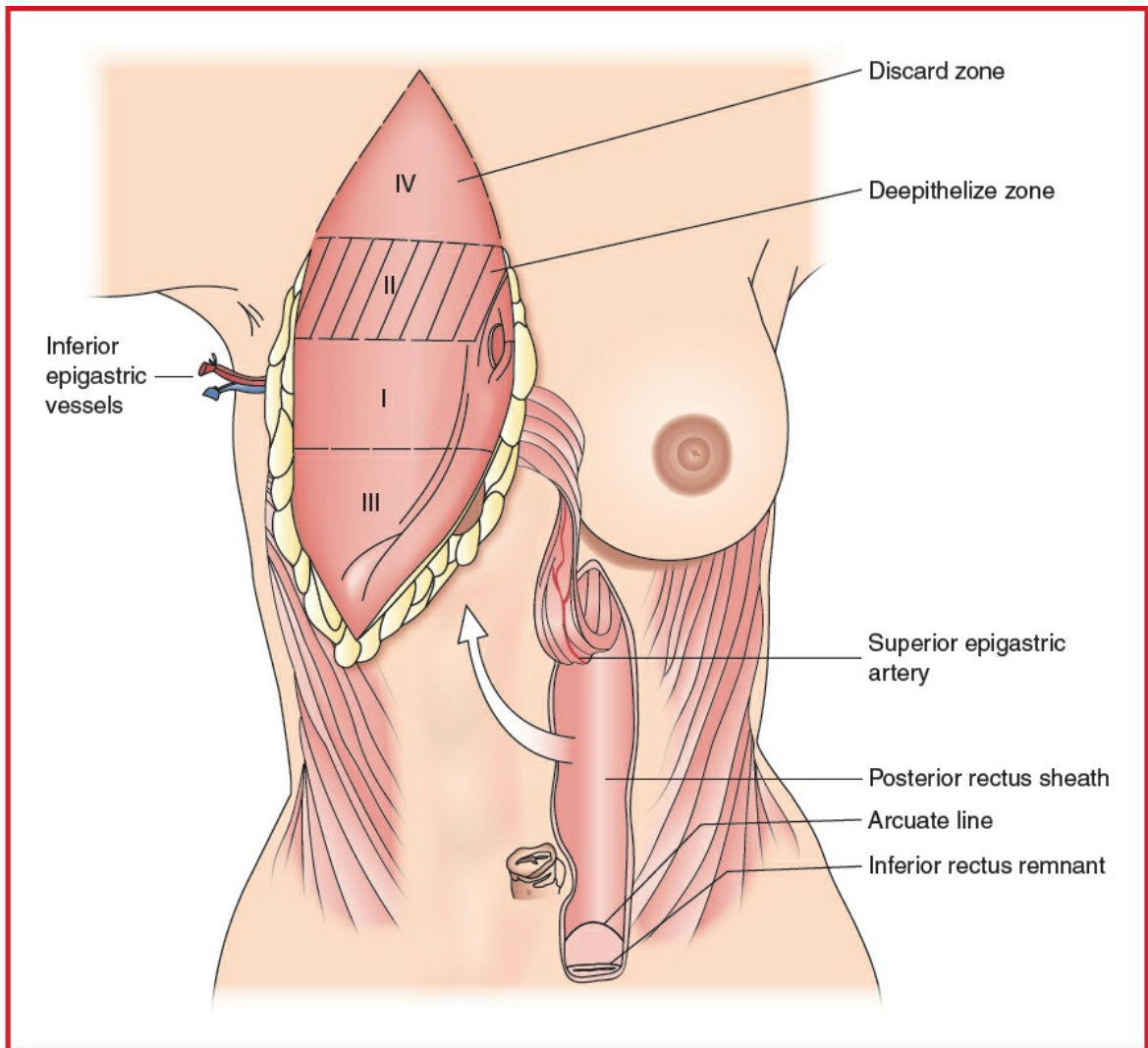


Figure 11.4-6 TRAM flap reconstruction of the breast. A contralateral pedicle TRAM flap has been elevated and will be tunneled under the skin to be inset at the mastectomy site. The zone of tissue furthest from the blood supply is discarded. A portion of the TRAM flap will be de-epithelialized and placed under the mastectomy skin flaps. (Reproduced with permission from Spear SL: *Surgery of the Breast: Principles and Art*. Lippincott-Raven, Philadelphia: 1998.)

Usual preop diagnosis

Carcinoma of the breast; radiation therapy; cardiovascular surgery; Poland syndrome

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status and possible benefits of weight reduction (time permitting), preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Tissue

Latissimus Dorsi

	Expander/Implant	Flap	TRAM Flap
Position	Supine	Lateral decubitus → supine	Supine
Incision	Breast	Posterior (supine); lateral (thorax)	Abdominal
Unique considerations	May have to place patient in sitting position during procedure; SCDs; Foley catheter if case >3–4 h	⇐	⇐ + Avoid use of N ₂ O.
Antibiotics	Cefazolin 2 g iv	⇐	⇐
Surgical time	1–2 h	4 h (+ mastectomy time)	6–8 h (free TRAM) 4 h (pedicled TRAM) (not including mastectomy time)
Closing considerations	Extensive dressing required	⇐	⇐ + Flex table for abdominal closure.
EBL	Minimal to 100 mL	200–300 mL	200–400 mL
Postop care	PACU → room	⇐	⇐ (free TRAM flap → ICU)
Mortality	Rare	⇐	⇐
Morbidity	Capsular contraction: ± 30%	—	—
	Decreased sensation: 15%	⇐	⇐
	Hematoma: 2.2%	⇐	⇐
	Fat/skin necrosis: 1.7–1.9%	⇐	⇐
	Nipple areola necrosis: 1.4%	—	—
		Flap loss: Rare	⇐
			Abdominal hernia:

Pain score	5	5	infrequent 5
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■ PATIENT POPULATION CHARACTERISTICS

Age range	30–70 yr
Male:Female	Mostly female
Incidence	Breast reconstruction is performed in 9% of female population.
Etiology	Cancer; trauma; idiopathic; radiation or postcardiovascular surgery
Associated conditions	Breast cancer; cardiovascular disease; S/P chemotherapy; pulmonary disease

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Breast and Chest Wall Reconstruction.

Suggested Readings

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CHEST WALL RECONSTRUCTION

SURGICAL CONSIDERATIONS

Description

Chest wall reconstruction is most commonly performed for infections, sternal dehiscence, tumor extirpation, and radiation injuries. Complications after median sternotomy make up a large number of these cases. The patients often have comorbidities and are at high risk for anesthetic and surgical complications. The goals of reconstruction are to provide a stable chest wall for respiration, to eradicate infection, and to obtain a healed wound.

Sternal wound infections and dehiscences

Wound complications after median sternotomy include dehiscence of the sternum and mediastinitis. In these cases, radical debridement of all devitalized tissue is the cornerstone to a successful outcome. The initial debridement is, therefore, performed in conjunction with the cardiovascular surgeons. During this debridement, blood loss may be extensive, requiring transfusion. The resulting dead space around the heart and great vessels is obliterated, most commonly with **pectoralis major or rectus abdominis muscle (TRAM) flaps**. For patients who have failed the initial reconstruction or are not candidates for these flaps, a **latissimus dorsi muscle flap** may be used. The omentum can also be used to help obliterate the dead space. Despite radical excision of the sternum, the respiratory function of these patients remains adequate, and no bony stabilization is required.

Tumor extirpation and radiation injury

Tumor resection or removal of osteoradionecrosis of the chest wall often involves the full-thickness removal of skin, muscle, and underlying rib cage. The rib cage may be reconstructed with prosthetic mesh or bone grafts. These structures are then covered with muscle flaps. The pectoralis major, rectus abdominis, and latissimus dorsi muscles are the muscle flaps most commonly used. After surgery, the patient's ventilatory capacity may be diminished by the rib resection, and this should be anticipated preop.

Usual preop diagnosis

Chest wall infections; tumor; sternal dehiscence; radiation injuries

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status and possible benefits of weight reduction (time permitting), preop carb loading, VTE

risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES				
	Pectoralis Flap	Latissimus Flap	Rectus (TRAM) Flap	Omental Flap
Position	Supine	Lateral decubitus → supine	Supine; table flexed	Supine
Incision	Chest	Posterolateral thorax	Abdominal	Abdominal
Antibiotics	Cefazolin 2 g	⇐	⇐	⇐
Surgical time	3 h	⇐	4–6 h	3 h
Closing considerations	✓ flap perfusion	⇐	⇐	⇐
EBL	200–400 mL	⇐	300–500 mL	200 mL
Postop care	PACU → room	⇐	⇐	⇐
Mortality	Rare	⇐	⇐	⇐
Morbidity	Thrombosis/flap failure	⇐	⇐	⇐
	Hematoma	⇐	⇐	Diaphragmatic hernia
	Fat/skin necrosis	⇐	⇐	Bowel obstruction
			Abdominal hernia: Infrequent	Bowel perforation
Pain score	4–5	4–5	4–5	4–5

■ PATIENT POPULATION CHARACTERISTICS	
Age range	30–80 yr
Male:Female	1:1
Incidence	Uncommon
Etiology	Cancer; trauma; radiation; postcardiovascular surgery
Associated conditions	Breast cancer; cardiovascular disease; S/P chemotherapy; pulmonary disease; mediastinitis

ANESTHETIC CONSIDERATIONS FOR BREAST AND CHEST WALL RECONSTRUCTION

PREOPERATIVE

These surgeries are performed most commonly for reconstruction following cancer

surgery, such as radical neck dissection and mastectomy (see Anesthetic Considerations for the primary procedure), as well as complications after median sternotomy. The following considerations focus on patients undergoing reconstruction postchemotherapy. Always enquire about any treatment side effects.

Respiratory	Pulmonary toxicity (e.g., pneumonitis, fibrosis) may complicate some chemotherapeutic agents including paclitaxel and trastuzumab. Bleomycin (>200 mg/m²) carries the greatest risk of pulmonary toxicity (10%), but alkylating agents (e.g., cyclophosphamide and melphalan) may cause a degree of pulmonary toxicity as well. Avoid FiO₂ $> 30\%$ in bleomycin patients (to prevent progressive pulmonary fibrosis and edema). Patients presenting with complications from a prior sternotomy may have ↓ respiratory function/reserve and will require further workup. Tests: CXR; ABG and PFTs as indicated from H&P
Cardiovascular	Cardiomyopathy and CHF may result from chemotherapy, especially with trastuzumab and anthracyclines (e.g., doxorubicin >550 mg/m²). Previous XRT increases risk of clinically significant cardiomyopathy and CAD. Patients with median sternotomy-related complications who have undergone prior cardiac surgery will require careful evaluation of their current cardiovascular status. Tests: ECG; echo, if indicated from H&P
Neurologic	Note any previous damage to long thoracic nerves, as evidenced by winged scapula deformity. Taxanes (e.g., paclitaxel) can cause a dose-dependent, cumulative neuropathy and pain syndrome.
Musculoskeletal	It is traditional (not evidence based) to avoid iv and BP cuff in the upper extremity on mastectomy side. Myalgias/arthralgias occur with paclitaxel.
Hematologic	Myelosuppression/toxicity from chemotherapeutic agents may be present. Tests: CBC; Plt count; coag profile; Hb/Hct

Renal/hepatic	Taxanes (e.g., taxlitaxel) are mildly hepatotoxic. Methotrexate can produce renal and hepatic dysfunction. Elevated alkaline phosphatase may suggest metastatic bone invasion. Tests: Electrolytes; BUN; Cr; LFTs
Gastrointestinal	Tamoxifen, paclitaxel, docetaxel, and carboplatin can cause preop N/V and dehydration.
Laboratory	Other tests as indicated from H&P, prior chemotherapy
Premedication	Midazolam 1–2 mg iv immediately preop
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss possibility of nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40–80 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	Standard induction and intubation. Avoid iv and NIBP monitoring on mastectomy side (a non–evidence-based tradition).
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia. Consider preincision placement of nerve blocks (see Appendix K , p. K-6).
Maintenance	Standard maintenance. Muscle relaxation is usually appropriate. These patients should be kept warm and hydrated to minimize peripheral vasoconstriction, which might impair graft perfusion.
Emergence	During some of the procedure and for application of dressing, patient may be moved to sitting position,

	with consequent coughing, bucking, etc. (Rx: deeper anesthesia, e.g., propofol 0.5 mg/kg or lidocaine 1 mg/kg). Watch BP carefully, and treat orthostatic hypotension if it occurs, usually with a pressor or fluid bolus if the patient is not volume sensitive (e.g., Hx of CHF or renal failure). Consider nerve block placement or reinforcement if initially placed >4–6 h ago (see Appendix K, p. K-5). Consider iv acetaminophen if po dose given >4–6 h ago.	
Blood and fluid requirements	IV: 16 ga × 1 BSS @ 4–6 mL/kg/h Warm fluids. Humidify gases.	Extensive blood loss may occur with debridement of sternum in cases of infection or dehiscence. Goal-Directed Fluid Management (see Appendix K, p. K-4) Keep patient warm and maintain a slightly positive fluid balance. Hypothermia may impair flap perfusion.
Monitoring	Standard monitors UO	Use invasive monitoring in cardiovascular-challenged patients or in cases with significant expected blood loss, where regular ABGs and blood chemistries will be useful.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Pneumothorax Decubitus ulcer	Pneumothorax should be considered with any ↑ lung inflation pressure, ↓ O₂ sat, or ↓ BP. Pressure necrosis can occur in as little as 2 h.

	Dextran reaction	Carefully pad and repeatedly ✓ pressure points. Dextran is now rarely used because risks (e.g., allergic reactions, ARDS, pulmonary edema) outweigh benefits.
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POSTOPERATIVE

Complications	Pneumothorax ↓ Flap perfusion	
Multimodal pain management	Acetaminophen NSAID Gabapentin Parenteral opiates. PCA Regional nerve blocks	Regional or local blocks. Continuous infusion catheters may be beneficial. Pain should be treated promptly to minimize reflex peripheral vasoconstriction and impaired graft perfusion. Paravertebral and TAP blocks are effective for pain control.
ERAS	PONV protocol	Encourage early enteral nutrition, Foley catheter removal, and early mobilization.

Suggested Readings

1. Clasen SC, Wald JW: Left Ventricular dysfunction and chemotherapeutic agents. Curr Cardiol Rep 2018; 20(4):20.
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4. Sanna S, Brandolini J, Pardolesi A, et al: Materials and techniques in chest wall reconstruction: a review. *J Vis Surg* 2017; 3:95.
5. Sato M, Shirakami G, Fukuda K, et al: Comparison of general anesthesia and monitored anesthesia care in patients undergoing breast cancer surgery using a combination of ultrasound-guided thoracic paravertebral block and local infiltration anesthesia: a retrospective study. *J Anesth* 2016; 30(2):244–51.

PRESSURE SORE RECONSTRUCTION

SURGICAL CONSIDERATIONS

Description

Pressure sores occur when constant pressure is placed on an area of the body. They tend to occur in debilitated, bedridden, paralyzed, and wheelchair-bound patients. Multiple factors—including altered sensory perception, poor nutrition, incontinence, moisture, and shear forces—also may contribute to the formation of the pressure sores. The most common locations are the sacrum, ischium, and greater trochanter regions, as well as the heel and scalp.

The management of pressure sores is multidisciplinary and involves more than just debridement and wound closure. Patient compliance is the most important factor of wound management, as recurrence rates after closure are high. All components that contributed to the formation of the ulcer must be addressed before wound closure. This is best done in a team setting, with the goal of preventing further sores. In this effort, nutrition must be optimized; infection, muscle spasm, and contractures controlled; pressure relief measures instituted; and the psychological issues addressed. Once these are in place, wound reconstruction may begin.

Reconstruction is based on adequate debridement, elimination of dead space and pressure points, and closure of the wound with healthy, durable tissue. Multiple flap designs have been described for the coverage of pressure sores, depending on their location ([Fig. 11.4-7](#)). An important point in the design of a flap is the consideration for future reconstructions because of the high incidence of recurrent wounds. The choice of flap also needs to take into consideration the donor site morbidity. Skin-only flaps, fasciocutaneous flaps, and musculocutaneous flaps are commonly used.

The patient is placed in the prone position and may be jackknifed to facilitate exposure of the wound. Debridement of the wound is performed; and the bursa and underlying

bony prominences may be removed. Hemostasis is obtained, and the wound is irrigated with pulsed lavage. The flap to be used is designed over the vascular pedicle. The donor site for sacral, gluteal, and greater trochanteric wounds are typically the tissues of the buttocks and posterior thigh. The skin is incised and the flap is raised to include the appropriate layers of tissue, depending on whether a skin, fasciocutaneous, or musculocutaneous flap is to be used. The flap is then rotated into position in the wound. A portion of the flap may be de-epithelialized and placed into the wound to eliminate dead space. Drains are placed into the wound, and the flap is sutured into its new location. The donor site defect that occurs after rotation of the flap may be closed directly or may require a skin graft. Once the operation is complete, the patient is transferred to a pressure-relief bed (e.g., Clinitron™ or air-fluidized bed) and extubated.

Usual preop diagnosis

Pressure sores

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, preop carb loading, VTE risk assessment, and postop pain management options. See [Appendix K, p. K-1](#).

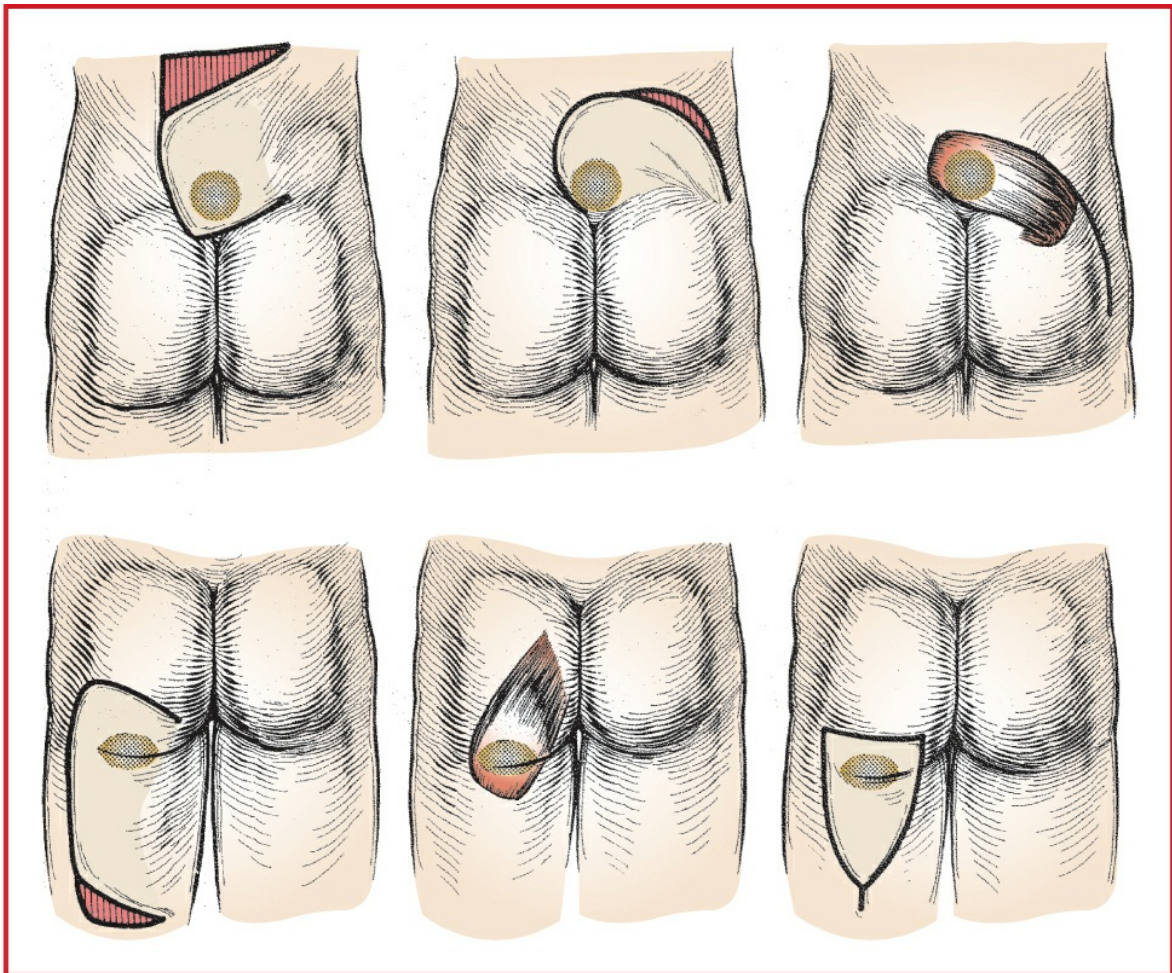


Figure 11.4-7 Commonly used flap designs for coverage of sacral and ischial pressure sores. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CHM, eds. Grabb and Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

■ SUMMARY OF PROCEDURE

Position	Prone or lateral decubitus
Incision	Gluteal or posterior thigh
Unique considerations	Foley catheter; SCDs
Antibiotics	Cefazolin 2 g iv. Some surgeons may prefer broader polymicrobial coverage.
Surgical time	3 h
Closing considerations	None
EBL	200–300 mL

Postop care	Strict bed rest on a pressure-relief bed for up to 3 wk. Care must be taken to ensure that the patient is turned every 2 h so that excessive pressure is not transferred to a new location. Antibiotics and the control of urine and stool are important to prevent soiling of the suture lines.
Mortality	0–1%
Morbidity	Infection: 2–3% Dehiscence: 1% DVT: 1% Recurrence
Pain score	4–6 (depends on spinal cord status)

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–70 yr
Male:Female	Predominantly male
Etiology	Immobility; unrelieved pressure; altered sensation
Associated conditions	Paraplegia; debilitation; quadriplegia

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Typically, these surgeries are carried out on nonambulatory patients, with the spinal cord injury patient constituting a large subset. Anesthesia for plegic patients may well present several challenges, as discussed below.

Respiratory	Plegic patients may have intercostal muscle weakness → atelectasis and ↓ clearance of secretions → recurrent URIs and V/Q mismatching → hypoxemia. Tests: PFT; ABG; others as indicated from H&P
Cardiovascular	Autonomic dysreflexia (AD) may present as acute episodes of uninhibited hyperactivity in the patient with an injury level of T10 or above. The main

	clinical signs are paroxysmal HTN and bradycardia, in response to stimulation below the lesion. Severe HTN can result in pulmonary edema, myocardial ischemia, and cerebral hemorrhage. Identify triggering stimuli (e.g., bowel or bladder distension, cutaneous stimulation). T4 or higher lesions may → ↓ BP on induction of GA or regional anesthesia, initiation of IPPV, or postural changes. Tests: ECG; others as indicated from H&P
Neurologic	✓ level of cord injury. AD in patient with spinal cord injury (see Cardiovascular, above) may manifest as headaches, sweating, facial flushing, or syncope. Hyperreflexia below injury level.
Musculoskeletal	Immobility → skeletal muscle atrophy, osteoporosis, and decubitus ulcer formation
Gastrointestinal	Spinal cord injury → ↓ GI function → constipation/full stomach
Renal	Chronic spinal cord injury → recurrent UTIs and calculi → renal failure. Foley catheter placement may → AD. Tests: UA; BUN; Cr; others as indicated from H&P
Laboratory	Immobility → ↑ Ca⁺⁺, → dysrhythmias and nausea. Tests: Others as indicated from H&P
Premedication	Standard premedication is usually appropriate. Patients with limited respiratory reserve should receive minimal sedation. Nifedipine (10 mg sublingually 5 min or po 20 min before induction) may be used to blunt AD.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: If flap donor and recipient sites are confined to the lower half of the body, regional anesthesia may be considered for short procedures. Spinal or epidural anesthesia will minimize AD; however, anesthetic level may be difficult to assess, and regional anesthesia may not be tolerated for prolonged surgery. Lighter levels of GA will not prevent AD.

Induction	Standard induction and intubation. Although the risk of $\uparrow$ K ⁺ 2° succinylcholine is reportedly decreased 6 mo after injury, NMRs are still preferred. AD may occur with Foley catheter placement.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia (Goal-Directed Management) and normothermia. Consider preincision placement of nerve blocks (see Appendix K , p. K-6).	
Maintenance	Standard maintenance. Muscle relaxation is usually appropriate and may be necessary to reduce muscle spasticity. Avoid drugs that are primarily renally excreted in patients with CRI. Direct arterial vasodilators and alpha-adrenergic blocking agents should be readily available. Plegic patients are prone to hypothermia. These patients should be kept warm and hydrated to minimize peripheral vasoconstriction, which might impair graft perfusion.	
Emergence	AD 2° distended bladder or rectum may occur on emergence from anesthesia. Consider nerve block placement or reinforcement if initially placed >4–6 h ago (see Appendix K , p. K-5). Consider iv acetaminophen if po dose given >4–6 h ago.	
Blood and fluid requirements	IV: 16 ga $\times$ 1 BSS @ 2–4 mL/kg/h Warm fluids. Humidify gases.	Keep patient warm and maintain a slightly positive goal-directed fluid balance. Hypothermia may impair

		flap perfusion. Initial debridement of ulcer may be extensive and significant blood loss can occur. T&C as appropriate. Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors UO ± Arterial line	An arterial line may be useful in patients susceptible to AH and for prolonged procedures where regular ABGs and blood chemistries will be useful.
Positioning	✓ and pad pressure points ✓ eyes	Many of these patients may be osteoporotic, so great care should be used in moving and positioning.
Complications	Hypothermia AD	Patients with spinal cord injury often have impaired thermoregulation. Maintain normal body temperature with warming blankets and fluid and airway warmers. Remove inciting stimulus. ↑ BP should be promptly controlled with NTG, SNP, or clevidipine bolus and infusion, while anesthesia is deepened.

	Decubitus ulcer	Pressure necrosis can occur in as little as 2 h. Carefully pad and repeatedly ✓ pressure points.
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▼ POSTOPERATIVE

Complications	Respiratory insufficiency AD	Quadriplegic patients may have ↓ VC and ↓ ERV and be uniquely susceptible to residual respiratory depressant effects. AD may occur 2° distended bladder or rectum. Rx: remove inciting stimulus and NTG, SNP, or clevidipine bolus/infusion.
Multimodal pain management	Acetaminophen NSAID Gabapentin PCA (see p. C-3) Regional nerve block	Regional or local blocks. Continuous infusion catheters may be beneficial. Pain should be treated promptly to minimize reflex peripheral vasoconstriction and impaired graft perfusion. Avoid respiratory depression. Useful for pain control in extremities
ERAS	PONV protocol	Encourage early enteral nutrition, Foley catheter removal, and early mobilization.

Suggested Readings

1. Cushing CA, Phillips LG: Evidence-based medicine: pressure sores. *Plast Reconstr Surg* 2013; 132(6):1720–32.
2. Israel JS, Carlson AR, Bonneau LA, et al: Reconstructive surgery and patients with spinal cord injury: perioperative considerations for the plastic surgeon. *J Plast Surg Hand Surg* 2016; 50(1):44–9.
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5. O'Neill DK, Tsui SM, Ayello EA, et al: Anesthesia protocol for heel pressure ulcer debridement. *Adv Skin Wound Care* 2012; 25(5):209–19.
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CHAPTER 11.5

Burn Surgery

ANESTHESIOLOGISTS

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We gratefully acknowledge the past contributions of surgeon, Kenneth K. Yim, MD, FACS.

FREE SKIN GRAFT FOR BURN WOUND (WITH TANGENTIAL EXCISION, EXCISION TO FASCIA, OR DEBRIDEMENT)

SURGICAL CONSIDERATIONS

Description

Until the mid-1970s, management of burn wounds involved daily debridement, hydrotherapy, and spontaneous eschar separation, with subsequent skin grafts applied to the granulated tissue. Operative management has become much more aggressive with the description of **tangential excision** by **Janzekovic**. In burn patients without inhalation injury, early excision and grafting (days 1–5) can reduce mortality and length of hospital stay. There are two surgical approaches to burn wounds—tangential excision and fascial excision.

Tangential excision (TE) is the more frequently performed procedure. The concept of TE is extremely simple but requires considerable experience and teamwork. Thin slices of burn eschar (burned, necrotic tissue) are shaved sequentially with manual or power dermatomes until a healthy wound bed is developed. Assessment of the wound bed is

done with visualization of bleeding and/or the clinical appearance of the excised bed. Blood loss is generally diffuse and can be massive; therefore, communication between anesthesiologist and surgeon is essential. In large excision, PRBCs should be available in the OR before excision so that the anesthesiologist does not get behind in blood and fluid replacement.

Diffuse bleeding, especially dermal, is controlled by laparotomy pads soaked with warm 1:100,000 epinephrine solution. These pads are replaced every 3–5 min and, after ~10 min, are removed one at a time, with persistent bleeding points controlled by electrocautery. Although very high plasma epinephrine levels have been reported after major burn excision, systemic manifestations are very rare in acute burn patients (probably 2° chronic high-level endogenous catecholamine secretion).

TE in the extremities usually is accomplished with a pneumatic tourniquet to minimize blood loss. In some centers, subcutaneous injection of a diluted (1:1,000,000) epinephrine solution under the burn wound also is used to minimize blood loss; however, the resulting vasoconstriction makes the end point of excision—that is, bleeding—difficult to ascertain.

Fascial excision involves removing the burn eschar and all underlying fat en bloc to the level of muscle fascia or beyond. Fascial excision can be performed more rapidly and with less blood loss than TE. Its disadvantages, however, are the marked cosmetic deformities and functional limitations that occur because of the loss of all soft tissue overlying the musculature. Because of its disadvantages, fascial excision is reserved for 4th-degree burns or for patients with very extensive, life-threatening, full-thickness (3rd-degree) burns.

In patients with serious burns (>40% total body surface area [TBSA]), excision usually commences on postop days 2–5, after completion of fluid resuscitation, and is performed every 2–3 d, as the patient's condition permits. If eschar excision can be completed before secondary sepsis supervenes, management of the patient is easier and the complications and morbidity are lessened considerably.

The end points for surgical excision in large burns are (a) operative time of 2–3 h, (b) core temperature of 35°C, or (c) blood loss of 10 U of PRBC. The violation of any of these parameters invites coagulopathy and increasing problems with hemostasis and VS stability. Adverse effects occurring after 3–4 h of operative time are usually the result of massive transfusion or hypothermia.

Due to loss of skin integrity and large exposed surfaces, these patients lose heat rapidly.

Fluids, gases, and the OR should be warm, although there is no demonstrable benefit to warming the OR past the point of isothermic neutrality (~82°F [28°C]). Many surgeons however, will maintain the room at ~100°F (38°C). All areas not in the operative field should be covered, and a warming blanket (Bair Hugger™) is used frequently.

Coverage

After excision of wounds and attainment of hemostasis, wounds are covered, using either an autograft or temporary coverage with an allograft, xenograft, or synthetic/biologic dressing. An autograft is used for coverage when the wound bed is deemed suitable, a donor site is available, and the patient is stable. A split-thickness skin graft (STSG) often is used for coverage of a burn wound. Because an STSG is harvested at the dermal level, bleeding also is controlled with topical epinephrine-soaked laparotomy pads before application of dressings. Depending on the location of donor sites, many surgeons use subcutaneous infiltration of diluted (1:1,000,000) epinephrine in saline solution to smooth out irregularities (e.g., underlying ribs) or to create a flat surface (e.g., scalp) to physically improve the ease of taking skin grafts. A substantial volume of saline may be infiltrated, and this should be added into the total fluids administered to the patient.

Intraop position change may be necessary between the burn excision and the STSG harvest. For example, donor skin may be harvested from the back for application to the chest or abdomen.

The STSG is held temporarily in place with staples or sutures. Uncontrolled patient movement may dislodge the graft. To protect against this, grafts are secured with circumferential bolsters, dressings, and/or splints. This procedure may be time consuming, and any uncontrolled patient movement should be avoided.

It has become apparent that early eschar excision is advantageous even if wounds are so extensive they cannot be covered with autografts. In this situation, temporary coverage of the excised wound is accomplished with the application of an allograft, porcine xenograft, or synthetic/biologic dressing. The wound is maintained in this way, with further debridement and biologic dressing changes as necessary, until autograft becomes available.

Usual preop diagnosis

Thermal, electrical, or chemical burn

ERAS: Since these procedures are often urgent and repeated at short intervals, the usual preop ERAS considerations are not often applicable. The maintenance of

normal temperature, volume, and nutritional status can be challenging.

■ SUMMARY OF PROCEDURES

	Tangential Excision	Fascial Excision
Position	Supine, prone, or lateral	⇐
Incision	Anywhere eschar is to be excised	⇐
Special instrumentation	Dermatomes, as determined by the surgeon—manual or powered. Humby and Goulian knives	—
Temperature considerations	Keep room T at ~82°F (28°C).	⇐
Unique considerations	Possible subcutaneous infiltration of diluted lidocaine-epinephrine-saline solution	None
Antibiotics	Cefazolin (adult = 2 g; child = 15 mg/kg) iv on induction of anesthesia	⇐
Surgical time	2–3 h	⇐
EBL	Massive; limit surgery to 10 U PRBC transfusion	250–500 mL
Postop care	Generally, patients with smaller burns can be extubated at the end of these procedures, recovered in the PACU, and returned to the burn center. Patients with larger burns or those with inhalation injuries will often remain intubated. These patients are transported	⇐

	directly to the burn center, where body T can be maintained more easily.	
Mortality	18–40% (Mortality is a function of burn size, plus other associated conditions, especially inhalation injury [Figs. 11.5-1 and 11.5-2; Table 11.5-1].)	⇐
Morbidity	Massive blood loss Sepsis Infection 2° to catheters and lines	— ⇐ ⇐
Pain score	2–10 (Most patients are maintained on opiates for periop pain control.)	⇐

■ PATIENT POPULATION CHARACTERISTICS

Age range	All
Male:Female	More commonly male
Incidence	45,000 burn center admissions per year
Etiology	Scald; flame burns; chemicals; electric burns
Associated conditions	Generally few, except in elderly patients. Mental illness and substance abuse have been increasing recently in this patient population.

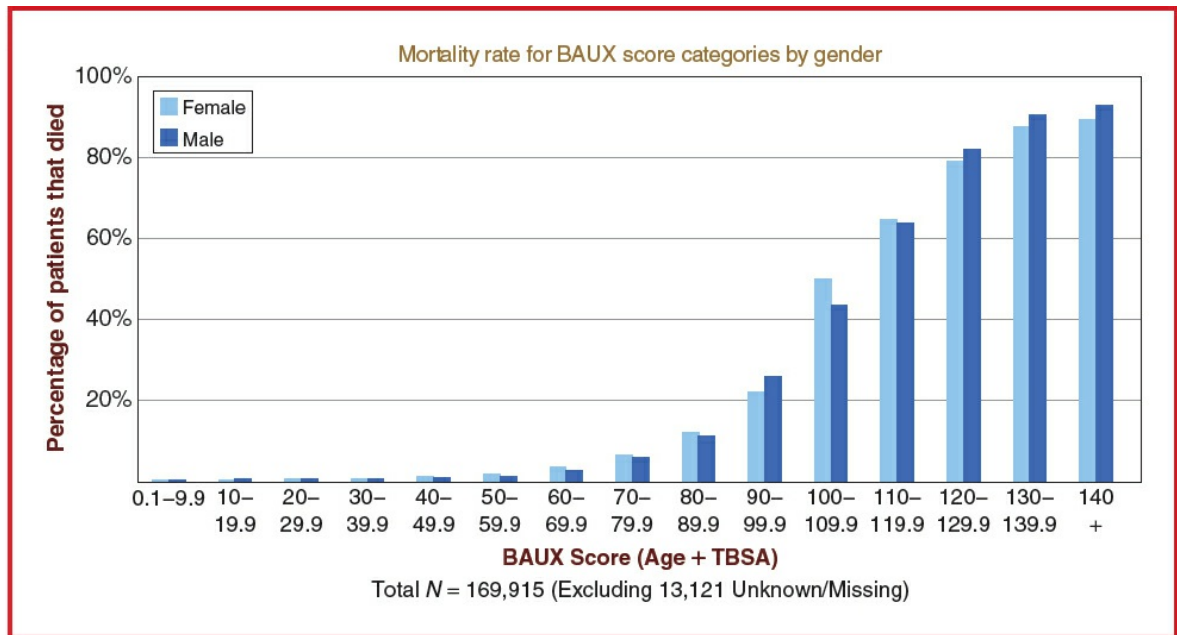


Figure 11.5-1 This figure demonstrates the relationship between death, increasing age, and burn size. The proportion of patients who died (case fatality) is plotted as a function of the sum of age and the total percentage of BSA burned, the so-called BAUX score. There is a strong association between BAUX score and case fatality for both men and women. Overall, women had a higher case fatality than did men (4.20% vs. 3.48%, $P < 0.01$). The sum of age and burn size (BAUX score) associated with a case fatality of 50% ($P = 50$) was 108. There was no significant difference between genders. A report from the American Burn Association, 2012.

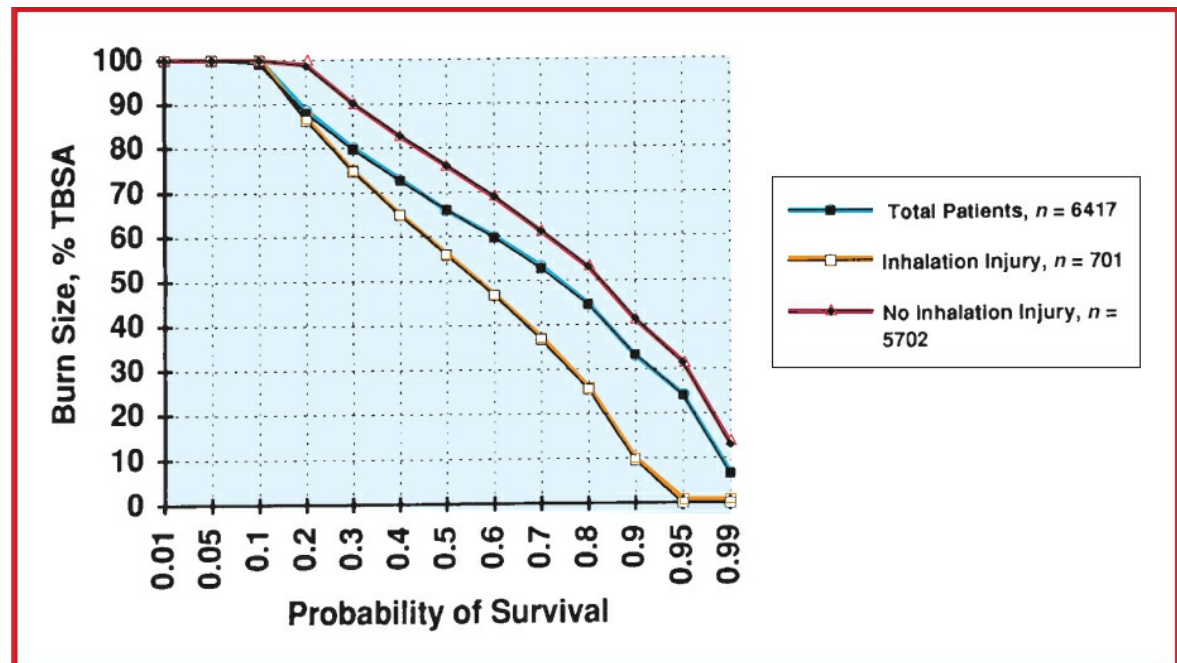


Figure 11.5-2 Probability of survival for patients with and without inhalation injury. (From Saffle JR, Davis B, Williams P, et al: Recent outcomes in the treatment of burn injury in the United States: a report from the American Burn Association Patient Registry. *J Burn Care Rehab* 1995; 16(3 Pt 1):219-32. Reproduced by permission of Burn Science Publishers, Inc.)

Table 11.5-1 Burn Wounds—Classified According to Depth of Burn

Degree	Burn Depth	Tissue Involved
1st degree	Superficial	Epidermis only
2nd degree	Partial thickness	Various thickness of dermis
3rd degree	Full thickness	Entire thickness of dermis
4th degree	To underlying tissue	Beyond dermis (e.g., subcutaneous fat, fascia, or muscle)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Burn injuries may result in a broad spectrum of physiologic impairments. These vary, depending on the percent of TBSA burned, location of burns, age of the patient, time elapsed since initial injury, and interim treatment. In the first hours following burn injury, patients may be affected by both carbon monoxide (CO) and cyanide (CN) toxicities. CO impairs O₂ transport, O₂ release at the tissue level, and mitochondrial function → tissue hypoxia + metabolic acidosis. Pulse oximetry measurement will be falsely high in the presence of CO. CN toxicity is usually the result of inhaled combustion products that include cyanide and hydrocyanic acid → metabolic acidosis, ↓ BP, and dysrhythmias. Ideally, burn patients are fluid resuscitated and stabilized before being brought to the OR. Typically, skin grafts to cover the burn area start on postinjury day 3. Blood loss and hypothermia are the predominant considerations during surgery on burn patients. Blood loss can be rapid and massive, as much as 8 U in 15 min, and can be difficult to estimate as it generally is not collected into the suction.

Respiratory

Upper airway: A patient with burns around the airway (e.g., singed nose hairs) should be intubated as early as possible. Direct inhalational/thermal injury and fluid resuscitation may make delayed intubation more difficult 2° upper airway edema.

Lower airway: Physiologic derangements may include pulmonary edema and ARDS even without inhalation injury. In addition, burn patients may be

	severely hypermetabolic (e.g., a patient with 40% TBSA burns may have twice the normal metabolic rate) with corresponding $\uparrow$ CO_2 production. These patients may have high PIPs and high minute ventilation requirements. Pressure control ventilation and high levels of PEEP may be useful. Other possible effects of severe burns include $\downarrow$ lung and chest wall compliance, $\downarrow$ FRC, $\uparrow$ A-a gradient, pneumonia, and pulmonary embolism. Tests: ABG, depending on pulmonary status; CXR, PRT
Cardiovascular	1–48 h postburn injury is associated with $\uparrow$ cardiac demand and a greatly elevated circulating levels of catecholamines; however, $\downarrow$ intravascular volume, myocardial depression, and $\uparrow$ SVR/PVR may $\rightarrow$ $\downarrow$ CO and $\downarrow$ urine output. 48–72 h post burn, the hypermetabolic/hyperdynamic phase begins. $\downarrow$ SVR may mimic sepsis. Tests: As indicated from H&P
Neurologic	Evaluate for burn encephalopathy. Characterize baseline mental status before anesthesia to allow evaluation of recovery postop.
Musculoskeletal	Damaged muscle $\rightarrow$ $\uparrow$ acetylcholine receptor density, resulting in $\downarrow$ sensitivity to nondepolarizing muscle relaxants and potentially fatal elevations of K^+ in response to succinylcholine. In burns $>5\%$ TBSA, avoid succinylcholine after 24 h post burn and for at least 1 yr thereafter for burns $>10\%$ TBSA ($\uparrow\uparrow \text{K}^+$). Recovery of normal response to muscle relaxants does not occur until burns have healed completely.
Hematologic	Coagulopathies may result directly from the burn injury, as well as from rapid replacement of blood loss during operative procedures. Tests: Hb/Hct; electrolytes; coagulation profile
IV access	May be difficult; assess preop. Consider central line placement with a large-bore catheter (e.g., Cordis®).

Fluid resuscitation guidelines	Parkland formula: LR @ 4 mL/kg/%TBSA/24 h (50% given in first 8 h); next 24h: 5% albumin @ 0.3–0.5ml/kg/%TBSA Brooke formula: LR @ 1.5 mL/kg/%TBSA + colloid 0.5 mL/kg/%TBSA (50% given in first 8 h); next 24h: 5% albumin @ 0.3–0.5ml/kg/%TBSA. UO goal: 0.5-1ml/kg/hr
Laboratory	Other tests as indicated from H&P. For recent injury, blood COHb should be measured. FENa < 1% or BUN/Cr > 20 suggests hypovolemia.
Premedication	Patients are commonly placed on high-dose narcotics after the initial injury; additional narcotics are frequently required to provide adequate analgesia for transport and movement to the OR table. It is important to identify opioid-tolerant patients to plan pain management.
Transport	For patients with severe ARDS, transportation from burn unit to OR may pose formidable challenges with regard to ventilation. Cardiopulmonary monitoring must be continued during transport; the ventilation system used in transport must be capable of delivering high minute volumes, PEEP, and inspiratory pressures. These requirements may not be satisfied by standard bag-valve systems and may require a high-quality transport ventilator.

INTRAOPERATIVE

Anesthetic technique

GETA: Regional techniques are rarely feasible, given the multiple surgical sites for harvesting and grafting. Some surgeons will employ tumescent infiltration of the burn and donor sites using a large volume of lidocaine-epinephrine-saline solution to facilitate debridement and graft harvest (↓ blood loss + ↓ postop pain). LMAs are not recommended, given the potential for significant fluid resuscitation and subsequent airway edema, as well as the frequent repositioning of patient intraop.

Induction	If the patient is adequately volume resuscitated, propofol (1.5–2.5 mg/kg iv) may be used. If the patient is intravascularly volume depleted, etomidate (0.3 mg/kg) or ketamine (1–3 mg/kg) is recommended. Patients with extensive burns (>30% TBSA) will develop a decreased sensitivity to nondepolarizing NMBs; therefore, 1.5× the usual intubating dose of NMB is recommended: vecuronium (1.5 mg/kg) or rocuronium (1.5 mg/kg). If rapid-sequence induction is indicated, succinylcholine should not be used if the initial injury is >24 h old. Under these circumstances, high-dose rocuronium (2 mg/kg) can be used. If the face is burned, awake FOI may be necessary and securing the ETT may be difficult. Alternatives to taping the ETT include wiring/suturing the tube to the teeth or using an umbilical tape strap.
Maintenance	Standard maintenance. Physiologic derangements of the respiratory system (ARDS, pulmonary edema) and a hypermetabolic state may require minute volumes >30 L/min and high inspiratory pressures and PEEP for adequate ventilation. Depending on the anesthesia machine, an ICU ventilator (capable of high-frequency oscillatory ventilation) may be necessary. Intraop, surgeons may use epinephrine-soaked sponges to ↓ blood loss. Systemic absorption of epinephrine will cause tachycardia and increase the probability of dysrhythmias; therefore, it is best to avoid halothane or desflurane. Isoflurane and sevoflurane are acceptable. Remifentanyl is associated with opiate-induced hyperalgesia in this patient population and should be avoided. A low-dose ketamine infusion (0.1–0.5 mg/kg/h) is useful, especially in patients with opiate tolerance.
Emergence	Estimation of an adequate dose of narcotic to provide postop analgesia may be difficult because these patients are often receiving high doses of narcotics preop. An aggressive dosing strategy may

be necessary. Ketamine has been used with success in providing improved analgesia in the setting of escalating opioid doses. If large-volume resuscitation has occurred intraop, there is the possibility of clinically significant airway edema; use caution before extubating to ensure a patent airway.

Blood and fluid requirements

Extensive blood loss
IV: 14–16 ga × 2 or a Cordis®
BSS @ 8–10 mL/kg/h
Keep UO @ 0.5–1 mL/kg/h.
Blood: ~200 mL/1% BSA excised and grafted
Fluid warmer
T&C 2–4 U PRBC (to keep ahead)

Blood must be in OR before induction. Major blood loss generally is associated with eschar excision, usually the first procedure. For patients without contraindications to hemodilution (e.g., CAD, anemia), it may be better to consider delaying PRBC transfusion until major blood loss is complete. IV hyperalimentation should be continued during surgery or replaced with 10% dextrose infusion to avoid hypoglycemia. If sudden ↓ BP occurs during very rapid infusion of blood (>150 mL/min), consider using Ca⁺⁺ to counteract the chelating effect of citrate. Avoid fluid overload, especially if patient has ARDS, is a small child, or is elderly. As the surgical site is superficial, there is not much 3rd-space loss.

Thermal

Room = ~82°F

Severe burn injury will

considerations	Warm all fluids. Humidify gases. Warming blanket Reflective head cover	disrupt thermal homeostasis. Temperature must be monitored throughout the case. The surgeon should be notified if patient's core T is dropping. 82°F is isothermic, however some surgeons will request substantially higher OR T, although there is no convincing evidence to support this practice.
Monitoring	Standard monitors + ART line ± CVP line ± PA catheter/TEE	ECG may require needle electrodes or alligator clip electrodes to skin graft samples if there is no skin availability to apply adhesive electrodes. The hemodynamically unstable patient should be monitored with a PA catheter/TEE and ART line.
Positioning	✓ and pad pressure points ✓ eyes	The burn patient may be uniquely susceptible to laryngeal or upper airway edema in the prone position; therefore, examination of the upper airway before extubation is recommended to avoid emergent reintubation.
Complications	Massive blood loss VTE	

POSTOPERATIVE

Complications	Hypothermia Coagulopathy	May occur as the result of massive blood loss and replacement
Transport	Continue cardiopulmonary monitoring. High minute ventilation requirements ↑ PIP ↑ PEEP	Verify adequacy of transport ventilation system before departing OR.
Multimodal pain management	Oral methadone or sustained-release morphine sulfate IV (PCA if possible) fentanyl or hydromorphone	Opiate requirements may become impressively large. IV or PO acetaminophen, NSAIDs, and gabapentin, often combined with a low-dose ketamine infusion, are all useful in acute burn and skin graft pain management.
Tests	Hct, ABG, electrolytes, PT, PTT, Plt, if massive transfusion given	

Suggested Readings

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3. Gillenwater J, Garner W: Acute fluid management of large burns: pathophysiology, monitoring, and resuscitation. *Clin Plast Surg* 2017; 44(3):495–503.
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6. Jones SW, Williams FN, Cairns BA, et al: Inhalation injury: pathophysiology, diagnosis, and treatment. *Clin Plast Surg* 2017; 44(3):505–11.
7. Moran KT, O'Reilly TJ, Furman W, et al: A new algorithm for calculation of blood

loss in excisional burn surgery. *Am Surg* 1988; 54(4):207–8.

3. Osler T, Glance LG, Hosmer DW: Simplified estimates of the probability of death after burn injuries: extending and updating the BAUX score. *J Trauma* 2010; 68(3):690–7.
4. Rizzo JA, Rowan MP, Driscoll IR, et al: Perioperative temperature management during burn care. *J Burn Care Res* 2017; 38(1):e277–83.
5. Zuo KJ, Medina A, Tredget EE: Important developments in burn care. *Plast Reconstr Surg* 2017; 139(1):120e–38e.